ROTATION INVARIANCE OF THE THREE-DIMENSIONAL ABELIAN HIGGS MODEL: THE CURVATURE SECTOR

Abstract. This document gives the proof details of the rotation part of the main paper. King gave a construction of the continuum limit and the infinite-volume limit of the lattice $U(1)$ Higgs model in three dimensions, with a massive non-compact photon, on lattices of a single orientation, and stated a theorem comparing two lattices whose orientations differ by $\theta_0 = \arctan(3/4)$ on periodic cubes. As printed that statement is false; a corrected form, for curvature sources read as 2-forms in each lattice's own frame and for the two lattices at the same spacing, is proved in the main paper. We prove that King's infinite-volume state on curvature sources, that is, the generating functional of all curvature Schwinger functions, built on the cubes on which King's renormalisation group is run, is $O(3)$-invariant, and then invariant under the full Euclidean group, on all of King's range of the couplings, every positive (e, λ, m, μ_0) at block factor 5. The proof passes through two statements, the corrected torus comparison on walled tori and a closing statement which says that King's state is the lower envelope of the walled tori. The first is the corrected torus theorem; the second follows from a heavy-shell decoupling estimate at one configuration, which in turn follows from four shell estimates for King's block-spin flow with the shell mass inside the covariance. Every input of the proof is stated in one numbered list, with the place where it is proved: here, in the main paper, or in the literature, cited as printed. Among the inputs proved in the main paper are the bounds at long trees for the graphs with block vertices in the families with one inserted external field, the fluctuation estimates for the in-block contours rooted at the block centres and for the contours of the hybrid flow, the positivity condition on the constant that relates the two counterterm conventions on the walled tori, and the large-field bound on the skew tori at the final scale of Bałaban's bounds; where the positivity condition fails, the invariance follows by right-continuity of King's state in the scalar mass. Further inputs proved there are a bound, at each renormalisation step and on the propagator domains of Bałaban's upper bound as well as on the torus, for the first-order term, in the photon fluctuation field, of the determinant of the step, uniform in the shell mass and with no bound on the size of the background, which has two independent proofs; on the skew tori of the corrected torus comparison, the decay, with the depth of its two points, of the difference of Bałaban's propagators on two nested domains; on King's cubes, the bound at each of two cutoffs, by the cut of the long lines, of the graphs whose trees are long, which gives their difference with no rate; and the transfer of the perturbative bound to Bałaban's propagator domains at order fourteen, for the families that the removal of the tail of the background produces. The steps that Bałaban and King assert without a printed argument, among them Bałaban's estimates at block factor 5, the renormalisation bound for the operator with a scalar wall included, King's cut of the long lines of his renormalised graphs on his cubes, and King's cube limit, are proved in the main paper, in the forms used, and so are the six readings of Bałaban's papers behind them, the bounds for the vector-field propagators and the vector half of one completion included (Section 8). Proved here are the following: an exact involution reduces the torus comparison to one-sided bounds at a single spacing, correlation inequalities compare King's cubes with walled tori, the photon boundary condition is changed only across a matter-free collar where the change is controlled uniformly in the spacing, a two-level sandwich turns the heavy shell into a higher wall, and a continuity lemma for convex functionals carries invariance from a dense subgroup to $O(3)$. The extension to scalar sources is stated as an open problem. An appendix lists the corrections to the literature that the argument needs.

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1. Introduction

1.1. **Background.** The lattice $U(1)$ Higgs model couples a two-component scalar field $\phi = re^{i\vartheta}$ to a non-compact abelian gauge field A. In three dimensions King gave a construction of its continuum limit [K1, K2], whose printed proof asserts several steps (items (5k) and (13) of Section 1.3), using the ultraviolet stability bounds of Bałaban [Ba1, Ba2, Ba3, Ba4], for the version of the model in which the photon is written in Feynman form and carries a mass $\mu_0 > 0$ [K2, (2.5)]. In [K1] the continuum limit is taken on tori. In [K2] it is taken in the cubes C_J with Neumann boundary conditions for A and Dirichlet boundary conditions for ϕ, realised on the lattice as the cubes C_J^K on which King's renormalisation group is run (Definition 2.15), and the infinite-volume limit $J \to \infty$ is then obtained from the correlation inequalities of Brydges, Fröhlich and Seiler [BFS], which make the cube expectations monotone in J [K2, Thm. 3.2] (with a J-independent Wick constant, in the form King's proof proves, for the boxes of that proof and for the cubes on which his renormalisation group is run; Lemmas 2.14 and 2.16). King states that the infinite-volume generating functional is translation invariant, reflection positive and entire in the coefficients of the sources [K2, Thms. 2.2–2.3]; on curvature sources we prove the parts that we use (Section 3).

Rotation invariance is not automatic, since all lattices in the construction have the same orientation. Theorem 2.4 of [K2] states that on the tori D_J the lattices $\varepsilon_k\mathbb{Z}^3$ and $R(\varepsilon_k\mathbb{Z}^3)$, with R the rotation by $\theta_0 = \arctan(3/4)$ in a coordinate plane, have the same continuum limit; since θ_0/π is irrational these rotations generate a dense subgroup of $SO(3)$. As printed the statement is false in two ways (Appendix A): the source it pairs with the plaquettes is one scalar function on both lattices, which couples the field to two different 2-forms, and the Wick constant it uses misses an order-e^2 logarithm. We therefore call the printed statement *King's torus conjecture*, and we use instead the corrected statement, for 2-form sources sampled in each lattice's own frame and for the two lattices at the same spacing, which is proved in [CT]. Moreover, even granting the torus comparison, two steps remain between it and rotation invariance of King's state: the infinite-volume limit along tori, where correlation inequalities give no monotonicity, and its identification with the limit along cubes, which King writes can be expected from a cluster expansion [K2, (2.24)]. The ultraviolet analysis of [K1, K2] needs boundaries that are lattice hyperplanes [K2, p. 326], and no bounded region has that property for both lattices [K2, p. 326]. Where to place the two-lattice comparison is therefore the whole question. This supplement, which gives the proof details of the rotation part of the main paper, places it on periodic tori carrying a smooth mass wall, and proves $O(3)$ invariance of King's state in the curvature sector. The inputs of the proof are listed in one numbered list (Section 1.3), each with a one-line mathematical statement and the place where it is proved: here, in the main paper, or in the literature, cited as printed; the list is the complete set of inputs that are not printed theorems whose proof has been read at the source. The theorem holds on all of King's range. Among the inputs of the shell estimates is the bound (DOM) of Section 1.4 for the first-order term of the step determinants on Bałaban's propagator domains, which has two independent proofs; three statements inside the proofs of the torus comparison, of King's cube

limit and of the shell estimates, (P7)–(P9) of Section 1.4, are proved in the main paper as well. Section 8 restates in plain words every input that is not proved here.

1.2. **The torus route.** On each lattice separately King's cubes are compared with tori carrying a *smooth mass wall*. Fix R in the generating set $\mathcal{R}_0 = \{R_{\mu\nu}(\pm\theta_0)\}$ and a real 2-form f, smooth with compact support. For a walled torus $\mathfrak{c} = (J', w)$, the periodic model on $D_{J'}$ in which the scalar mass squared is raised by $2w$ with $w \geq 0$ smooth and vanishing near the support of f, the chain is

$$Z_{C_L}(\mathrm{i}\,R\cdot f, 0) \;=\; Z^{R^{-1}\Lambda}_{R^{-1}C_L}(\mathrm{i}f, 0) \;\leq\; T^{R^{-1}\Lambda}_{\mathfrak{c}}(f) \;\xrightarrow[\varepsilon\to 0]{}\; \mathcal{W}_{\mathfrak{c}}(f,0) \;\xleftarrow[\varepsilon\to 0]{}\; T^{\Lambda}_{\mathfrak{c}}(f). \tag{1.1}$$

Here Z_{C_L} is King's cube functional on the axis lattice Λ, the first equality is a passive rotation, the inequality is a correlation inequality on the rotated lattice $R^{-1}\Lambda$, for which $R^{-1}C_L$ is a cube with faces on lattice hyperplanes, and the two continuum limits agree by the torus comparison. What is missing is an upper bound for walled tori on the axis lattice by King's state:

$$\text{for every } \eta > 0 \text{ there is an admissible } \mathfrak{c} \text{ with } \mathcal{W}_{\mathfrak{c}}(f,0) \leq Z_\infty(\mathrm{i}f, 0) + \eta. \tag{1.2}$$

This is Hypothesis (TIGHT_0) of Section 5. It involves one lattice and one orientation, and the reverse inequality holds in the limit (Lemma 5.2), so (TIGHT_0) says that King's state is the lower envelope of the walled tori. Granting it, letting $L \to \infty$ in (1.1) gives $Z_\infty(\mathrm{i}\,R\cdot f, 0) \leq Z_\infty(\mathrm{i}f, 0)$, and applying this to $R^{-1} \in \mathcal{R}_0$ gives equality. The passage from the dense subgroup to $O(3)$ uses a continuity lemma for convex functionals (Lemma 3.2), so no regularity of the Schwinger functions is needed.

We isolate the local estimate that implies (TIGHT_0). In the walled torus let the source sit in a box H and let U be a thick shell around H inside the heavy plateau, where the wall is at least 1. Raise the scalar mass squared on the shell further by $2K$, through a second smooth profile, and let $\Psi_k(n, K)$ be the resulting expectation at lattice spacing ε_k, with $\Psi_k(n, \infty)$ the expectation with the matter removed on the shell. The hypothesis, (RES_0) of Section 7, asks that for one such configuration and one finite K

$$\liminf_{k\to\infty} \big[\Psi_k(n, K) - \Psi_k(n, \infty)\big] \leq \eta,$$

with the width of the shell chosen so that a photon boundary term is below η as well. At every fixed spacing the bracket tends to 0 as $K \to \infty$ (Lemma 7.4); the content of (RES_0) is that the heavy-shell limit and the continuum limit can be interchanged, at one geometry and with no rate. A two-line sandwich (Lemma 7.6) shows that the heavy shell is bounded below by a higher wall, and once the matter on the shell is dead the photon on the far side can be pinned and then replaced by King's Neumann photon on a larger cube at a cost exponentially small in the width of the shell, uniformly in the spacing (Lemma 7.1); so (RES_0) implies (TIGHT_0) (Theorem 7.8). In the curvature sector (RES_0) at one configuration is itself proved (Theorem 7.10): the shell mass κ is put inside the matter covariance of King's block-spin flow, and four estimates on that flow, two-sided bounds on the stopping-scale integral uniformly in κ, a disc bound for the heavy shell, a κ-Cauchy property of the reduced small-field action, and a threshold clause, give the tail bound at one configuration. Each of the four is proved in the main paper [SW1, SW3]; the statements used inside those proofs are listed in Section 1.3, each with the place where it is proved.

The torus comparison is reduced to one-sided bounds at a single lattice spacing (Section 4). The passive map $S = R^{-1}\sigma_\nu$ is an involution that exchanges the two lattices and maps the torus D_J to the torus $R^{-1}D_J$. A forward bound of King's type on each of the two tori therefore gives both directions of the comparison on D_J. The forward bounds and the existence of the axis-lattice limits are the content of [CT], whose own closing supplies them with the large-field clause of King's upper bound in place of King's torus bound [K2, (4.2)], which is vacuous at a non-zero wall; we state its

theorem and its inputs. An alternative placement of the comparison, on a Neumann–Dirichlet box aligned with one lattice and a staircase for the other, is possible in principle and is not used here.

1.3. **Inputs.** We fix $e > 0$, $\lambda, m, \mu_0 > 0$ in King's range and King's block factor 5 (King's L; below, L denotes only the index of a King cube C_L). The proof of the theorem of this supplement runs through two statements about King's walled tori, (K_0) and (TIGHT_0) (Hypotheses 4.1 and 5.4):

(K_0) *The torus comparison, curvature sector*: for every $R \in \mathcal{R}_0$, every walled configuration $\mathfrak{c}$ of the class (6) below and every real smooth periodic 2-form G, $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are equal.

(TIGHT_0) *The closing statement, curvature sector*: $\mathcal{W}^{\mathrm{env}}(f,0) \le Z_\infty(\mathrm{i}f,0)$ for every real 2-form $f \in C_c^\infty$; that is, King's state is the lower envelope of the walled tori.

Both are proved: (K_0) is proved from the inputs (1)–(5) of the following list, on the class (6), by the corrected torus theorem (Theorem 4.8); (TIGHT_0) is proved from (K_0), the heavy-shell estimate (RES_0) at one configuration and the dead-shell chain (10c) at that configuration (Theorem 7.8; also written out in [TE]), and (RES_0) is proved from the inputs (8)–(11), with their literature inputs (12), by Theorem 7.10, which is item (7). The list (1)–(13) and (15), together with the statements of Section 1.5 (Propositions 1.2, 1.3, 1.4, with their inputs), is the complete set of inputs of the theorem that are not printed theorems whose proof has been read at the source; item (13), King's cube limit, is an input because the theorem's object is defined through it, item (14) is proved here from (13), and item (15), the right-continuity of King's state in the scalar mass, is used at the points of E_W only. The objects are defined in Sections 2–7 and in [CT, H6, RS, SW1, SW3], whose notation is used in (1)–(12). Every item of the list is proved here, proved in the main paper, or is a statement of the literature printed with its proof; where a proof uses a statement of the literature that is asserted in print without a complete argument, the item names the part of the main paper in which that statement is proved. No item depends on a hypothesis of the theorem beyond King's range. Section 8 restates in plain words every input that is not proved here.

(1) **$(\mathbf{BAL})_5$.** Every estimate of [Ba1, Ba2, Ba3, Ba4] that is asserted there without a printed argument (Section 8.2) holds as asserted at blocking factor 5, with per-step constants that depend on the step only through the running scale (scale-local), hence uniformly in the initial spacing, including the large-field clause of King's upper bound [K1, (3.4)] on the skew torus: the large-field contribution to the partition function at the stopping depth $r(k)$ of the chain, $k \ge k_0$, where $k - r(k)$, the number of steps left to the chain, is a fixed fraction, less than one ninth, of k, rounded down (and at every admissible depth for the standard flow on D_J), is at most $\exp(-c_* p(\delta_{r(k)})^2 + C|T| + o(K^2))$, with $p(\delta) = b_0(1 + \log \delta^{-1})^p$ at the fixed integer p of Convention P (Section A.1) and a constant $c_* \in (0,1]$, $c_* = 1$ at zero source, every $c_* < 1$ at a non-zero curvature source (with constants depending on c_*) and $c_* = 1 - 1/p_H$ at a non-zero wall (every consumer uses only $c_* > 0$: $p(\delta_{r(k)})^2$ grows at least quadratically in k). Asserted in print without a complete argument; proved in [AS], at zero scalar source, from Bałaban's printed proofs, at every background, the replacement of the propagators in the interaction terms of [Ba2, Prop. 2.6] included (Section 8.2; on the skew tori of [CT] through the change of domain (P7) of Section 1.4), in the forms used by (K_0) and by the shell estimates, among them the renormalisation bound for the operator with the scalar wall in the form (P1) of Section 1.4; six readings of Bałaban's papers, (B1)–(B6) of Section 8.2, are not covered by [AS] and are proved in [SUP]. The dependence on the blocking factor, and the repair at block factor 5 of four printed steps (one of which is not covered by its printed proof at any block factor), are proved in [CT] (the two repair steps (5l)(v), (vi) proved); the re-runs of the same steps for the model with the shell mass inside the covariance are item (8e). The repair at block factor 5 of Bałaban's cube step [Ba3, p. 433] uses an end-functional bound of those re-runs at zero

shell mass, and through it item (1), and with it (K_0), rests on the readings (B3), (B5), (B6) and on the regrouping of the cut families, all proved. At a non-zero wall the large-field clause is proved in [SUP], from the clause at zero scalar source, an upper bound for the walled partition function obtained by exact shifts of the scalar mass to other masses, among them a mass M_1 with (e, λ, M_1, μ_0) in King's range, and correlation inequalities, and a Hölder bound for the large-field part, with King's upper bound at fixed sources in its proved form (Section 8.3); the mass M_1 exists at every point of King's range at which the positivity condition of Section 1.5 holds, and at the remaining points, a set that is a graph over the couplings (e, λ, μ_0) up to ties, Theorem A is obtained by right-continuity in the scalar mass (Section 1.5). For the rotated lattice, whose flows in [CT] use in-block contours rooted at the block centres, and for the composite contours of the hybrid flow, the estimates of [Ba1, Ba2, Ba3] and of King's upper bound are proved in [SUP], by a census of every occurrence in the printed proofs of the operators built from the contours (Section 1.5); on the skew torus the clause holds at every J of the class (6) under the cap on the coarsest scale that the class carries (Remark 5.5).

(2) **(H1*).** For some θ_1 with $0 < \theta_1 < \min(\alpha, 1)$ and every $0 < \theta \le \theta_1$, the two-flow comparison bound of [CT]: for every renormalised family of [K1, §3.5] of order at most $\bar{n}$, in per-flow normal form and with the h-linear field-independent families excepted, the difference between its values on the standard and on the hybrid flow is bounded by the right side of Bałaban's renormalisation bound [Ba3, (1.33)] times η'^{θ}, η' the ratio of the fine spacing to the block side at the scale in question, times a factor polynomial in the number of scales, in $p(\delta_k)$, in $\delta_k^{-1/2}$ and in the number of vertices, uniformly in the scale, the torus and the masses; any $\theta > 0$ with any polynomial factor suffices for the consumers. Bałaban's bound with one element weakened by θ_1 in all five sectors for the two-flow families is not an input. Proved in [CT], from (3), (5) and:

   (2a) the first-layer slice-kernel bound $|\nabla_r \partial_b S_j^y(r, b)| \le C\ell_j^{-3} e^{-\delta|r-b|/\ell_j}$ for the slice kernels of [CT] at scale j with two derivatives, which [CT] obtains from the cubic-Hermite cardinal interpolation of the lattice kernels: proved in [CT] (the extension by the sharp box-zone formula, whose decaying form is false, is not used);

   (2b) the restricted contraction rule and the local-term chain of the triangle and overlapping-triangle compounds under it: in the contraction estimate for the two-flow families, the moment-removal contraction at a bare vertex, or at a vertex of an unbundled split unit, is applied only when the vertex's photon field is external or a photon line outside the component; when it is a line inside, the component is closed by the common-moment bracket instead, for the triangle compounds after a first-order bundling at a leg: proved in [SUP].

   The one-flow instance is what King asserts in the proof of [K1, Prop. 3.6], p. 665.

(3) **($\mathbf{B}_\Delta$).** The hybrid block-spin kernels of [CT] obey the decay and regularity bounds of the standard kernels [K1, Thm. 3.3], in the periodic form (on the infinite lattice and on periodic cells fitted by both lattices, with the two-torus clause in the smaller quotient metric; the Neumann variant on digitised rotated domains is neither claimed nor used), with the Hölder constant $C_\alpha(1 + \log 5^k)^{3\alpha}$, and the difference elements of the two flows (including contour lines, contour vertices and the constant-background phases) carry a positive power of the scale ratio. Proved in [CT], with the step (F2) of Section 8.5 in its completed form, the steps (5l)(viii), (ix) included; the slice-kernel decay rests on a decay lemma, proved in the clauses consumed (the kernel of each slice at a regular background, its covariant derivatives at one or both ends, and their transported Hölder quotients).

(4) **(H6W).** For every configuration $\mathfrak{c} = (J, \psi, n)$ of the class (6) and every real smooth periodic 2-form G the axis-lattice limit $\lim_K \langle e^{tF(G)} \rangle_{\mathfrak{c}, X_K}$ exists for all real t, where $X_K :=$

Λ_K is the axis lattice of spacing ε_K (Section 2), and it is the same number in the free-boundary conventions and in King's torus conventions at the shifted constant source. This is Theorem 2.1 of [K1] with the wall as King's scalar source, read on the normalised ratio, in which the field-independent constants linear in the source cancel, and carried from King's torus conventions to the free-boundary conventions, a further step written out in [H6] (the printed statement is affected at $\int h \neq 0$, Appendix A); the statement is not an input, its proof is written out in [H6]: proved there from (1), (5), the convention bridge (CT) of [CT], the polynomial lower bound at the wall sources ((H5) at the wall as proved in [CT], or, as [H6] writes it, Bałaban's lower bound [Ba1, (1.14)] with Jensen's inequality and the Wick bound (4c) under the wall condition of (6)), the restricted form of King's renormalisation theorem [RS] (item (5h)) and King's estimate [K1, Prop. 3.6] in its one-flow instance (item (5g)). Of its literature inputs, (1) and King's asserted steps (5k) are proved in [AS]; two further statements of King are proved in [SUP]: his readings of the sources, in the form of the small-field half of his upper bound [K1, Thm. 3.1] at every fixed source of bounded size, for King's flow and for the centred flow, and the source-free strict degree property of King's renormalised graphs as rewritten by Bałaban ([Ba3, (3.5)], [K1, (3.78)]), which King asserts at [K1, p. 668], in the componentwise form of [Ba3, (3.5)] (Section 8.3). Its residues, each proved [SUP], are:

(4a) the Taylor step for the self-energy in the order-$(2,0)$ vacuum family, applied with its counterterm partner although the whole graph has degree 0, in its position-summed form [RS, Thm. R-iii(b)];

(4b) Bałaban's ordering-restricted rewriting applied to the field-free members of the classes R-ii and R-iii(a) of [RS] (on the classes with an external leg or a curvature vertex it is proved in [RS]);

(4c) the torus one-point Wick bound beyond one loop, at every coupling in King's range, with the uniformity of its constants on a compact interval of masses. The rate $|\Delta_T(s) - \Delta_T(0)| \leq Cs$ of the convention bridge, which (4) consumes and no other item does, is proved at one and at two loops (which exhaust King's counterterm, [K1, (3.32)]) ([CT], (CT)(b); [H6]);

(4d) the change between two cutoffs of the non-polynomial terms of King's effective action [K1, (3.14)] at the frozen wall, at the rate that (4) uses, for which [H6] cites lines of [K1, (3.96), (3.98)] that are false as printed; these lines are not used.

Item (4) uses the large-field clause of (1) at the wall, on the axis lattice with King's own blocks; through it (4) is proved at every point of King's range at which the positivity condition of Section 1.5 holds; at the remaining points Theorem A is obtained by right-continuity in the scalar mass, and item (4) is not used there.

(5) **Literature inputs of the torus theorem**, each with the place where it is proved:

(5a) decay and regularity of the block-spin operators of the standard flow, [K1, Thm. 3.3], proved in [Ba4, Lemmas 2.1, 2.2, 2.4, Cor. 2.3]: printed with proof, except three steps that are asserted without an argument, the scale-uniformity of the analyticity neighbourhood in [Ba4, Lemma 2.4] (p. 586) and the identities [Ba1, (2.39)–(2.42)] (p. 612) behind the zero-background case of [Ba4, Lemma 2.2], which are proved at block factor 5 under (1), by written derivations [AS], and the photon clause of [K1, Thm. 3.3] (p. 656, "of course with $A = 0$"), a specialisation of two lines on the torus that is not written in print, proved under (5k); the use of [Ba4] at the coefficients $a_k \in [a(1-5^{-2}), a]$ and uniformly in a constant $m^2 \geq 0$, which the printed statement leaves implicit, is proved under (1) as well; the transfer of [Ba4, §§2–4] to the tori of [CT] is (5l)(iv);

(5b) the polynomial lower bound at zero wall [Ba1, (1.14)], whose proof in [Ba1, §§3–4] is a frame with asserted steps: asserted in print without a complete argument; proved at block factor 5 under (1), given (CT);

(5c) the zero-momentum counterterm conditions [K1, (3.32)–(3.33)] through order four (definitions), the Ward identities and the vanishing of the one-loop vacuum-polarisation moments for full propagators [Ba3, (2.18)–(2.28)], and Proposition 2.1 and (2.8) of [Ba3] with the ordering device (2.7) there: printed with proof;

(5d) the local limit law for the simple random walk [LL, Chap. 2]: a textbook theorem, cited by name;

(5e) Reisz's power-counting and renormalisation theorems [Re1, Thm. 1], [Re2, Renormalization Theorem, p. 92], in the corrected three-dimensional form of [CT]: three printed statements need repair, and the naturalness clause of the printed theorem is false as printed and is replaced by a condition proved for the graphs consumed; the forest identity [Re2, Lemma 5.1], whose proof there is a pointer to Zimmermann's paper, is Zimmermann's forest formula (Section 8.6), with its hypothesis [Re2, (2-24)] checked; and the membership of the subtracted integrands in Reisz's class, asserted at [Re2, (5-1)] and on p. 97 there, is proved in Remark 8.1. Proved in the main paper, in a derivation of [CT] from the printed proofs [SUP];

(5f) the uniformity of the constant of King's upper bound [K1, Thm. 3.1] under a shift of the scalar source by a bounded constant (King's statement that the constant depends on g, h and $|T|$), used by the convention bridge (CT)(b) for (H3), (H4): proved under (1) and (5k), from the proof of [K1, Thm. 3.1], which refers to [Ba1, Ba2, Ba3, Ba4] and to its Appendix; King's statement that the small-field analysis includes the sources [K1, p. 677] is proved [SUP], in the consumed form: the small-field half of [K1, Thm. 3.1], in King's vertex form [K1, (3.14)–(3.21)], at every fixed source of bounded size, on King's aligned tori and on the rotated torus, for King's flow and for the centred flow; for the hybrid flow the lower half, which is the half consumed, with a cap depending on the number of scales;

(5g) King's estimate [K1, Prop. 3.6], stated on p. 662, whose proof on pp. 663–665 ends in an assertion (p. 665) and uses [K1, Props. 3.8, 3.9, Lemma 4.5]: its one-flow instance is proved in [CT, H6], from [Ba3, (3.5)], [K1, Props. 3.7–3.9] (item (5k)) and a degree count of the subgraphs, for the classes with block vertices with the split count of (5l)(x); its degree count is proved: for King's flow the strict degree property named in (4), in the componentwise form of [Ba3, (3.5)], is proved in [SUP] (the componentwise replacement of the sources by their values at the coarse points, p. 665, costs $O(5^{-k})$, by a summation by parts for the curvature source and the mean value theorem for the scalar source and the partition of unity, the sources entering through their supremum norms only: proved in [SUP]); for the hybrid flow the unweakened form (degree $D \geq 0$ with a bounded count of the zero-degree subgraphs) is proved, and the strict form $D' > 0$ is not consumed;

(5h) King's renormalisation theorem [K1, Thm. 3.5(i), (3.38)] is false as printed on the h-linear vacuum family of order $(2, 0)$, and is not used; its restricted form [RS], Theorems R-i–R-iii, is proved there, with the degree count of its merged counterterm vertices, with the split count (5l)(x) for the classes with block vertices, and with the residue (4b), both proved;

(5i) the hybrid lower bound (H4) of [CT], the analogue of [K2, Prop. 4.1] for the hybrid flow (whose printed (4.5) has a sign slip and is replaced in [CT]): proved in [CT], from (1), (3), (CT), the skew instance [SKEW-L], its last input being Bałaban's lower-bound argument at block factor 5 on the torus of the step, with its own steps (5l)(i)–(iii),

and the lower half of King's small-field bound for the hybrid flow at the fixed sources it uses (item (5f)); the bridge from King's vertex form of that bound to the Cameron–Martin form of the effective action in which [CT] poses (H3) and (H4) is proved, with the insertion bound for the background legs at every scale, which has two independent proofs, both in [SUP];

(5j) the further residues of [CT], proved in [CT, SUP]: the standard bound on the free massive lattice propagator and its first three differences (on the infinite lattice, uniformly in the spacing and in the mass); the completions of the places where [Ba1, Ba2, Ba3] sketch (the depth inventory, the cube count and the constants of [Ba2, §3C], whose values at block factor 5 are (5l)(vii)), which are part of (1); and the decay bound for the coarse covariance, whose derivation in [CT] by the Paley–Wiener theorem is incomplete as written and which holds by a second derivation without that theorem ((F1) of Section 8.5);

(5k) **King's asserted steps of the torus construction**, each named in Section 8.3: [K1, Thm. 3.1], whose proof refers to [Ba1, Ba2, Ba3, Ba4] and to its Appendix; the propagator bounds [K1, Props. 3.7–3.10, Lemma 4.5, (3.72), (3.88)] and the remainder bounds of [K1, Thm. 3.3], of which [K1, Prop. 3.7] is a printed sketch and the proofs of [K1, Props. 3.8, 3.9] are printed for the last line of (3.71) and the first line of (3.73) only; the photon clause of [K1, Thm. 3.3] (p. 656); and King's readings of the sources, the componentwise replacement of the sources (p. 665) and their inclusion in the small-field analysis (p. 677), with the uniformity of his constants over the source family (pp. 664–665). Asserted in print without a complete argument; proved in the forms consumed in [AS] ([K1, Prop. 3.10, (3.72), (3.88)] by written derivations); King's readings of the sources (the componentwise replacement, p. 665, the inclusion in the small-field analysis, p. 677, and the uniformity of his constants over the source family, pp. 664–665) are proved in [SUP], in the consumed forms of item (5f) and Section 8.3. Of the steps behind [K1, Lemma 4.5], the coercivity bound (4.33) is proved here (Proposition A.4).

(5l) **The own steps of** [CT]**.** The closing implication of [CT], from the inputs (H1)–(H6), (SK), $(\mathrm{BAL})_5$ and (CT) of Section 4.3 to (K_0), is proved there. The own steps of [CT] on the chain of that implication are the following, together with the Taylor step (4a) and the Hölder bound (F3); each of them is proved in [SUP]:

- (i) the cumulant bound that replaces the cumulant lemma of Benfatto et al. (Section 8.2), a bound on the cumulants of the fluctuation integral of one renormalisation step with constants free of the spacing and of the shell mass, with its consumed forms on cubic tori, with and without the shell mass, and its form on the skew tori of [CT]. It is also an input of (8);
- (ii) in the hybrid lower bound (H4), the absorption of the additional lower-bound term of (i) into the factor $e^{-C\delta_k^{\sigma}|T|}$;
- (iii) Bałaban's renormalisation bound [Ba3, Prop. 1] for the hybrid flow, with its instances on D_J and on RD_J used in (H4), in the form consumed (the hybrid-flow lemma as printed in [CT] has a gap and is not used); its degree input for the hybrid flow is the unweakened form of the degree count, proved (item (5g));
- (iv) the transfer of Bałaban's bounds for the lattice Green's functions [Ba4, §§2–4] to the admissible lattice tori of [CT], read at the printed page;
- (v) in the repair at block factor 5 of the steps of [Ba3, pp. 433–445], the independence of the domain after the split;

(vi) in the repair of the identification step, the two identities that identify Bałaban's all-small-field term with King's effective action, the bound for the determinants that enter them included;

(vii) the constants of the completion of [Ba2, §3C] at block factor 5, and the uniformity in the scale index of the bounds that [Ba2] writes as $O(1)$ without qualification; they are also inputs of (8). Their vector half is the reading (B4) of [Ba2, p. 591], proved in [SUP] (Section 8.2). The count [Ba2, (3.45)] and the first inequality of [Ba2, (3.46)] are false as printed, at every block factor, and are not used: a repaired count of the cubes replaces them;

(viii) two statements of (3): the double contour lines of [K1, (3.26)], with both ends on contours, as a further type of difference element of the two flows, and the bounds of [K2, Prop. 5.6] for the inverses of the hybrid operators restricted to a union of large blocks, which (H4) uses;

(ix) four steps of lower weight in (3): the cover of the skew tori by cubes; the polynomial size of the constant in the convergent remainder of the degree count of the appendix of [CT]; the domain difference of the hybrid kernels in the second clause of (3), in the form consumed (a lift-distance clause printed with it is false and is not used); and the first clause of (3) at the first two levels of the hybrid flow, where [CT] states it for the later levels only; and with them the regularity across the seam of the periodic cell, in the form consumed;

(x) for the classes of graphs with block vertices, the degree counts behind the restricted renormalisation theorem (5h), the ordering-restricted rewriting (4b) and the one-flow instance of King's estimate (5g): King's credit of one degree for each photon field contracted outside a subgraph [K1, p. 664, (3.68)–(3.70)] is false as printed for the counts of subgraphs, and the conclusions are carried by a count that splits the degree, with the step that the split degree governs the multiscale bound. These counts enter (4), and through the restricted theorem also (10d) and (10g)(i).

(4a) enters (K_0) through (4), and (F3) through (iii).

(6) **The configuration class.** A definition, printed because the infimum in (TIGHT_0) is taken over it: $J \in 5^{k_*}\mathbb{N}$ with $J \geq \max(J_0(\varepsilon_0), J_{\mathrm{pos}})$ (k_* the number of large-field scales of the rotated flow, $J_0(\varepsilon_0)$ the non-wrapping threshold of the skew torus at the coarsest scale, J_{pos} the threshold of the positivity condition of Proposition 1.4, above which its margin holds; the consumer takes J above it); $\mathfrak{c} = (J, \psi, n)$ with $\psi \geq 0$ smooth and D_J-periodic, $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$, and $n > 0$ (a mass-increasing wall); the wall condition $n\langle\psi\rangle_{X_K} > \Delta_*|T|/2$ for every K, with Δ_* the bound of the convention bridge (CT)(b) of [CT]; block alignment: H' and T are unions of blocks of the final scale, whose side we denote ℓ_f; and the grid offset: with $M = 5M'$ and the prime $\mathfrak{p} \neq 5$ dividing M' of Definition 2.1, every corner coordinate of H' is $c\,\ell_f$ with $c \equiv M'/\mathfrak{p} \pmod{M'}$ (the sides $50MJ$ of D_J are multiples of $5M\ell_f$). Theorem 7.8 produces configurations in this class (Remark 5.5).

(7) **The heavy-shell theorem.** For King's walled torus with the shell $U = T \setminus H'$ (the whole exterior of the source box) and the shell mass $\kappa\mathbf{1}_U$ inside the matter covariance, the four estimates (8)–(11) imply (RES_0) at one two-level configuration $(\mathfrak{s}, \chi)$ with plateau height $n \geq n_*$ (n_* the threshold of Section 7.4) and one finite K (Theorem 7.10, proved in Section 7.4).

(8) **$(\mathbf{SW1}^\circ)_{\mathrm{int}}$.** For every stopping scale $\eta_0 \leq \eta_1$ there are $a(\eta_0), b(\eta_0) < \infty$, with $a(\eta_0), b(\eta_0) \to 0$ as $\eta_0 \to 0$ at fixed T, such that, uniformly in $\kappa \geq c_0/\eta_0^2$ and in $\varepsilon \leq \varepsilon_0(\eta_0)$, for $z \in [-1, 1]$ and $g' \in \{0, g\}$, the partition function with sources (zf, g') and the capped stopping-scale

integral I_κ^{cap} of [SW1] satisfy $e^{-a} I_\kappa^{\mathrm{cap}}(zf, g') \le Z_\kappa(zf, g') \le e^{a} I_\kappa^{\mathrm{cap}}(zf, g') + b\, Z_\kappa(0,0)$, for the κ-model of Definition 2.4 (free-boundary δm_ε^2), that is the model of [SW1] with its counterterm offset by $\Delta_T(\varepsilon)$, $|\Delta_T| \le \Delta_*$ (the shell-side bridge $(\mathrm{CT})_{\mathrm{sh}}$, [SW1, Lemma M(d)]). Proved in [SW1], at block factor 5, with the steps (F2), (F3), (F5) of Section 8.5 in their completed forms, from the following residues, each proved; the determinant clause of (8k)(ii) uses for the first-order term of the step determinants the bound (DOM) of Section 1.4, on the torus and, on the histories of the upper bound with proper propagator domains, on those domains; on the same histories the removal of the tail of the background uses the sub-domain form of the perturbative bound at order fourteen, (P9) of Section 1.4, through (8i) and the difference form of (P2):

(8a) the frozen-scale sub-domain bound in scale-free form, [SW1, Lemma subdom(a)(iii)], with a constant independent of c_0 for every $c_0 > 0$ and a rate depending on c_0 (the form with the constant 16 and the rate s'' in the penetration depth is not used);

(8b) the statements written in [SW1] as short extensions of proved arguments: the face clause at one block [SW1, Lemma subdom(b)]; the tangentially localised amplitude on every sub-domain [SW1, Lemma DVlocN]; the sub-domain form of the perturbative bound with its boundary-difference clause, in Bałaban's restricted form [SW1, Prop. TDsub, Lemma dG]; the legs in the half-amplitude norm and the derivative clause for small-field data [SW1, Prop. legs(a),(d)]; the norm form and the mismatch factor of the perturbative bound [SW1, Prop. TD(b),(c)]; King's Appendix class for the shell model with the background-gradient and block-mismatch clauses [SW1, Lemmas appclass, G, P], with the indicated remainder of the expansion in the background field; the e^2 and two-loop parts of the mismatch vertex and its third clause [SW1, Lemma M]; the light-scale constant on far blocks; the payment of the seventh family past the stopping scale under block alignment; the dominated-convergence step at the endpoint $\kappa = \infty$; the image barrier with Neumann faces; and the use of the external field of the Appendix class in the proof of the finite-domain perturbative bound (F5): each proved in [SUP], or not used (the image barrier, which the tangentially localised amplitude supersedes); the endpoint step at $\kappa = \infty$, which Theorem 7.10 does not use, enters item (15) with the counterterm offset graded in the couplings and is proved there; the payment of the far blocks [SW1, D7(c)] is (8k)(vi);

(8c) the near-face argument of [K2, §5.2], printed as a one-page sketch, transplanted to the shell model with the contour run from the first light layer: proved, with Bałaban's [Ba2, Lemma 2.7, (2.113)], whose proof is omitted in print, proved at block factor 5 under (1);

(8d) the second part of the proof of the perturbative bound, Bałaban's enumeration [Ba3, §§2–3] used as a frame: asserted in print without a complete argument; proved in [AS]; its load-bearing first clause is proved, with two residues, the constants of the pairing lemma and the shell side of the photon two-leg lemma of [SW1], proved in the main paper, in derivations alongside [SW1] [SUP];

(8e) the re-runs, for the model with the shell mass inside the covariance, of the four steps of [Ba2, Ba3] that are repaired at block factor 5 (item (1)): proved, with the renormalisation bound for the walled operator at zero shell mass in the form (P1) of Section 1.4, and with the bound for the mixed classes with two scalar legs and one photon leg, which rests on the sub-domain form (8k)(i) and on the local coefficient (B3), proved on the torus (Section 8.2) (the order of these steps is stated at the end of Section 1.4);

(8f) the counterterm offset: if the counterterm is $\delta m^2 + c(\varepsilon)$ with $\sup|c| \le \Delta_c$, graded by coupling order, everything holds with constants depending also on Δ_c [SW1, Lemma M(d)]; it is used with $c = \Delta_T$, $\Delta_c = \Delta_*$ (no sign of Δ_T and no equality of states across the two conventions is used).

(8g) **(K-pt)**, the pointwise difference bounds of the line kernels: for the lines G of step j, uniformly in κ on both sides of ∂U, $|D_A G(x,y)| \le C|x-y|_\eta^{-2} e^{-c|x-y|}$ and, for the terms of the current two-point function with both differences on one line, $|D_A^x D_A^y G(x,y)| \le C|x-y|_\eta^{-3} e^{-c|x-y|}$. With the undifferentiated bound $|G(x,y)| \le C|x-y|_\eta^{-1} e^{-c|x-y|}$, which is proved in [SW1] with an argument of [SW3], they give the line–line part of the bound $|\Pi_j^\kappa(x,y)| \le C|x-y|^{-4}$ on the remainder of the photon Taylor step [SW1, Lemma ward(b)] (the seagull and block-vertex terms being carried by the printed argument of [SW1]), and through it the norm form of the perturbative bound [SW1, Prop. TD(b)]. The bounded-source estimates of [SW1] do not imply these pointwise bounds. They are proved by two independent arguments on the torus, at every scale, for every $\kappa \in [0,\infty]$ and on both sides of ∂U; on the Neumann sub-domains used in [SW1] they hold in the restricted form used there, with both points at distance at least one block from the complement of the sub-domain, by the corollary of the argument through the exact scale decomposition into leg pairs, the second argument of (8g), at such points (near a re-entrant edge the first-difference bound fails, and it is not used there); the proofs use the interior, first-difference and amplitude estimates of [SW1], a bound for the mixed differences at the far end of a line, Kato's inequality, the random-walk estimates of [LL] with the gradient bound (8k)(iii), and [K1, (2.17), (3.5)]; [SW1] cites both arguments;

(8h) The regularity of the reflected gauge field [SW1, Lemma refl(a)] is not an input. The statement made there, that for a Neumann face F of a sub-domain the gauge field reflected across F (tangential components even, the normal component odd, and 0 on the bonds crossing F) is regular at its scale in the sense of [K1, (3.5)], which bounds every first difference and not only the curvature, is false: across F the normal difference is of size $|A_m|/\eta$, and a constant normal component is a counterexample. What [SW1] uses is that the reflected field is regular after a gauge transformation, which a smoothed gauge on the face layer supplies, and the coercivity of the doubled operator for the mirrored field, which holds whether that field is regular or not: both are proved, and [SW1] states the lemma in this form. The interior estimates up to flat faces and convex corners [SW1, Lemma refl(b)] are used only at zero gauge field, in the proof of (8i), where the reflected field vanishes and they reduce to the interior estimates of [SW1] for the operator without gauge field;

(8i) **(BAL)$_5^\kappa$**, the clause of (1) for the model with the shell mass inside the covariance (this is (8e), named in [SW1] as an input of its main theorem): Bałaban's steps of the lower and the upper bound, and the cube step of the finite-domain perturbative bound, re-run for that model on the four objects that [SW1] replaces. That the four objects do not depend on κ is proved; the transfer is proved, all its clauses included: the identification of Bałaban's all-small-field term with King's action, with bounds for the derivatives in the gauge field of the terms of that identification in the shell model, and its block-vertex part, which is moved into a determinant on the block lattice. Its steps of the lower and the upper bound expand the determinant of each renormalisation step through the determinant clause of (8k)(ii), with the bound (DOM) of Section 1.4 for its first-order term, on the torus and on the propagator domains of the upper bound, and on the histories of the upper bound with proper propagator domains its removal of the tail of the background uses, for the families of order fourteen that it produces, the sub-domain form (P9) of Section 1.4; its cube step and its identification step do not;

(8j) **(LOC-R)**, the light-side regularity estimates [LOC]. On the torus, at frozen scales, they are consumed by the derivative clause for small-field data [SW1, Prop. legs(d)]

and by the near-face argument (8c): proved, with the reading (8k)(iv) of the wall inside them. On sub-domains $\Omega \neq T$, at points whose distance from the complement of Ω is at least a fixed multiple of the block size and near which every cube of Bałaban's decomposition that meets the boundary of U is a flat-face cube (the flat-face condition (L_x)), they are supplied by [LOC, Prop. global(b),(c)], which is stated for every union of large blocks with Neumann boundary conditions, with no condition on the faces of the sub-domain, at such points, proved; [SW1] consumes them on no sub-domain (every former consumer there uses the far-block bound of (8k)(vi) or the localisation lemma below). Near the edges and corners of H', where (L_x) fails, the forms that the consumers use are supplied by a localisation lemma [LT], which compares, at a point of depth at least 4ρ in any Neumann union of blocks, Bałaban's local inverse with the torus inverse for the data restricted to the cube of side 8ρ around the point, with an error $Ce^{-\delta\rho}$ times the size of the data in the supremum and in the first difference, uniformly in κ, the spacing, the scale and the domain. It covers every consumer, the payment of the far blocks near the edges of H' included, (8k)(vi);

(8k) **The further steps of** [SW1], besides (8e) = (8i), each proved in [SUP], the determinant clause of (ii) with the bound (DOM) of Section 1.4:

(i) the sub-domain form of the finite-domain perturbative bound [SW1, Prop. TD-sub]: the transfer of its torus form to the propagator domains, at vertices of depth at least Bałaban's radius, and its norm form on those domains, together with the tangentially localised amplitude and the bounds of the amplitude theorem at depth at least Bałaban's radius, uniformly in κ and in the spacing, on the domains every face of which lies at distance at least two blocks of the scale from every face of ∂U parallel to it. For the large-block integer M of Definition 2.1 and the corners of H' of (6) every propagator, splitting and localisation domain of every scale is of this kind; for other choices, for instance M a power of 5, domains with a face on a plane of ∂U occur, generically at the scales at which $M\ell_k$ divides ℓ_f, and on them the bound is not proved. Its difference steps for several cut lines regroup the cut families of Bałaban's renormalised classes by cut datum, into products of the full classes of each label of the components, which are unions of orbits of the completion moves; at a non-zero background a regrouping along single orbits does not suffice. This regrouping is proved, and the closure of the per-label full classes under the completion moves, with the bound for every closed family, on which it rests, is proved (Section 8.2). (8k)(i) uses (ii) only through the norm form and the even sector of the mismatch clause at zero background, which contain no determinant; it uses the readings (B5) and (B6) of Bałaban's construction, proved (Section 8.2). A localisation statement of the first written proof is false as worded and is replaced by a weaker form that suffices. The transfer is stated for the families of order at most 13; at order 14, for the families with one more vertex that the removal of the tail of the background produces on the histories of the upper bound with proper propagator domains, it is (P9) of Section 1.4;

(ii) in the norm form of the perturbative bound [SW1, Prop. TD(b)], the bound on the members cut by the resummation in the vanishing of the zeroth moment of the photon two-leg kernel along a line, which also enters its block-data form, and the mismatch clause [SW1, Prop. TD(c)] in both of its sectors, at King's backgrounds (a form for every regular background is not claimed); the block-data form [SW1, Prop. TD(a)] is obtained from the norm form and the mismatch clause, so that the cut members that carry the mismatch vertex on window lines are used by no consumer. The block-data form holds on King's

small-field set [K1, (3.2)] as printed, which gives no bound $|A_k| \le Cp$ uniform in the scale: the background enters the families only as the field inside the lines, the transporters and the block averages, through its first differences and their Hölder quotients, which the derivative clause of [K1, (3.2)] bounds, and, through the first clause of [K1, (3.2)], in King's mass correction [K1, (3.44)–(3.45)] and in the far region of the leg lemma of [SW1]; and the photon legs that occur are Bałaban's fields, the legs of his cube step after the constant part of the background is removed by a gauge transformation, and small differences, for which the light-side first moment of the photon two-leg kernel [SW1, Lemma ward(c)] holds with the printed constants, up to factors of order one that depend on μ_0. The frame in the odd sector is (P3), the vacuum families with the mismatch vertex at King's backgrounds are bounded by (P2), and the device for divergent subgraphs with two scalar legs uses (P1) (Section 1.4); the mixed classes with two scalar legs and one photon leg use Bałaban's local coefficient (B3), proved on the torus for the window coefficient (Section 8.2). The determinant of the matter integral of a renormalisation step, which is kept exactly in the effective action, enters the bounds only per step, in Bałaban's form [Ba1, (3.46)–(3.47)], [Ba2, (2.99)–(2.107)]: at each step the determinant at the background plus the fluctuation is the determinant at the background, carried to the next scale, times a factor that is expanded in the photon fluctuation field, and the coefficients of that expansion enter as kept vertices, with bounds per block (the *determinant clause*). The determinant clause is proved, on the torus and on the propagator domains of the upper bound. The first-order term, which on a propagator domain is the first-order coefficient of Bałaban's kept polynomial, composed over the earlier steps [Ba2, (2.104)–(2.107)], is bounded block by block, uniformly in the shell mass and with no bound on the size of the background, by (DOM) of Section 1.4, on the torus and on those domains alike; its zero-shell-mass part is also Bałaban's per-step bound [Ba1, Prop. 3.2, (3.58)], [Ba2, pp. 578–579] under (1). The second-order term, whose two legs are fluctuation fields, is bounded as the photon two-leg classes of the norm form, the third-order term on the frame (P3), and the terms of order at least four have positive degree; on the propagator domains the terms of order at least two are bounded in the same way, with (8k)(i) in place of the torus forms. The absolute ratio of the determinants at the background and at zero is used only at the final scale, at base point 0, where the background is bounded and the odd orders vanish by charge conjugation [Ba1, (3.66)–(3.69)]; King's interpolation of the determinant [K1, (3.94)–(3.98)] is not used (Remark 1.1);

(iii) the gradient bound for the simple random walk behind the pair estimates of [SW1]: its written proof uses a bound at each fixed number of steps that fails by parity; the statement holds, by an argument that pairs consecutive steps;

(iv) in [LOC], the reading of the wall $n\psi$ as constant on each cube of Bałaban's decomposition, with the profile varying inside the cubes;

(v) the cumulant bound (5l)(i), the Hölder bound (F3) and the constants (5l)(vii), which (8) uses as well;

(vi) the payment of the far blocks at frozen scales on the propagator domains [SW1, D7(c)]: its radius R_0 is taken from a bound free of the large-block integer, the far-block lemma of [SW1]; and at the points near the edges and corners of H' where (L_x) fails and whose distance from U exceeds R_0, the smallness

$|a_i[G_i^\kappa - G_i^0]Q_i^*\phi_i(x)| \le \frac{1}{4}|\phi_i(x)|$ of the κ-difference of the background is supplied by the localisation lemma of (8j).

Two further points of [SW1] are settled and are not inputs. The order-$(2,0)$ h-linear vacuum family, on which [K1, (3.38)] fails (item (5h)), carries the source g', so it does not occur among the families consumed in [SW1, Cor. TDcons(v)] at $g = 0$, where the chain of Theorem A runs. King's identities [K1, (A.3)–(A.6)], which are photon identities, are cited in [SW1] for matter backgrounds ((F5)(f)) but carry no load there: the matter content at those places is Bałaban's [Ba2, Lemma 2.4], whose derivative clauses are printed as following similarly and are completed in [CT], with a comparison for the wall profile; one display of [SW1] at that place is false as written for a non-constant wall and has an exact replacement.

(9) **(SW2$^{\min}$).** The disc bound for the heavy shell: $\sup_{|z|\le 2}|Z_\kappa(zf,g)/Z_\kappa(0,0)| \le M < \infty$ uniformly in $\kappa \in [0,\infty]$ and in ε. At $g = 0$ its matter moment factor (the supremum over κ of the expectation of the Wick-source weight) equals 1 and the bound is the Gaussian domination of Lemma 2.13(a), with no input: proved in Step 0 of the proof of Theorem 7.10.

(10) **(SW3°)$_{\rm int}$.** On the capped stopping-scale set, for every $\eta_0 \le \eta_2$, every κ, κ' with $\max(c_0,1)/\eta_0^2 \le \kappa \le \kappa' < \infty$, every $z \in [-1,1]$ and $g' \in \{0, g\}$, the mixed differences of the reduced small-field action satisfy $\sup_{b,b'}|\Delta_z(b) - \Delta_0(b')| \le 2R(\kappa,\kappa') \le 2r(\kappa;\eta_0)$ with $r(K;\eta_0) \to 0$ as $K \to \infty$, uniformly in $\varepsilon \le \varepsilon_0(\eta_0)$, for the same model ((CT)$_{\rm sh}$, [SW3, Lemma mismatch]). Proved in [SW3], with the step (F6) of Section 8.5 in its completed form, from (10a), (10b), (10d)–(10g), and from the statements of [SW1] that it cites, among them the perturbative bound and the photon Taylor step, which carry the inputs (8g), (8i)–(8k) at the places where they rest on them; each of its residues below is proved:

(10a) the tail of the block-free determinant factor on the torus [SW3, Thm. LD$_T$]: for $\kappa \ge 2m_V^2$, uniformly in $\varepsilon \le \varepsilon_0 \le w_U$ with $\varepsilon_0 \le \ell_H/2$, ℓ_H the smallest side of H' (both automatic under the block alignment of (6)), and in the background A with $|\partial_\mu A_\nu| \le C_A$ and any torus holonomy, $0 \le \Phi^{\rm bf}(\kappa) - \Phi^{\rm bf}(\infty) \le \tau(\kappa)$ with

$$\tau(\kappa) := Ce^2C_A^2|U|\kappa^{-1/2} + C(T,H')\,e^2C_A^2\,m_V^{-1}\kappa^{-1/2}\big(1+\log(\kappa/m_V^2)\big)^c + C(T,H',m_V)\Big[e^{-c\,w_U\sqrt{\kappa}} + \big(1+\tfrac{\varepsilon_0^2\kappa}{8d}\big)^{-w_U/(2\varepsilon_0)}\Big],$$

$c \le 5$, $d = 3$, m_V the mass on the source box, w_U the width of the shell; the bracket collects the loops winding through U and, since $w_U \ge \varepsilon_0$ under the block alignment of (6), is at most $C\varepsilon_0^{-1}\kappa^{-1/2}$, so τ is bounded by an ε-free nonincreasing function of κ that tends to 0 as a power of κ at fixed (T, H', m_V, η_0), which is all that (10) uses (the per-unit one-loop reading of the same theorem, with $|A_\nu| \le C_a$ in the given gauge, is consumed in the per-unit bound on the block-free determinant of [SW3], inside its majorant): proved, with the step (F6) in its completed form; inside it the bookkeeping of the crossing and winding loops, whose bracket is the form uniform in ε in the frozen regime (the first term alone is not), is proved as well. The form without the winding bracket and with a right side proportional to C_A^2 is false on the torus (a constant background has $C_A = 0$ while the winding loops contribute a positive tail) and is not used [SW3, Remark LDfalse];

(10b) the heavy-kernel bounds for the inverse G^κ of the matter operator with the shell mass $w = \kappa\ell^2$ in block units [SW3, Lemma Yp(a)–(g), Prop. p39(a)–(d)]: the kernel and mass bounds, the far pairs and the heavy far field in both regimes, the bounds on the inserted line, and the differentiated light estimate with its lattice-regime display, together with the near pairs of the full covariance with a heavy endpoint in the form with $\mathrm{dist}(x,y)^{-2}$ and in L^1 over the heavy end: proved from the operator bounds of [SW1]. Not used, and not claimed: a sup bound for the full covariance next to ∂U in the lattice regime and a form of the near pairs with $\mathrm{dist}(x,y)^{-3}$ (the former

is false), and the Yukawa form of the heavy-pair clause, which is false for first-layer pairs [SW3, Remark Yrev];

(10c) the steps of the dead-shell chain of Theorem 7.5 for the shell $U = T\backslash H'$ with one convex face: proved. This item is not an input of [SW3]: it is consumed by Theorem 7.8, through Theorem 7.5 at the configuration of Theorem 7.10;

(10d) the κ-derivative bound for the deep-U linear part of an insertion on a line used as a leg, for deep clusters closed in matter [SW3, Lemma Σ(iv′)(d)]. For deep clusters with a scalar connection the deep payment holds; for a cluster all of whose connections are photonic it fails, because the photon factors do not see the shell mass, and the heavy-matter photon subfamily containing the insertion must be bounded as a whole, with the insertion unsplit. The bound is: for the members of King's renormalised graphs with one insertion of the κ-derivative of the matter operator whose matter component through the insertion is closed (no vertex of it carries a scalar leg), a class that contains the deep clusters closed in matter, the sum over the members, less its field-free vacuum part, is at most $C\,|U|\,\hat\kappa^{-3/2}(1+\log\hat\kappa)^{\bar m}$ per unit of $\hat\kappa = \kappa\eta_0^2$, in the units of the stopping scale, uniformly in the spacing and in both regimes, so that it is integrable in $\hat\kappa$. It is proved, from the inputs that [SW3] applies to every graph with one insertion, among them (8g), (8i), (10g)(i), (10g)(vi), the split count (5l)(x) and the enumeration (10f); the proof applies one gauge identity to each class of members with the same full lines, and expands the classes into King's members only after that, with the background rebased for each member. A route through a photon Ward identity for each fixed ordering, or fixed scale assignment, of King's graphs is not available: that identity is false (it holds for full lines and in Bałaban's resummed organisation, with a convergent remainder). The step (10g)(viii) is settled;

(10e) the further inputs of [SW3]: the restricted form of [K1, (3.38)] for the shell model, with the degree count of [RS] re-run for the shell vertex set and the shell model's propagator bounds; the Taylor-leg placement for lines used as legs at vertices in H' or near ∂U; the source-decaying clause of the inserted-leg lemma on U; the positions in U of the one-loop z^2-term in the lattice-frozen regime; and the membership of the capped set in the class of the lower bound [SW1, Lemma appclass(d)(i)]. All five are proved, the restricted form of [K1, (3.38)] and the positions of the one-loop z^2-term being (10g)(i), (ii);

(10f) Bałaban's all-orders enumeration [Ba3, §§2–3] to order 13 with one inserted vertex, as in the proofs of [K1, Thms. 3.4, 3.5]: asserted in print without a complete argument; proved in [AS], on tori and on King's cubes, with King's vertex list including the seagull vertex of order two (Section A.6);

(10g) **The further steps of** [SW3], besides the winding bookkeeping inside (10a), each proved [SUP]; those that rest on (8k)(ii) use its forms for families of graphs, which contain no determinant:

- (i) the restricted form of [K1, (3.38)] for the shell model (10e), with King's bound read clause by clause at the ends of the lines, the members produced by interpolating the single-scale inserted covariance and the grouping of near pairs included; the winding of the shifted constant on the torus is handled by a rebasing at a fixed radius;
- (ii) the positions in U of the one-loop z^2-term (10e), in the lattice-frozen and in the continuum regime, including the part of the one-loop z^2-term and of the tadpole in g' carried by King's block fluctuation line; one input of the block part, in both regimes, is the bound for the vector-field propagators [Ba3, p. 426], the reading (B1) of Section 8.2, proved on the torus [SUP];

(iii) the Hölder remainder of the Taylor step on the cubes of U for lines used as legs [SW3, Prop. taylor(ii)], when the matter component through the insertion is not closed;

(iv) the photon-leg classes with an insertion [SW3, Lemma photon(b),(c)]: the mixed class with two scalar legs and one photon leg (the two-photon-leg families carrying the insertion lie in the closed class of (10d) and are bounded there as a class; a per-member first-moment bound with one more insertion is not integrable in the shell mass and is not used); this class and the Hölder remainder (iii) use Bałaban's local coefficient (B3), proved on the torus for the window coefficient (Section 8.2);

(v) the scale sum of the Hölder-leg remainder terms [SW3, Lemma Σ(iii)] in the lattice regime;

(vi) for the divergent subgraphs without insertion whose external photon fields are lines, in the model with the shell mass inside the covariance, the bound that (10d) uses, the renormalised-degree clause, uniform in κ, for the photon-end families without insertion, obtained by splitting the kernel into its part at zero shell mass (Bałaban's, its p. 441 step being (B3)) and the shell-mass difference; a standalone first-moment bound for the two-photon kernel built from full lines is neither proved nor needed, and the bound on the remainder of Bałaban's regrouped Ward identity with the gain in degree claimed for it, whose form at $\kappa = 0$ is Bałaban's [Ba3, p. 441], is false as so posed and is not used;

(vii) the bound for the members of a source vertex with no open photon end, which are linear in the source and so are not part of the field-free vacuum part, at every placement, the light local vertex of the Taylor split at all orders and in the lattice regime included, when the support of g' has distance at least η_0 from U;

(viii) the field-dependent members of the closed class of (10d) that the light local vertex of (vii) may produce: they are bounded by the count of (10d).

(11) **(SWT).** The threshold clause between adjacent representations of the stopping-scale integral; for the capped object there is one test set on $[\max(c_0, 1)/\eta_0^2, \infty)$ (the threshold κ_0 of Section 7.4) and the clause is empty.

(12) **Literature inputs of the shell estimates**, each with the place where it is proved:

(12a) the correlation inequalities [K2, Prop. 3.1, (3.24)–(3.26)], the monotonicity in the cube [K2, Thm. 3.2] (with a J-independent Wick constant, in the form its proof proves, for the boxes of that proof and for King's cubes, Lemmas 2.14 and 2.16), [BFS, Thms. 6.1, 6.2(1)–(2)] and [BFS, §3.3, (3.7), (3.16)–(3.17)]: printed with proof (the extension of the last to a site-dependent mass, used in [SW1], is the step (F5)(f) of Section 8.5, proved there); the reversed inequalities [BFS, Thm. 6.2(3)–(4)], whose printed proof is a sketch, and [K2, (3.28)], which is stated there without a proof and is not among the inequalities printed in [BFS], are proved here (Lemma 2.12(d));

(12b) the infinite-volume limit [K2, (3.40)–(3.44)] is not an input: at $h = 0$ what is used is proved here from (13) and Lemmas 2.14 and 2.16 (Theorem 3.3(i)–(iv)), where [K2, (3.44)] is the monotone limit of quantities bounded by 1 and [K2, (3.42)–(3.43)] are not used; at $\int h_+ > 0$ the bound [K2, (3.43)], asserted in print by "a standard argument (see [GJ1])" (not printed), fails in King's Wick ordering at order e^2 in perturbation theory (Proposition A.3), and [K2, (3.44)] rests on it; neither is used;

(12c) the estimates of Bałaban's lower and upper bounds that [SW1] re-runs, [Ba1, (3.26)–(3.69)], [Ba2, (2.2)–(3.54)], [Ba3, Prop. 1, (1.33)] and [Ba4, (1.8), (1.15)–(1.16), (1.20), (2.5)–(2.33)]: each range contains steps asserted without a printed argument (among

them [Ba3, Prop. 1], whose printed proof (§3, pp. 421–445) consists of sketched steps, [Ba1, Props. 3.1–3.2, (3.66)–(3.69)], [Ba4, (1.8)] on general domains, whose proof is not printed and is completed in [CT, SW1], and the transfer of [Ba4, Cor. 2.3] to the mixed operators with its consequences [Ba4, (1.15)–(1.16), (1.20)], the domain-difference clause (1.20) not being argued): asserted in print without a complete argument; their asserted steps are proved at block factor 5 under (1), the readings (B1)–(B6) of Section 8.2 proved in [SUP]; within [Ba4, (2.5)–(2.33)] the print is complete except the two asserted steps named in (5a), on which the proof of [Ba4, Lemma 2.2] rests and which are proved likewise (the lower bound π^2 for the gap in [Ba4, (2.27)] is false as printed, and the correct value $4N^2\sin^2(\pi/2N)$ suffices); for the transfer of [Ba4, §§2–4] to the tori of [CT] see (5l)(iv);

(12d) King's near-face analysis [K2, (5.41)–(5.49)]: asserted in print without a complete argument; its transplant to the shell model is (8c), and its use on King's cubes, at the first scale included, is part of (13), both proved;

(12e) King's Appendix [K1, (A.1)–(A.8)]: a printed argument with asserted steps. Its argument asserts the expansion to order 13 with the claim that (3.54) still suffices (p. 676) and the inclusion of sources (p. 677), and (A.5)–(A.6) use the photon clause of [K1, Thm. 3.3]; the expansion and the photon clause are proved in [AS] (items (5k) and (10f)), the other steps of (A.2)–(A.8) are read, two of them (the regularity of the background in (A.8), replaced by the two consequences, a block average and a gradient bound, that the lower bound consumes, the pointwise Laplacian clause being consumed by no step, and "the other bounds in (A.1) follow similarly") being completed in [SW1, Lemma appclass] and proved (item (8b)), and the inclusion of sources is proved in the form consumed (item (5f), Section 8.3); for the shell model the class is [SW1, Lemma appclass] (item (8b)). The photon legs of the per-step constants of the field-independent part of the perturbative bound of the shell model [SW1, Prop. TD(d)] are handled by (KA*): the photon slice lines and King's photon legs are Bałaban's external fields in the norm [Ba3, (1.32)], proved from the bounds for the vector-field propagators on the periodic torus (the reading (B1) of Section 8.2, itself proved in [SUP]), componentwise and uniformly in the spacing, the scale and the scale mass, in that norm [SUP]; no moment bound of [K1, (A.2)–(A.6)] is read;

(12f) the all-orders enumeration with one inserted vertex: item (10f).

(13) **King's cube limit for 2-form sources.** [K2, Thm. 2.1] at $h = 0$ on King's cubes: for every King cube C_J^K (Definition 2.15) and every finite family of real smooth 2-forms supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, sampled at plaquette centres in the lattice's own frame, the limit $\lim_k \langle e^{\sum_j z_j F_{\Lambda_k}(f_j)} \rangle_{k,J}^K$ exists in $(0,\infty)$ for real z, with an a priori bound $e^{\pm\beta_f |z|^2/2}$ on the limit at real z whose constant β_f depends on the sources only through $\|\nabla f_{j,\mu\nu}\|_\infty$ and the cube (Remark 2.9). Its printed proof [K2, §3.1] has asserted steps: it transfers the torus proof of [K1] to Neumann–Dirichlet cubes and states that "all the results of [K1] go through" (p. 332), the boundary analysis being printed for one counterterm graph only and, for the photon, one ordering of the renormalised graphs at a time [K2, pp. 331–332], a step replaced in [CA] (Section 8.3); the transfer uses the image-sum identity [K2, (3.9)], which together with [K2, (3.3), (3.8)] is false as printed for the weight printed there and holds with the Dirichlet weight f_D, which suffices for the transfer and with which King's flow on C_J^K is run. Proved in [CA]: the limit follows from the ultraviolet construction on C_J^K with the flow of weight f_D, the convergence of the scalar boundary remainders, the line bounds and the source-free strict degree property on that flow [K2, App. A], King's transfer [K1, (3.90)–(3.98)] of the quadratic form and of the determinants to the cube, in the form of the difference at two cutoffs (the middle line of

(3.98) and the even-n line of (3.96), false as printed, are not used; item (4d)), and the cube clause of the split degree count with its difference clause, whose part for the long trees is (P8) of Section 1.4, each proved from (1), (5k) and (10f), with the bound at long trees for the families with one inserted external field whose graphs contain block vertices, which the ultraviolet construction on C_J^K uses and which is proved in [SUP] (Section 1.5). The bounds for the classes with curvature vertices on the cube, for the tails of the localisations near the faces of the cube and for the source-free classes of the restricted theorem [RS, Thm. R-i] on the cube are proved in the same way for the localisations, at a scale of the flow with blocks of size δ, whose tree has length at most $p(\delta)$ in units of these blocks, p being King's small-field function, and away from the faces up to a length of order $\delta^{-1/2}/(e\,p(\delta))$, and at greater lengths by the cut of the long lines (P5) of Section 1.4. The regrouping of the cut families along the orbits of the completion moves of Bałaban's classes, which the difference steps for several cut lines use, is proved as well, with the closure of his per-label classes under the moves. The ultraviolet construction on the cube uses the subdomain form (8k)(i) of the perturbative bound at zero shell mass, and with it the readings (B5) and (B6); the boundary analysis uses Bałaban's bounds (B2) for the massive lattice Green's function and its differences; all three readings are proved (Section 8.2). The photon boundary step uses the zeroth moment of the full photon kernel of the cube and no identity for a fixed ordering. Behind it stand [K1, Thm. 3.1, Prop. 3.6] (items (1), (5k), (5g)), the strict degree property named in (4), for one class with curvature and scalar vertices, and King's componentwise treatment of the sources [K1, p. 665], both proved (items (4), (5g), (5k)). The further statements [K2, Thms. 2.2, 2.3(iii)] and the parts of [K2, (3.40)–(3.44)] at $h = 0$ that we use are not inputs: they are proved here from (13) and Lemmas 2.14 and 2.16 (Theorem 3.3(i)–(iv)); [K2, (3.42)–(3.43)] are not used. On the boxes $C_J \cap \Lambda_k$ of Definition 2.5(a) the fixed-J limit is neither known nor used; Lemma 2.16 gives the same infinite-volume state from them (Corollary 2.17).

(14) **Translations.** Theorem 3.3(v), proved here from (13), the cutting lemma and Lemma 2.16; King's printed argument [K2, Thm. 2.3(i)], which appeals to uniqueness of the limit and is written for centred cubes only, is not used. Used for the $E(3)$ clause only.

(15) **Right-continuity in the scalar mass.** At every point (e, λ, m, μ_0) of the set E_W of Proposition 1.4, for every finite family of real 2-forms $f_j \in C_c^\infty$, the functional $Z_\infty(\sum_j z_j f_j, 0)$ of item (13) at the scalar mass M converges to its value at m, locally uniformly in $z \in \mathbb{C}^N$, as M decreases to m, and the nearby masses M lie off E_W. Proved in [SUP], from: the monotonicity of the walled and the cube states in the bare mass coefficient, [BFS, Thm. 6.1] and [K2, (3.25)], and in the cube, [K2, Thm. 3.2] (item (12a)), printed with their proofs; the existence of the limit of the differences $\alpha_k^{\mathrm{f}}(M) - \alpha_k^{\mathrm{f}}(m)$ of the free-boundary bare mass coefficients of Proposition 1.4, equal to $\alpha_\infty(M) - \alpha_\infty(m)$ for a function α_∞ real-analytic in M, which puts the nearby masses off E_W; a comparison of King's cube with the dead-shell torus, through the photon boundary switch across a matter-free collar (as in Lemma 7.1) and the correlation inequalities, and of the dead-shell torus at a fixed stopping scale with the model at the nearby mass, through the endpoint step at $\kappa = \infty$ of [SW1] (item (8b)) with the counterterm offset graded in the couplings and the comparison of the counterterm offsets at a fixed stopping scale on the dead-shell torus; and item (13) at the nearby masses. Used for Theorem A at the points of E_W only, for the $E(3)$ clause as for the rotations; it does not enter (K_0), (TIGHT_0) or (13).

Not carried: King's expected identity [K2, (2.24)], which is never used. The extension to scalar sources, which needs statements outside the chain of Theorem A, is the open problem of Section 10.

1.4. **The step determinants on the propagator domains, and seven further inputs.** This subsection states, in the forms in which the chain uses them, eight inputs that are proved in the

main paper: first (DOM), a bound for the first-order term of the determinant of a renormalisation step on the propagator domains of Bałaban's upper bound and on the torus, uniform in the shell mass, an input of the shell estimates (8); then four inputs of the shell estimates and of King's cube limit, (P1)–(P3) and (P5); then three statements inside the proofs of item (1) on the skew tori, of King's cube limit (13) and of the upper bound of (8), (P7)–(P9); and last the order in which the residues of (8) are proved.

The first-order term of the step determinants. In Bałaban's upper bound at scale i the factors $Z^{(j)}$, $j < i$, of the earlier steps are kept on the domains of the kept terms, composed there and expanded in the photon fluctuation field [Ba2, (2.104)–(2.107)]; the term of first order is the object of the following bound.

(DOM) *The first-order term of the step determinants on the propagator domains* [SUP]. In the model with the shell mass $\kappa \mathbf{1}_U$ inside the matter covariance, let i be a scale of the flow, at most the final scale, and use the units of the blocks of scale i, of side ℓ_i; let $e_i^2 := e^2 \ell_i$ be the coupling at that scale and p King's small-field function. Let Ω be a union of blocks of scale i with Neumann conditions on $\partial\Omega$: one of the two domains of the kept terms of Bałaban's upper bound at scale i, $B^i(\Lambda_0^{(i)})$ [Ba2, (2.104)–(2.106)] and $B^i(\Lambda_2^{(i)})$ [Ba2, (2.107)], or the torus T, whose sides are at least M/ℓ_i blocks [K1, (2.21)]. On Ω the factors of the earlier steps compose [Ba2, (2.105)]: in the shell model their product is $c\,(\det M_\Omega^\kappa(A))^{-1/2}$, where $M_\Omega^\kappa(A)$ is the operator of the composed step on Ω at the photon field A (the covariant Laplacian with Neumann conditions on $\partial\Omega$, the scalar potential with the wall and the shell mass, and the block term of scale i) and c depends neither on A nor on κ. Put $F_\Omega^\kappa := \ln\det M_\Omega^\kappa$; the term of first order in the fluctuation field a of the kept polynomial at the background B is $-\frac{1}{2} DF_\Omega^\kappa(B)[a]$, with $(B,a) = (\tilde{B}, \theta_{i+1} A'^{(i)})$ on $B^i(\Lambda_0^{(i)})$ and $(B,a) = (B^{(i+1),\eta}, g_i A'^{(i)})$ on $B^i(\Lambda_2^{(i)})$ in the notation of [Ba2, p. 579]. Let B be a background on Ω whose first differences are at most $Cp(\ell_i)$ on Ω [Ba2, (2.98)], with no bound on $|B|$ itself, and let $|a| \le Cp(\ell_i)$ [Ba2, (2.86)], with the first differences of a at most $Cp(\ell_i)$ [Ba3, p. 412]. Bałaban's final block size being taken small enough (so that the first differences of B obey the regularity bound of [K1, (3.5)]), there are $C, C_\sigma, \delta > 0$, independent of $\kappa \in [0,\infty]$, of the spacing, of i, of Ω and of B (they may depend on the parameters of the model, among them m, the height n of the wall and μ_0, and C_σ on σ), such that:

(a) $DF_\Omega^\kappa(B)[a] = \sum_{x\in\Omega} d_x^\kappa$, with $d_x^\kappa := -\int_0^\infty Dg_x^{\kappa,s}(B)[a]\,ds$, where $g_x^{\kappa,s}(B) := (\varepsilon/\ell_i)^3 \operatorname{tr}(M_\Omega^\kappa(B) + s)^{-1}(x,x)$ is invariant under every gauge transformation of B on the sites of Ω; and for every block Q of scale i in Ω, the blocks at the boundary of Ω, next to its holes and at its re-entrant edges included, every $r \ge 1$ and every $\sigma > 0$,

$$\sum_{x\in Q} |d_x^\kappa| \le C\, e_i^2 \Big[\sup_{Q_r} |\partial B| \sup_{Q_r} |\partial a| + e^{-\delta r/2} \sup_{\Omega} |\partial B| \sup_{\Omega} |\partial a| \Big] + C_\sigma\, e_i^2\, p(\ell_i)^2\, \ell_i^\sigma ,$$

Q_r being the set of points of Ω joined to Q by a path in Ω of length at most r; in particular the left side is at most $Ce_i^2 p(\ell_i)^2$. The same holds for the shell-mass residual, with $d_x^\kappa - d_x^0$ in place of d_x^κ. No Hölder quotient of B or of a enters.

(b) On the two domains of the kept terms write $DF_\Omega^\kappa(B)[b] = \sum_{y\in\Omega} b(y)\cdot j_B^\kappa(y)$. Every vertex of the kept terms lies in a block Q of scale i at distance greater than $r(\ell_i)-1$ from the complement of $B^i(\Lambda_0^{(i)})$, respectively at least $5r(\ell_i) - 5$ from the complement of $B^i(\Lambda_2^{(i)})$ [Ba2, pp. 578–579], where $r(\cdot)$ is Bałaban's localisation function [Ba2, (2.7)]. At every such block Q, if in addition the Hölder quotients of the first differences of B are at most $Cp(\ell_i)$ at the points at distance less than $r(\ell_i)$ from Q, as they are for

Bałaban's background at that depth ((F3)), then for every $\sigma > 0$

$$\sum_{y \in Q} \left| j_B^{\kappa}(y) - j_B^{0}(y) \right| \leq C\, e_i^2\, p(\ell_i) + C_\sigma\, \ell_i^{\sigma},$$

and the zero-shell-mass current j_B^0 obeys Bałaban's per-step bound [Ba1, Prop. 3.2, (3.58)], [Ba2, pp. 578–579], under (1).

(c) On the torus T, besides (a), which holds there at every background as above, the bound of (b) holds at every block of T at King's backgrounds, whose size is bounded by [K1, (3.2), (A.1)].

The term of first order is paired with the fluctuation field, so (a) bounds the first-order vertex of the determinant, and its shell-mass residual, by $Ce_i^2 p(\ell_i)^2$ per block, at every block of the domain; this is the form in which the determinant clause of (8k)(ii) uses it, the photon legs being measured there in a norm that dominates their first differences. (b) gives the same bound through the L^1 norm of the current, at the blocks that carry the vertices. (DOM) bounds the first-order term on each of the two domains; the replacements of the domains in the kept terms, of the domains of the factors by $B^i(\Lambda_0^{(i)})$ [Ba2, p. 578] and of $B^i(\Lambda_0^{(i)})$ by $B^i(\Lambda_2^{(i)})$ [Ba2, (2.107)], are terms with one difference of two propagators at vertices deep inside the domains, bounded by the boundary-difference clause of the sub-domain form of the perturbative bound ((8b), (8k)(i)), and, for the first-order term, also by the comparison with the torus of the argument for (b) below.

Part (a) and part (b) are proved in [SUP] by two independent arguments; each part alone gives the bound that the determinant clause of (8k)(ii) uses, which therefore has two independent proofs, and a third argument, which compares the domain with the torus at the blocks of the vertices with no use of (a), gives (b) again. None of them uses a photon Ward identity for a fixed ordering of King's graphs. The arguments for (a) and for (b) both start from the covariance of M_Ω^κ under every gauge transformation on the sites of Ω, whatever the shape of Ω: the Neumann form sees only the bonds inside Ω, and the blocks of Ω are whole blocks. The argument for (a) writes the first-order term as an integral over a uniform mass shift $s \geq 0$ of the derivatives of the diagonal of the resolvents of $M_\Omega^\kappa + s$; at each site it subtracts from the background a constant field that is a large gauge transformation, so that the value of the background at the site becomes small, removes the rest near the site by a gauge transformation that is affine there, and bounds the columns of the resolvents at every point of Ω, its boundary included, by an energy estimate from Bałaban's coercivity bound on Ω [Ba4, (1.8)] under (1) and an on-diagonal bound for the reflecting random walk on Ω; the shell mass enters only as a non-negative potential. The argument for (b) transforms the background, on a cube of side of order $r(\ell_i)$ around Q, by a gauge transformation into a field that vanishes at a point of Q and is bounded on the cube through its first differences alone, cuts the result off smoothly to a field on the torus that is regular everywhere, compares the coefficient on Ω with the coefficient on the torus at that field up to an error smaller than every power of ℓ_i, and bounds the torus coefficient through the first moments of the two-photon kernel [SW1, Lemma ward(c)], with the kernel bounds (8g), (8k)(iii), the Hölder bound (F3) and the frame (8d), (P3); on the torus at King's backgrounds this last step is the whole argument. In the torus bound that the arguments for (b), the third one included, and for (c) use, the far region, at distance more than a quarter of the side $N \geq M/\ell_i$ of the torus from Q, is summed slice by slice over the scales $j \leq i$ of the lines: the slice of scale j decays at a rate at least $c\,5^{i-j}$ in the units of the blocks of scale i (the bounds (8g) for the lines of step j, whose rate is in the units of that step, with the uniform exponential decay of the step propagators [SW1]), so that the sum over the slices is at most $CN^3 e^{-cN/4}$ times the size of the background, with constants C, c independent of the scale i and of the slice and with no factor of the number of scales, and this part is at most $C_\sigma \ell_i^\sigma$, uniformly in the spacing, as in (b) and (c) (a bound of each slice at the rate c alone would

leave a factor of the number of scales, which is not bounded at a fixed block size). The argument for (a) has no such sum.

(DOM) is used at one place, the determinant clause of (8k)(ii): on the torus by the lower bound of (8) and by the upper bound on its other histories, and on the domains of the kept terms by the upper bound of (8) on its histories with proper propagator domains, at the steps of Bałaban's bounds re-run in (8i). The corrected torus comparison (K_0), with the repaired cube step of (1), and King's cube limit (13) are statements at zero shell mass, where $d_x^\kappa - d_x^0 = 0$ and only Bałaban's per-step bound enters. The other residues of the shell estimates, (8k)(i) with its form (P9) at order fourteen, the bound for the mixed classes, the cube step and the identification step of (8e) = (8i), (P2), (P3), (10d), (10e) and (10g), use the forms of the perturbative bound for families of graphs, which contain no determinant; the residues of (8) among them come before the determinant clause in the order of proof stated at the end of this subsection.

Remark 1.1 (King's interpolation of the determinant). King controls the ratio $\ln[Z_k(A^{(k)})/Z_k(0)]$ of the determinants of the matter integral of a renormalisation step at the background $A^{(k)}$ and at zero by interpolating between them in $R_k := 1 + [(\mu_0 \ell_k)^{-1}]$ steps [K1, (3.90), (3.94)–(3.97)]. This serves his convergence theorem [K1, Thm. 3.4]: [K1, (3.98)] bounds only the difference of these ratios at two cutoffs. In the model with the shell mass the first-order terms summed over the interpolation points equal, up to second-order corrections, the absolute ratio at the stopping scale, which no step of the shell estimates bounds at a scale before the final one: the chain keeps the determinant exactly in the effective action, expands it per step in the fluctuation field (the determinant clause of (8k)(ii)), and expands the absolute ratio only at the final scale, at base point 0. In particular no bound is needed, at King's interpolated backgrounds, for the part of the shell-mass residual of the two-photon kernel that is odd under the exchange of its two points; a bound of that part through the supremum of the background would lose the factor R_k, since $|A^{(k)}|$ can be as large as a constant times $p(\ell_{k-1})(\mu_0 \ell_k)^{-1}$ on King's small-field set.

Seven further inputs proved in the main paper. The following statements are proved, in the forms stated here, which are the forms in which the chain uses them.

(P1) *The renormalisation bound for the operator with the scalar wall* [AS]. Consider the model at zero shell mass whose scalar field carries a potential $0 \le v \le \bar{v}$, among them the wall $m^2 + n\psi$, and whose photon carries none, with the model's own counterterms (Bałaban's rule applied to the graphs with the walled lines) and bounded finite vertices.

(a) Bałaban's renormalisation bound [Ba3, Prop. 1, (1.33)] at block factor 5 holds for every renormalised family of Bałaban of order at most 13 with at least one external field, at every regular background, the vacuum families being cancelled by the model's own vacuum counterterm up to bounded image terms, with constants polynomial in $\bar{v}$ and independent of the spacing, of the scale, of the domains, of the background and of the torus, Bałaban's depth constant for the scalar-propagator bounds being enlarged by a fixed amount (ten blocks), which the margins of item (8)'s parameter list absorb.

(b) At zero background, on an aligned torus of side at least one and at every step, the bound (1.33) holds also for the classes of graphs closed under Bałaban's completion moves at each node (passing to the counterterm partner of a divergent subgraph, permuting the external scalar legs of the divergent subgraphs with two scalar legs and one vector leg [Ba3, (2.18)–(2.22)], and forming the classes with two vector legs [Ba3, p. 431]), with constants polynomial in $\bar{v}$ and independent of the spacing, of the scale, of the labels and of the torus. This includes the bounds for the zeroth-order coefficients of the parts with two scalar legs and one vector leg whose leg vertex is pinned at the root; for such a part with a bridge the bound fails for the single member under the wall and holds for the member taken together with its counterterm partner.

Part (a) is used by the block-data form of the perturbative bound (8k)(ii), at the background of the block data, by the transfer (8e) = (8i) at the two places where it evaluates walled objects (Bałaban's [Ba2, Lemma 2.4] at the walled background, in its walled form, which holds under the caps on Bałaban's final block size of item (8) that depend on the wall height, and the cube step with walled lines), and, in its stated form, by the device of (8k)(ii) for divergent subgraphs with two scalar legs. Part (b) is used only through the bounds, at every step, for the finite parts of the counterterm vertices of the walled model, which enter (a) as sizes of its finite vertices and the wall term at zero shell mass of the mismatch bound [SW1, Lemma mismatch(b)]. Through these uses (P1) reaches (8k)(i), (10g)(i) and King's cube limit (13).

(P2) *The vacuum families with the mismatch vertex at King's backgrounds* [SUP]. In the model with the shell mass $\kappa \mathbf{1}_U$ inside the matter covariance, let F be a renormalised vacuum family of [SW1, Prop. TD(d)] of order at most 13 that contains the mismatch vertex M of [SW1, Lemma mismatch], for instance one in which a tadpole carrying M is joined by a photon line to a tadpole without it, and let Q be a cube of scale k, of side $\ell_k = 5^k\varepsilon$ (the δ_k of (P5)); $\beta_0 > 0$ and $\delta_0 > 0$ are the exponent of the coefficient bounds and the decay rate of the perturbative bound [SW1, Prop. TD]. *King's class* at scale k is the set of backgrounds whose first differences and their Hölder quotients are at most $Cp(\ell_{k-1})$ and whose supremum is at most $Cp(\ell_{k-1})(\mu_0\ell_{k-1})^{-1}$, in the units of the blocks of scale k; it contains King's backgrounds $uA^{(k)}$, $u \in [0,1]$, Bałaban's cut background and the segment between them. On the torus, at King's backgrounds $B = uA^{(k)}$, $|F(B;Q) - F(0;Q)| \le C\ell_k^{\beta_0}$, and for two backgrounds B, $B+b$ of King's class, $|F(B+b;Q) - F(B;Q)| \le C\ell_k^{3/2}N_k(b)$, where $N_k(b)$ is the sum of the suprema of the first differences of b, of their Hölder quotients and of $e^{-\delta_0 N/16}|b|$, N the side of the torus in blocks of scale k. On a propagator domain Ω of Bałaban's upper bound, with the lines of F taken on Ω at the background of the step and with Q at the depth at which the sub-domain form (8k)(i) holds, the first bound holds in the same form, $F(0;Q)$ being the value of the torus family at zero background: the cube step of (8e) = (8i), applied on Ω to the whole vacuum class, writes $F(B;Q)$ on Ω as the value of the torus family at zero background plus a sum of classes at zero background with at least one external field, which the norm form of the perturbative bound controls, and the values at zero background on Ω and on the torus are compared through (8k)(i). The constants do not depend on $\kappa \in [0,\infty]$, on the spacing, on the scale, on the domain or on the torus. The value $F(0;Q)$ at zero background is carried as it is, and no bound on it uniform in the spacing is used. These families are used in [SW1]: in the absolute form on the propagator domains of the upper bound, by [SW1, Cor. TDcons(iii)] on the collars at every scale and by [SW1, Cor. TDcons(v)] at the final scale (the final step of the lower bound, on the torus, expands at zero background and meets them only through classes at zero background), and in the difference form by two steps of the upper bound, the replacement of the cut-off photon background by King's background in the identification step and the removal of the tail of the background at each step; on the histories with proper propagator domains the second of these uses (8k)(i) at order 14, for the families with one more vertex that it produces: this is (P9). The torus bound is not used on the propagator domains. The proof uses (P3) only for the difference form, and uses the cube step of the re-runs (8e) = (8i), carrying the readings on which these rest. Bałaban's sentence on the divergent vacuum graphs [Ba3, p. 445] is not used (Section 8.2).

(P3) *Bałaban's enumeration block by block at higher orders* [SUP]. In the odd sector of the mismatch clause [SW1, Prop. TD(c)], after the closed odd matter blocks of a graph are rewritten by charge conjugation on full lines, the resulting families have order at most $14+n_o$, n_o the number of rewritten blocks, and block backgrounds in the class of fields whose

first differences are bounded as in [K1, (3.5)] and whose Hölder quotients are bounded as in [Ba3, pp. 412–413]. For these families Bałaban's bound for generalised expressions [Ba3, Prop. 2.2] and his enumeration [Ba3, §§2–3], read block by block, hold with constants that depend on the order, uniformly in $\kappa \in [0, \infty)$, in the spacing and in the interpolation parameters. The proof is from the frame (8d), with the cube step of (8e) = (8i), (P1) and the bound for the mixed classes, which uses the local coefficient (B3), and it carries the readings (B3), (B5), (B6) on which these rest, each proved.

(P5) *King's cut of the long lines on King's cubes* [CA]. On King's cube C_J^K with the flow of weight f_D, the free-boundary δm_ε^2 and the free Wick constant, let $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$ and $0 \le k \le K - J_*$, where J_* is a fixed integer, the largest of the finitely many thresholds for the smallness of δ_k that the proof uses (each use of (P5) takes its range of scales inside this one), and let $p(\delta)$ be King's small-field function. Let τ be a tuple of localisation points whose tree has length $d_\tau > p(\delta_k)$ in units of the blocks of scale k, and let H be a member of one of the following classes: the classes of the restricted theorems [RS, Thms. R-i, R-ii, R-iii(a)] on the cube at zero scalar source, that is, those that carry no vertex of a scalar source; the classes with curvature vertices that King's cube limit uses, those with curvature and scalar vertices included; and the source-free field-dependent classes. Then, at both cutoffs of a diagonal pair, the sum $E(H_{\mathrm{ren}}; \tau)$ of the renormalised class of H at τ obeys King's bound [K1, (3.38)] in its restricted form,

$$|E(H_{\mathrm{ren}}; \tau)| \le C\,(1 + K - k)^{p_0}\, \delta_k^{\rho(H)}\, p(\delta_k)^{n_1}\, e^{-\delta d_\tau},$$

with $\rho(H)$, n_1, p_0, δ and C as in the torus form [RS, Thm. R-ii], independent of k and K. On the torus this is King's treatment of every line longer than a fixed multiple of the block size as an external line [K1, p. 666], made exact: the tuple splits into clusters of bounded size, each cluster is reduced on its own box of size $O(p(\delta_k))$, and the lines between clusters are bounded by their decay [AS]. On the cube one box containing the tree suffices only for trees of length at most $p(\delta_k)$, and away from the faces up to a length of order $\delta_k^{-1/2}/(e\,p(\delta_k))$, below which the expansion of the lines in the background as a Neumann series, the reading of [K1, (3.52)] used on the torus, converges on it; beyond that length it fails in general, and King's finite form of [K1, (3.52)] leaves remainder lines at the full background, which the one-box bounds do not cover. The proof on the cube cuts the tuple into clusters as on the torus, reduces the clusters away from the faces as there, on the flow with the weight f_D, and the clusters at the faces by the boundary bound of the cube with the lines between clusters as external legs; it uses Bałaban's bounds (B2), proved (Section 8.2). (P5) is used in the proof of (13) in [CA], for the classes with curvature vertices on the cube, the tails of the localisations near the faces and the source-free classes of the restricted theorem. Its use for the localisations with long trees in the difference clause of the split degree count of (13), at the faces of the cube and away from them, whose clusters at the faces carry the boundary bound in the forms that this clause uses, is (P8). The torus statements, and with them (K_0), do not use it.

(P7) *The change of domain of Bałaban's propagators on the skew tori* [AS]. On each skew torus of [CT], at every scale k of the flow and in the units of the blocks of scale k, let $\Omega_2 \subset \Omega$ be two unions of large blocks of scale $k-1$ (Ω_2 may be of scale k), A a background regular on Ω in the sense of [K1, (3.5)] (King's backgrounds A_θ are regular in this sense on the torus and on the cube, with the exponent $\beta \in (0,1)$ of [K1, (3.5)] and constants depending on β and e, proved in [SUP]), $G_k(\Omega, A)$ and $G_k(\Omega_2, A)$ Bałaban's propagators of scale k on the two domains, and $d(x)$ the distance of $x \in \Omega_2$ from the complement of Ω_2. There are $C, \delta > 0$, independent of k, of the spacing, of the domains and of A, such that for all $x, x' \in \Omega_2$ with $d(x), d(x') \ge D - 1$, where $D \ge R_0 + 2$ is a fixed integer and R_0 the

constant of Bałaban's localisation function [Ba2, (2.7)],

$$\big|\partial_x^\sigma \partial_{x'}^{\sigma'} \big[G_k(\Omega, A) - G_k(\Omega_2, A)\big](x, x')\big| \le C\, e^{-\delta(d(x)+d(x')+|x-x'|)}, \qquad |\sigma|, |\sigma'| \le 1,$$

and the Hölder quotients of exponent α_0 of these first differences obey the same bound. On the cubic tori, and on the domains of the shell estimates, this bound is proved in [AS], from Bałaban's bounds for the lattice Green's functions and the block operators [Ba4, §2] at every scale and the scope statements of his construction. On the skew tori it is proved by two independent arguments. The first transfers that proof line by line: the proof uses no symmetry of the cubic lattice and no aligned axis or periodicity of the torus, its axial gauge and its contours lying inside single blocks, which are cubes of the lattice on the skew tori as on the cubic ones; and its inputs, Bałaban's bounds for the Green's functions and the block operators and the scope statements of his construction, hold on the skew tori (item (5l)(iv)). The second obtains the bound on a skew torus from the bound on the aligned cubic torus that covers it, with a group of deck transformations of order 125, with the same rate and the constant multiplied by 125. (P7) is the factor that, with the renormalisation bound [Ba3, (1.33)] for the full classes of the cut components, gives Proposition 2.6 of [Ba2] at a non-zero background in the form stated in Section 8.2; the change of the background on the smaller domain is exact there. It enters only the torus comparison: through Proposition 2.6 of [Ba2] in item (1) on the skew tori, in the skew instance [SKEW-U] of King's upper bound, and through it (K_0).

(P8) *The long trees in the difference clause of the split degree count on King's cubes* [CA]. In the proof of King's cube limit (13), on C_J^K with the flow of weight f_D, $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$ and $0 \le k \le K - J_*$ as in (P5), the cube clause of the split degree count (5l)(x) for the classes with block vertices has a difference clause: the change of the sum of such a class over its localisations between the cutoffs (K, k) and $(K+1, k+1)$ of a diagonal pair is at most $C[5^{-\gamma' k}\delta_k^{-\beta'} + \delta_k^{\sigma'}]|C_J|$ on King's small-field set, with $\gamma', \beta', \sigma' > 0$ as in the torus form [RS, Thm. R-ii]. At the localisations τ whose tree has length $d_\tau \le p(\delta_k)$, in units of the blocks of scale k, the clause holds with its rate on one box of size of order $p(\delta_k)$. At those with $d_\tau > p(\delta_k)$, in the zone of the faces of the cube and away from it, each of the two cutoffs obeys the bound of (P5),

$$|E(H_{\mathrm{ren}}; \tau)| \le C\,(1 + K - k)^{p_0}\, \delta_k^{\rho(H)}\, p(\delta_k)^{n_1}\, e^{-\delta d_\tau},$$

with constants independent of k and K, the clusters of the tree at the faces being bounded by the boundary bound of the cube for the scalar boundary remainders in the forms that the difference clause uses; with the bound $e^{-\delta d} \le C\delta_k^{3+\sigma} e^{-\delta d/2}$ for $d > p(\delta_k)$ [K1, (3.41)] and the sum over the trees of the remaining factor $e^{-\delta d/2}$, these localisations contribute at most $C_N \delta_k^N |C_J|$ at each cutoff, for every N, so that the difference clause holds there with no rate in k. The clause is proved; its part for $d_\tau > p(\delta_k)$ has three proofs. (P8) enters King's cube limit (13), and through it the definition of King's state (Definition 3.4), (TIGHT_0) and Theorem A; it does not enter (K_0) or the shell estimates (8)–(11). The families with one inserted external field whose graphs contain block vertices are not classes of (P8); their long trees are the bound of Section 1.5, proved.

(P9) *The sub-domain form of the perturbative bound at order fourteen* [SUP]. At each step i of the upper bound of (8), on its histories with proper propagator domains, Bałaban's background is freed of its tail [Ba2, (2.89)–(2.90)], written $\tilde{B}^{(i)}$ as in [SW1] (this tail is not the background $\tilde{B}$ of (DOM)), and the change of the perturbative part is written as $P^{(i)}[\mathrm{bg} + \tilde{B}^{(i)}] - P^{(i)}[\mathrm{bg}] = \int_0^1 \frac{d}{ds} P^{(i)}[\mathrm{bg} + s\tilde{B}^{(i)}]\, ds$, with bg the configuration of the step without the tail: a sum of families with one more vertex, whose additional external photon field is the tail, which lies at distance greater than $r(\ell_i)/2$, in units of the blocks

of scale i, from every place where the operators of the step on the propagator domain differ from those on the torus. These families have order at most 14, the vacuum families of (P2) with the inserted vertex among them. The transfer (8k)(i) of the torus form of the perturbative bound to the propagator domains, in its norm form and in its mismatch clause [SW1, Prop. TD(c)], holds for them, with and without the mismatch vertex, with constants independent of $\kappa \in [0, \infty)$, of the spacing, of the scale and of the domain. This is proved by two arguments. The first proves the transfer at every fixed order, with the depth condition of (8k)(i) taken at that order, that is, with Bałaban's radius and the smallness of his final block size chosen for order fourteen, and with constants that depend on the order only through the size of the families, by the order-uniform form of Bałaban's enumeration that (P3) uses; it covers all the families above, those with a vertex of a source included. The second shows that for these families the depth and the constants of order 13 suffice: it runs the cube step of (8e) = (8i) on the propagator domain at the background $\mathrm{bg} + s\tilde{B}^{(i)}$, after which the perturbative bound is used only with the lines at zero background, where the odd sector of the mismatch clause is empty and its even sector is bounded at zero background at the size of these families; the families with a vertex of a source are covered by the first argument. (P9) enters only the upper bound of (8), through (8i) and the difference form of (P2), and through (8) it reaches (RES_0), (TIGHT_0) and Theorem A; (K_0) and (13) do not use it.

None of the statements of this subsection uses a photon Ward identity for a fixed ordering or a fixed scale assignment of King's graphs.

The order of proof. The residues of (8) that cite one another are proved in the following order, so that no statement is used before it is proved: first the norm form of the perturbative bound (8k)(ii) and the even sector of its mismatch clause at zero background, with the frame (8d), (P1) and the cube step of the re-runs (8e) = (8i); then the sub-domain form (8k)(i), which uses these, with its form (P9) at order fourteen, and the bound for the mixed classes, which uses (8k)(i); then the identification step of (8e) = (8i), which uses the mixed classes; then (P3), the odd sector of the mismatch clause at King's backgrounds, which uses (P3), and (P2); then the block-data form and the determinant clause of (8k)(ii), the latter with the bound (DOM), whose argument for (a) uses no residue of (8); and last the steps of Bałaban's lower and upper bounds in (8). The sub-domain form (8k)(i) and the bound for the mixed classes use the perturbative bound only through its norm form and the even sector of its mismatch clause at zero background, (P2) uses the mismatch clause only for families with an external field, and (P3) is the frame of the odd sector only.

1.5. **Four further statements, and the odd-block clause of the degree census.** The following four statements, proved in [SUP], enter the chain through King's cube limit (13), the torus comparison (K_0) and the walled limit (4); they are recorded here in the forms in which the chain uses them, because the plain-words restatements of Section 8 and the table of Section 9 refer to them. The odd-block clause of the degree census of the shell model, which enters the κ-Cauchy estimate (10), is recorded at the end of this subsection.

Proposition 1.2 (block vertices at long trees in the families with one inserted field)**.** *On King's cube C_J^K with the flow of weight f_D, let $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$, $0 \le k \le K - J_*$ and $p(\delta)$ be as in (P5), and let the scalar source be zero. Consider the families of the perturbative part at a background $B = B' + \sigma X$ on C_J^K, $\sigma \in [0,1]$, whose first differences and their Hölder quotients of exponent α_0 are bounded by $C_1 p(\delta_k)$ (the class of* [Ba3, pp. 412–413]*), among them the backgrounds $A_\theta + \sigma X_0$, with A_θ Bałaban's cut-off photon background and X_0 its difference from King's background, that carry one inserted vertex whose additional leg is the external field X (the field X_0 in that case), and whose graphs, with every block vertex split into its two summation points as in the split count (5l)(x), contain a block vertex. Then at every localisation τ of such a family*

whose tree has length $d_\tau > p(\delta_k)$ in units of the blocks of scale k, in the zone of the faces of the cube and away from it, the sum $E(\tau)$ of the renormalised family at τ obeys at each of the two cutoffs of a diagonal pair the bound of (P5) multiplied by $\|X\|_{1,\alpha_0}$, the supremum of X, of its first differences and of their Hölder quotients of exponent α_0, with constants independent of k, K, B' and X (within the class, at fixed C_1); so that, by [K1, (3.41)]*, these localisations contribute at most $C_N \delta_k^N \|X\|_{1,\alpha_0} |C_J|$ for every N.*

Proved in [SUP]: the bound is (P8) for these families, with the three further rows that the inserted field requires, one cluster for each product of block vertices, whose block points stay inert and whose cut lines are paired at the ends of the products, the cut photon lines with an end on a contour, and the ordering bound with the split count, each carried at the per-tuple bound of (P5) times $\|X\|_{1,\alpha_0}$ and summed by [K1, (3.41)]. For the same families without block vertices the bound is proved in [SUP], and for King's classes with block vertices, which carry no inserted field, it is the part of (P8) for long trees. The proposition enters the ultraviolet construction on C_J^K of King's cube limit (13), through the norm form of the perturbative bound for the derivative of the perturbative part in the photon background at Bałaban's cut-off background, and hence the definition of King's state (Definition 3.4), (TIGHT$_0$) and Theorem A. It does not enter (K$_0$) or the shell estimates (8)–(11).

Proposition 1.3 (the fluctuation estimates for centre-rooted and hybrid in-block contours)**.** *In Bałaban's construction the fields are transported inside each block along a contour $\Gamma_{r,x}$ from a point x of the block to its reference point r* [Ba1, (2.1)–(2.3)]*, and King's flow uses axial contours rooted at a corner of the block* [K1, (2.12)]*. The flows of* [CT] *use in-block contours of two other families: the axial contours rooted at the centre of each block, of length at most $\frac{3}{2}$ times the side of the block, the coarse variable being attached to the centre; and the composite contours of the hybrid flow, of length at most* 2.1 *times the side of the block of each level. The statement: every estimate of the fluctuation integrals of* [Ba1, Ba2, Ba3]*, among them* [Ba3, Prop. 1]*, and King's upper bound* [K1, Thm. 3.1]*, in the forms that* [CT] *consumes, holds for these two families in place of King's, with constants that depend on the family through the bound $c_\Gamma \in \{\frac{3}{2}, 2.1\}$ on the length of its contours in units of the block side and, for the hybrid family, also through the hybrid kernel bounds (3) and the geometry of the tiles.*

Proved in [SUP]. The printed proofs use the contour family through a short list of its properties, which the three families share (each contour lies in its block; the block average built from the contours is gauge covariant, satisfies $QQ^* = 1$ and, at a constant field, does not depend on the contours; and the line bounds see the family only through the length of the contours). The proof is a complete census of the occurrences, in the printed proofs, of the contours and of the operators built from them, the block averages, their adjoints and the propagators and block operators of the flow, each occurrence being checked against the list; for the composite contours of the hybrid flow the census finds one estimate used beyond the list, Bałaban's bound for the fluctuation field ([Ba2, (2.65)–(2.66)]) in the regime of [Ba3, p. 421], which is proved for that family directly, by two written derivations; for the centre-rooted family there are three written derivations. The bounds for the Green's functions and the block operators [K1, Thm. 3.3], [Ba4] are printed for contours rooted at the corner of the block [Ba4, (1.4), p. 572] and are carried to the centre-rooted family by a change of root, which leaves the block projection invariant [SUP]; they are not part of the proposition. For the hybrid family the constants depend also on the hybrid kernel bounds (3) and on the geometry of the tiles. It enters (K$_0$): through King's upper bound (H3) for the rotated lattice on both tori, the large-field clause of (1) at the wall for that lattice, the hybrid lower bound (H4) and the comparison of the effective actions of the two flows in [CT]. It does not enter (13), the walled limit (4), whose flow uses King's contours, or the shell estimates (8)–(11).

Proposition 1.4 (the positivity condition on the constant of the convention bridge)**.** *For a scalar mass $M > 0$ let $\alpha_k(M) := M^2 + \delta m^2_{T,k}(M)$, where $\delta m^2_{T,k}(M)$ is King's mass counterterm* [K1, (3.32)–(3.33)] *at the mass M, computed with the propagators of the torus T at spacing ε_k, and let $\Delta_T(s)$, $s \le s_0$, be the constants of the convention bridge (CT)(b) on T. For each $R \in \mathcal{R}_0$ and each of the three pairs of a lattice and a torus used, the axis lattice on D_J and the lattice rotated by R on D_J and on RD_J, there are $p_H > 1$, a mass $M_1 > 0$ with (e, λ, M_1, μ_0) in King's range and a threshold, depending on the couplings, the block integer M, p_H, M_1 and R (equal to 1 if $\Delta_T(s) \ge 0$ for all $s \le s_0$ and every J), such that for every J above it either $\Delta_T(s) \ge 0$ for all $s \le s_0$, or*

$$\liminf_{k\to\infty} \big(\alpha_k(m) - \alpha_k(M_1)\big) > p_H \sup_{s\le s_0} \max\{0, -\Delta_T(s)\};$$

J_{pos} denotes the maximum of these thresholds over the finite set $\mathcal{R}_0$ and the three pairs. This holds at every point of King's range, $e > 0$, outside the set E_W of the points at which the scalar mass m minimises, at fixed (e, λ, μ_0), the function α_∞ of the mass defined, up to an additive constant, by the limits $\alpha_\infty(M) - \alpha_\infty(m) = \lim_k \big(\alpha^{\mathrm{f}}_k(M) - \alpha^{\mathrm{f}}_k(m)\big)$, where α^{f}_k is α_k with King's counterterm computed in the free-boundary conventions; E_W is the set of the masses minimising that function: for fixed (e, λ, μ_0) these masses form a non-empty finite set, and for fixed (λ, m, μ_0) the set E_W contains at most two values of e^2 (both proved from the closed form of α_∞ [SUP]*).*

Proved in [SUP], by three independent arguments, one of which finds M_1 without the lower bounds at the auxiliary mass. The constant c_* of the large-field clause of (1) at a non-zero wall is then $1 - 1/p_H$ ($c_* = \frac{1}{2}$ if $\Delta_T \ge 0$). The condition is needed only where the constant $-\frac{1}{2}\Delta_T$ that the bridge adds to the wall is positive, that is, where the bare mass is lowered: an upper bound there is obtained from a mass M_1 whose bare mass squared lies below that of m. At order λ the constant Δ_T is positive and the condition is void; the counterexample at $e = 0$ lies outside King's range. At the points of E_W the chain of (K_0) is not run: there Theorem A follows from its statement at the nearby scalar masses by right-continuity of King's state in the scalar mass (item (15); the nearby masses approaching from above and lying off E_W), which is proved in [SUP] from the stability of the ultraviolet construction under a change of the scalar mass at a fixed stopping scale (its last input, the comparison of the counterterm offsets at a fixed stopping scale on the dead-shell torus, is proved). The proposition enters the large-field clause of (1) at a non-zero wall on the three pairs, and through it King's upper bound (H3) and the walled limit (4); hence (K_0), and through (4) and Theorem 7.8 also (TIGHT_0). It does not enter (13) or the shell estimates (8)–(11).

The final-scale cap. Bałaban's lower and upper bounds [Ba1, (1.14)], [Ba2, (1.3)] for the partition functions at the fixed masses end at a final spacing $t_K \in (\varepsilon_0/5, \varepsilon_0]$, and the large-field clause of (1) at the wall on the skew torus D_J takes one step at that spacing. A cap that does not depend on J and is closed downward ensures that the skew tori do not wrap at that scale: the coarsest scale ε_0 of the flow is at most a threshold $\varepsilon_{\mathrm{sys}}(M, R, \rho, \delta_0)$ fixed by the large-block integer M, the rotation R, Bałaban's radius ρ and the finest block size δ_0; under it the non-wrapping threshold at the final scale equals 1 for every J in the residue class of (6), so that every configuration of the class (6) meets it (proved in [SUP]). The cap is a proviso of the corrected torus theorem of [CT] on its parameters, carried by the class (6) (Remark 5.5); its consumers are the fixed-mass bounds on the tori of the comparison, King's upper bound at the wall on D_J (item (1), (H3)) and the skew-torus instances (SK). It is not a hypothesis of Theorem A.

The odd-block clause of the degree census. The κ-Cauchy estimate of [SW3] (item (10)) uses a degree census of the renormalised graphs of the model with the shell mass inside the covariance, at zero source, component by component. Its clause for the closed matter blocks with one photon end, the blocks of degree -2 or -1 with one photon line attached, states that after the rewriting these blocks appear as classes with two photon-type ends, a background leg and a line slot, of degree exactly -1 or 0, imported as black boxes into the enumeration, and that, after the summation by

parts on the trees with one background slot, every proper prefix sub-component of a renormalised member has integer degree at least one or lies inside one of the black boxes, which enter the enumeration as local vertices at the window scale. This clause is proved in [SUP]: the census clause itself, and the tail of the coarse counterterm at orders two to four that its one-loop form leaves. It enters item (10) only, hence (RES_0) and (TIGHT_0); it does not enter (K_0) or King's cube limit (13).

1.6. **Results.** Let $Z_\infty(\cdot,0)$ denote King's state on curvature sources (Definition 3.4, Theorem 3.3): the entire function $z \mapsto Z_\infty(\sum_j z_j f_j, 0)$ of the coefficients of real 2-forms $f_j \in C_c^\infty$, whose derivatives at $z=0$ are the curvature Schwinger functions $\langle F(f_1)\cdots F(f_N)\rangle$. For $R \in O(3)$ and $b \in \mathbb{R}^3$, sources are transported as 2-forms, $R\cdot f$ and $\tau_b f$ (Definition 2.8).

Theorem A ($O(3)$ invariance of the curvature sector)**.** *Let $e > 0$ and $\lambda, m, \mu_0 > 0$ be any point of King's range, at block factor 5. Then* (K_0) *and* (TIGHT_0) *hold outside the set E_W of Proposition 1.4, and King's state on curvature sources (Definition 3.4) is $O(3)$-invariant at every point of King's range, at the points of E_W by right-continuity in the scalar mass: for every $R \in O(3)$, all real 2-forms $f_1, \dots, f_N \in C_c^\infty$ and all $z \in \mathbb{C}^N$,*

$$Z_\infty\Big(\sum_j z_j\, R\cdot f_j,\, 0\Big) = Z_\infty\Big(\sum_j z_j f_j,\, 0\Big),$$

so all curvature Schwinger functions are $O(3)$-covariant. With the translations (14), proved from (13), the same holds for every Euclidean motion $x \mapsto Rx + b$, with $\tau_b R\cdot f_j$ in place of $R\cdot f_j$, at the points of E_W again by right-continuity in the scalar mass (item (15)): the curvature sector is $E(3)$-invariant. For the curvature two-point function the weaker closing hypothesis (TIGHT_2) *suffices (Corollary 6.5).*

Theorem A has no hypothesis beyond King's range and the block factor 5. Its proof rests on the inputs of Section 1.3, each proved here or in the main paper, and on statements of the literature printed with their proofs. The four statements of Section 1.5 enter as follows: the bound at long trees for the families with one inserted field enters King's cube limit (13), which defines the object; the fluctuation estimates for the two further contour families and the positivity condition enter (K_0), through item (1) and the inputs (H3), (H4) of Theorem 4.8, the positivity condition also through (4) and hence (TIGHT_0); the final-scale cap is a proviso of the torus theorem on its parameters, met by the class (6). At the points of E_W, where the positivity condition fails, the chain of (K_0) is not run and the invariance follows from the nearby scalar masses by right-continuity of King's state in the scalar mass (item (15), Proposition 1.4). The implication (1)–(5) $\Rightarrow$ (K_0), on the class (6), is Theorem 4.8, the implication (8)–(11) $\Rightarrow$ (RES_0), with the literature inputs (12), is Theorem 7.10 (item (7)), (RES_0), (K_0) and the dead-shell chain (10c) give (TIGHT_0) by Theorem 7.8, and (K_0) and (TIGHT_0) give the invariance by Theorem 6.4; item (13) defines the object, King's state on curvature sources on King's cubes (Definition 3.4, Theorem 3.3). Among the inputs proved in the main paper, the change of domain (P7) of Bałaban's propagators on the skew tori enters (K_0), through item (1) on the skew tori; the long trees (P8) in the difference clause of the split degree count enter King's cube limit (13), which defines the object; and the sub-domain form (P9) of the perturbative bound at order fourteen enters the upper bound of (8), and through (8) (RES_0) and (TIGHT_0). The bound (DOM) for the first-order term of the step determinants on Bałaban's propagator domains and on the torus, uniform in the shell mass, has two independent proofs (Section 1.4); it enters (8) through the determinant clause of the perturbative bound (8k)(ii), and its shell-mass residual does not enter (K_0) or King's cube limit (13), which are statements at zero shell mass. King's cut of the long lines on his cubes, which King's cube limit (13) uses, is proved, (P5). No statement of the literature is used in a form asserted in print without a printed argument or read off a printed construction: the six readings (B1)–(B6) of

Bałaban's papers (Section 8.2), with the closure of his renormalised classes under the completion moves, King's two statements, the source-free strict degree property of his renormalised graphs as rewritten by Bałaban [K1, p. 668] and his treatment of the sources [K1, pp. 664–665, 677], and the photon legs of King's Appendix (Section 8.3) are proved in [SUP], in the forms consumed; (K_0) and (8) use (B4), and (10) uses (B1). Section 8 restates all the inputs in plain words, with the steps of the other parts of the main paper that are completed in [SUP].

Theorem A is Theorem 6.4 with Theorems 4.8, 7.10 and 7.8. The inputs of Section 1.3 are discussed in Remark 4.9 and Remark 7.11, and tabulated in Section 9. In one sentence: (K_0) is a one-sided comparison of effective actions at equal spacing of the kind sketched in [K2, §4] for King's torus conjecture, carried out in [CT] from (1)–(5), on the class (6), with Bałaban's asserted estimates (1), the reading (B4) included, the estimates (1) on the skew tori with the change of domain (P7), and the steps of [CT] completed in [SUP] (Section 8), outside the set E_W of Proposition 1.4; (RES_0) is the uniform-in-the-spacing decoupling of a heavy weakly coupled shell from a light core across a heavy plateau, which is proved from four estimates on King's block-spin flow with the shell mass inside the covariance, (8)–(11), whose own residues (8a)–(8k), (10a)–(10g) are proved in [SW1, SW3, SUP], the determinant clause of (8k)(ii) with the bound (DOM), and the upper bound of (8) with the sub-domain form (P9).

Definition 3.4 states the object, and the scope of the paper, in one place. The relation of King's state to infinite-volume limits along periodic boxes, King's expected identity [K2, (2.24)], is never used.

1.7. **Organisation.** Section 2 defines the model, the states and the sources, collects the correlation inequalities, and defines King's renormalisation-group cubes, which it compares with the boxes. Section 3 constructs King's state on curvature sources. Section 4 states the torus hypothesis, reduces it to one-sided bounds at one spacing, and recalls the corrected torus theorem with its inputs (1)–(5), on the class (6). Section 5 introduces walled tori and the closing hypothesis (TIGHT_0), and Section 6 proves the conditional invariance theorem. Section 7 contains the heavy-shell hypothesis (RES_0), the two-level sandwich, the proof that (RES_0) implies (TIGHT_0), and the heavy-shell theorem which proves (RES_0) from the shell estimates (8)–(11). Section 8 restates the inputs in plain words, Section 9 is the table of inputs, and Section 10 states the open problem. Appendix A collects the corrections to the literature.

2. The model

2.1. Geometry.

Definition 2.1. Let $\theta_0 \in (0, \pi/2)$ with $\cos\theta_0 = \frac{4}{5}$, $\sin\theta_0 = \frac{3}{5}$. For $1 \le \mu < \nu \le 3$ let $R_{\mu\nu}(\theta)$ be the rotation with $Re_\mu = \cos\theta\, e_\mu + \sin\theta\, e_\nu$, $Re_\nu = -\sin\theta\, e_\mu + \cos\theta\, e_\nu$, $Re_\rho = e_\rho$ for $\rho \ne \mu, \nu$. Put $\mathcal{R}_0 := \{R_{\mu\nu}(\pm\theta_0) : 1 \le \mu < \nu \le 3\}$ and let G_0 be the group generated by $\mathcal{R}_0$. Let $\varepsilon_k := 5^{-k}$ and $\Lambda_k := \varepsilon_k \mathbb{Z}^3$, $k \ge 0$. Let M be the large-block integer of [K1, §2.3], fixed throughout; we take $M = 5M'$ with M' divisible by a prime $\mathfrak{p} \ne 5$ and $M' \ge 2\mathfrak{p}$, for instance $M = 20 \cdot 5^b$ with b large. This is admissible: every condition that the constructions of [K1, K2, Ba1, Ba2, Ba3, Ba4] place on M is a lower bound, apart from divisibility conditions on the sides of King's cubes and tori, such as that the side $2MJ$ of C_J be a multiple of every block side [K2, (3.4)], which hold for every M by the definitions of C_J and D_J below; it does not change King's state (Definition 3.4), and it is used, with the grid offset of the box H' in the class (6) of Section 1.3, by the sub-domain bound (8k)(i). The *King cubes* are $C_J := \{x \in \mathbb{R}^3 : -MJ \le x_\mu < MJ,\ \mu = 1, 2, 3\}$, $J \in \mathbb{N}$ [K2, (2.7)], and the *King tori* are $D_J := \mathbb{R}^3/(50MJ\,\mathbb{Z})^3$ with fundamental cell $\Gamma_J := [-25MJ, 25MJ)^3$ [K2, (4.1)]. We write $Q_b := [-b, b]^3$, $N_r(X) := \{y : \mathrm{dist}_\infty(y, X) \le r\}$, and $\bar{B}_\rho$ for the closed Euclidean ball of radius ρ about 0.

Lemma 2.2. (a) *For $R \in \mathcal{R}_0$ the matrix $5R$ has integer entries, $R^{-1} \in \mathcal{R}_0$, and $\sigma_\nu R_{\mu\nu}(\theta)\sigma_\nu = R_{\mu\nu}(-\theta)$, where σ_ν is the reflection $x_\nu \mapsto -x_\nu$.*

(b) *For $R \in \mathcal{R}_0$, $k \geq 0$ and $J \geq 1$ the lattices $R\Lambda_k$ and $R^{-1}\Lambda_k$ are invariant under translations by $50MJ\,\mathbb{Z}^3$, so they descend to D_J.*

(c) *For $R \in \mathcal{R}_0$, $|Rx|_\infty \leq \frac{7}{5}|x|_\infty$, and $Q_{5b/7} \subset R^{-1}Q_b \subset Q_{7b/5}$ for $b > 0$.*

(d) *θ_0/π is irrational, and G_0 is dense in $SO(3)$.*

Proof. (a) The entries of R are $0, 1, \pm\frac{3}{5}, \pm\frac{4}{5}$, and $R_{\mu\nu}(\theta)^{-1} = R_{\mu\nu}(-\theta)$; conjugation by σ_ν reverses the orientation of the (x_μ, x_ν)-plane and fixes the third axis. (b) For $p \in 50MJ\,\mathbb{Z}^3$ we have $R^{\pm1}p = \frac{1}{5}(5R^{\pm1})p \in 10MJ\,\mathbb{Z}^3 \subset \Lambda_k$. Hence $R^{-1}\Lambda_k + p = R^{-1}(\Lambda_k + Rp) = R^{-1}\Lambda_k$, and likewise for $R\Lambda_k$. (c) The sums of the absolute values of the entries in each row of R are at most $\frac{7}{5}$. If $Rx \in Q_b$ then $|x|_\infty = |R^{-1}(Rx)|_\infty \leq \frac{7}{5}b$, since $R^{-1} \in \mathcal{R}_0$; if $|x|_\infty \leq \frac{5}{7}b$ then $|Rx|_\infty \leq b$. (d) If $\theta_0 \in \pi\mathbb{Q}$, then $e^{\mathrm{i}\theta_0} = (4 + 3\mathrm{i})/5$ is a root of unity, hence an algebraic integer. Its minimal polynomial over $\mathbb{Q}$ is $x^2 - \frac{8}{5}x + 1$, which does not have integer coefficients; this is a contradiction (compare [K2, pp. 337–338]). Hence $\{n\theta_0\}_{n\in\mathbb{Z}}$ is dense modulo 2π, and the closure of $\{R_{\mu\nu}(\theta_0)^n\}_{n\in\mathbb{Z}}$ is the group of all rotations in the (x_μ, x_ν)-plane. Every element of $SO(3)$ is a product $R_{12}(\alpha)R_{23}(\beta)R_{12}(\gamma)$ (Euler angles), so $\overline{G_0} = SO(3)$. □

2.2. **Lattices, fields and action.**

Definition 2.3 (lattices and fields)**.** A *lattice of spacing* ε_k is $\Lambda = O\Lambda_k$ with $O \in O(3)$, equipped with the frame $f_\alpha := Oe_\alpha$, $\alpha = 1, 2, 3$, and with the nearest-neighbour bonds $\langle x, x + \varepsilon_k f_\alpha\rangle$. A field configuration on a set X of sites consists of $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and $A_\alpha(x) \in \mathbb{R}$, $x \in X$; the bond $\langle x, x+\varepsilon f_\alpha\rangle$ carries $A_{\langle x,x+\varepsilon f_\alpha\rangle} := A_\alpha(x)$ and the reversed bond carries $-A_\alpha(x)$ (Feynman form of the vector field, [K2, (2.5)]). A plaquette is $p = (x; \alpha, \beta)$ with $\alpha < \beta$; its centre is $c(p) := x + \frac{\varepsilon}{2}(f_\alpha + f_\beta)$ and its field strength is

$$F(p) := \varepsilon^{-1}\big(A_\alpha(x) + A_\beta(x + \varepsilon f_\alpha) - A_\alpha(x + \varepsilon f_\beta) - A_\beta(x)\big) \qquad \text{[K2, (2.11)]}.$$

Definition 2.4 (King's action)**.** Write $\varepsilon = \varepsilon_k$. The formal action on a lattice Λ of spacing ε is [K2, (2.5)]

$$\begin{aligned} S(A,\phi) = {} & \frac{1}{2}\sum_{\alpha,\beta}\sum_{x} \varepsilon\,\big(A_\alpha(x+\varepsilon f_\beta) - A_\alpha(x)\big)^2 + \frac{\mu_0^2}{2}\sum_{\alpha}\sum_{x}\varepsilon^3 A_\alpha(x)^2 \\ & + \frac{1}{2}\sum_{\beta}\sum_{x}\varepsilon\,\big|e^{\mathrm{i}e\varepsilon A_\beta(x)}\phi(x+\varepsilon f_\beta) - \phi(x)\big|^2 + \sum_{x}\varepsilon^3\Big[\tfrac{1}{2}\big(m^2 + \delta m_\varepsilon^2\big)|\phi(x)|^2 + \lambda|\phi(x)|^4\Big], \end{aligned}$$

where $|e^{\mathrm{i}e\varepsilon A}\phi(y) - \phi(x)|^2 = r_x^2 + r_y^2 - 2r_x r_y\cos(\vartheta_x - e\varepsilon A - \vartheta_y)$. The mass counterterm δm_ε^2 is King's [K1, (3.32)–(3.34)], computed with propagators on the infinite lattice $\varepsilon\mathbb{Z}^3$ (free boundary conditions, as in [K2, p. 330]); it is a polynomial in e^2 and λ whose coefficients depend only on ε, m and μ_0. King's action also carries the normalisation constant E_0 and the vacuum-energy counterterm E_1 [K1, (2.4), (3.35)], [K2, p. 325]; both are field-independent, they cancel in every normalised expectation, and we omit them. The *free Wick constant* is $\mathsf{C}_\varepsilon := (-\Delta_{\varepsilon\mathbb{Z}^3} + m^2)^{-1}(0,0)$.

Definition 2.5 (states)**.** Let Λ be a lattice of spacing ε, and $d\nu := \prod_x \prod_\alpha dA_\alpha(x)\, d^2\phi(x)$ over the sites in question.

(a) *Neumann–Dirichlet boxes.* For a finite set X of sites of Λ, $\langle\cdot\rangle_{\Lambda,X}^{\mathrm{ND}}$ is the normalised measure $Z^{-1}e^{-S_{\Lambda,X}^{\mathrm{ND}}}d\nu$ on the fields on X, where $S_{\Lambda,X}^{\mathrm{ND}}$ is obtained from S by setting $\phi = 0$ at all sites outside X (Dirichlet condition for the scalar field), and by keeping the photon gradient terms only for pairs $x, x+\varepsilon f_\beta$ that both lie in X and the photon mass term only for $x \in X$ (Neumann condition for the vector field). For $\Lambda = \Lambda_k$ and $X = C_J \cap \Lambda_k$ this is King's state $\langle\cdot\rangle_{N,D;C_J}^{\varepsilon_k}$ of [K2, (2.9)–(2.10)] in the convention of King's proof of [K2, Thm. 3.2]

(pp. 333–334); we write $\langle\cdot\rangle^{\mathrm{ND}}_{k,J}$. The cube on which King's renormalisation group is run differs from it by the matter on one face layer (Definition 2.15).

(b) *Tori.* If Λ descends to D_J, $\langle\cdot\rangle_{\Lambda,D_J}$ is the normalised measure with the action S summed over the sites and bonds of the finite graph $\Lambda/(50MJ\,\mathbb{Z})^3$ (periodic conditions for both fields).

(c) *Walled tori.* A *walled configuration* is $\mathfrak{c} = (J', w)$ with $J' \in \mathbb{N}$ and $w \in C^\infty(D_{J'}; [0,\infty))$, the *wall*. If Λ descends to $D_{J'}$, $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ is the torus state of (b) with the density multiplied by $\prod_x \exp(-\varepsilon^3 w(x)|\phi(x)|^2)$. The wall raises the scalar mass squared by $2w$ and acts in the single-site measures. We write $(J', \psi, n) := (J', n\psi)$ for $n \geq 0$.

(d) *Killed sites.* For any of the states above and a set U of sites, the state *with the matter killed on* U is obtained by replacing the single-site measure $d^2\phi(x)$ times its weight, for $x \in U$, by the point mass at $\phi(x) = 0$. All hopping terms with an endpoint in U then reduce to their diagonal parts at the other endpoint; the photon variables are unchanged.

For a lattice Λ descending to a torus T we also use the *unnormalised* generating functional $\mathcal{Z}_{\Lambda,T}(G,h) := \int e^{-S(A,\phi)+F_\Lambda(G)+:\phi^2:_\Lambda(h)}\,d\nu$ (Definition 2.8), with King's constants E_0, E_1 omitted. Two lattices of the same spacing descending to T have the same number of sites in a fundamental cell, so any field-independent constant per site multiplies both by the same factor.

The definition of δm^2_ε with free boundary conditions is the one King uses in cubes [K2, p. 330]; it makes all states on all lattices of spacing ε carry the same local counterterm.

Remark 2.6 (the counterterm). King's δm^2_ε is a single number: it is fixed by King's two-point graphs of order at most 4 in (e,λ) on the infinite lattice $\varepsilon\mathbb{Z}^3$ at zero momentum [K1, (3.32)–(3.33)], with the propagators of the infinite lattice [K2, p. 330], and it is the same at every site of every box and every torus. We use only this; it is the reading under which King's cut of a cube out of a larger cube is "exactly" the smaller Neumann–Dirichlet state [K2, p. 334]. The theorems of [K1] are proved for the torus conventions of [K1, (3.33), (2.8), (3.21), (3.35)]; the bridge between the two conventions is the input (CT) of [CT] (Section 4.3) on the King side and $(\mathrm{CT})_{\mathrm{sh}}$ ([SW1, Lemma M(d)]: the offset Δ_T is a constant added to the counterterm-mismatch vertex; no cross-convention equality of states is used) on the shell side (items (8f) and (10) of Section 1.3).

Lemma 2.7 (killing)**.** *Let $\langle\cdot\rangle$ be a state of Definition 2.5 (a)–(c), U a set of sites, $\chi : U \to (0,\infty)$, and $\langle\cdot\rangle_{U,s}$ the state with the single-site measures at $x \in U$ multiplied by $e^{-s\varepsilon^3\chi(x)r_x^2}$, $s \geq 0$. Then for every continuous function Θ of the fields with $|\Theta| \leq e^{C\sum_x r_x^2}$, $\langle\Theta\rangle_{U,s} \to \langle\Theta\rangle_U$ as $s \to \infty$, where $\langle\cdot\rangle_U$ is the state with the matter killed on U.*

Proof. At fixed lattice spacing all integrals are finite-dimensional. Substituting $\phi(x) = (s\varepsilon^3\chi(x))^{-1/2}\zeta(x)$ for $x \in U$, the normalised single-site measures at $x \in U$ converge weakly to the point mass at 0, the hopping terms $\varepsilon r_x r_y \cos(\cdot)$ with $x \in U$ are bounded by $\frac{\varepsilon}{2}(s\varepsilon^3\chi(x))^{-1}|\zeta_x|^2 + \frac{\varepsilon}{2}r_y^2$, and the quartic terms at the other sites dominate; dominated convergence applies to numerator and denominator. □

2.3. **Sources and observables.**

Definition 2.8 (sources)**.** A *curvature source* is a real 2-form $f \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$, and a *scalar source* is a real $h \in C^\infty_c(\mathbb{R}^3)$; on a torus, smooth periodic sources are used. On a lattice Λ of spacing ε with frame (f_α) put

$$F_\Lambda(f) := \sum_p \varepsilon^3\, f_{c(p)}(f_\alpha, f_\beta)\, F(p),$$

$$\mathsf{S}_\Lambda(h) := \sum_x \varepsilon^3 h(x), \qquad :\phi^2:_\Lambda(h) := \sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2 - 2\mathsf{C}_\varepsilon\big),$$

the sums running over the plaquettes $p = (x; \alpha, \beta)$, respectively the sites, of the region (of one fundamental cell on a torus). For finite families and $(z, w) \in \mathbb{C}^{2N}$ put

$$\Theta\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big) := \exp\Big(\sum_j z_j F_\Lambda(f_j) + \sum_j w_j {:}\phi^2{:}_\Lambda(h_j)\Big).$$

For $O \in O(3)$ and $b \in \mathbb{R}^3$ let $(O \cdot h)(x) := h(O^{-1}x)$, $(O \cdot f)_x(u, v) := f_{O^{-1}x}(O^{-1}u, O^{-1}v)$, $(\tau_b h)(x) := h(x - b)$ and $(\tau_b f)_x := f_{x-b}$. The *cone* is

$$\mathcal{K} := \{(f, g) : \ f \text{ a real 2-form}, \ g \geq 0, \ \text{both in } C_c^\infty\},$$

with the observable $\Theta(\mathrm{i}f, g)$. $\mathcal{K}$ is invariant under $O(3)$, under translations and under $(f, g) \mapsto (tf, tg)$, $t \geq 0$, and under sums.

All states of Definition 2.5 are invariant under $(A, \vartheta) \mapsto (-A, -\vartheta)$, which maps $F_\Lambda(f)$ to $-F_\Lambda(f)$ and fixes r. Hence $\langle \Theta(\mathrm{i}f, g)\rangle = \langle \cos F_\Lambda(f)\, e^{{:}\phi^2{:}_\Lambda(g)}\rangle$ is real. For a walled configuration $\mathfrak{c} = (J', w)$, a lattice Λ descending to $D_{J'}$ and a real 2-form f supported in one fundamental cell we write

$$T_{\mathfrak{c},\Lambda}(f) := \langle \Theta(\mathrm{i}f, 0)\rangle_{\mathfrak{c},\Lambda} = \langle e^{\mathrm{i}F_\Lambda(f)}\rangle_{\mathfrak{c},\Lambda} = \frac{\mathcal{Z}_{\Lambda, D_{J'}}(\mathrm{i}f, -w)}{\mathcal{Z}_{\Lambda, D_{J'}}(0, -w)}$$

for the *walled expectation* of f (the Wick constants cancel in the ratio); it is real.

Remark 2.9 (the frame convention). In [K1, K2] the curvature source is written $F(g) = \sum_p \varepsilon^3 g(p) F(p)$ with one function g on all plaquettes [K2, (2.12)]. On one lattice a 2-form sampled as above is a function on plaquettes, and King's proofs treat the source terms componentwise ("the sources $g_{\mu\nu}(x'), h(x')$ ... can be replaced by $g_{\mu\nu}(x), h(x)$", [K1, p. 665]; the sentence is asserted there without an argument). We use King's single-lattice statements at $h = 0$ for 2-form sources in this sense: Theorem 2.1 of [K2] holds, on King's cubes C_J^K (Definition 2.15), for real smooth 2-form sources sampled at plaquette centres in the lattice's own frame, periodic or compactly supported, with an a priori bound on the limit that depends on the source only through $\|\nabla G_{\mu\nu}\|_\infty$ and the cube (item (13) of Section 1.3, asserted in print without a complete argument and proved in [CA], with the bound at long trees of Proposition 1.2); the statements of [K2, Thms. 2.2, 2.3, (3.40)–(3.44)] that we use, all at $h = 0$, are proved here from it and from Lemmas 2.14 and 2.16 (Section 3), and [K2, (3.42)–(3.43)] are not used, and the torus statement [K1, Thm. 2.1] at a wall is item (4). The two-lattice statement is different: Proposition A.2 shows that it fails for a single scalar function on all plaquettes, and Hypothesis 4.1 is formulated for 2-forms sampled in each lattice's own frame.

Lemma 2.10 (passive covariance)**.** *Let $O \in O(3)$, let Λ be a lattice of spacing ε with frame (f_α), and let X be a box, a torus or a walled torus for Λ. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions and wall $O \cdot w$. Then*

$$\big\langle \Theta(z\, O \cdot f, w\, O \cdot h)\big\rangle_{O\Lambda, OX} = \big\langle \Theta(zf, wh)\big\rangle_{\Lambda, X}, \qquad \mathsf{S}_{O\Lambda}(O \cdot h) = \mathsf{S}_\Lambda(h),$$

and likewise for finite families. Moreover, on every torus the periodic model on Λ does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, for a permutation π and any choice of signs.

Proof. The map $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$, with components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving $d\nu$. The action is written in the frame, and δm_ε^2 and C_ε depend on ε only. Plaquettes, their centres and field strengths are transported, and $(O \cdot f)_{Oc}(Of_\alpha, Of_\beta) = f_c(f_\alpha, f_\beta)$. For the second statement, a permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x - \varepsilon f_\alpha)$, a bijection of the torus variables that preserves the action, and changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha, f_\beta)$ while preserving the set of plaquette centres. □

2.4. **Correlation inequalities.**

Definition 2.11 (BFS measures)**.** A *BFS measure* on a finite set V of sites is a probability measure of the form

$$Z^{-1}\exp\Big(\sum_{\{x,y\}\in B} c_{xy}\, r_x r_y \cos(\vartheta_x-\vartheta_y-\ell_{xy}(A))\Big)\prod_{x\in V} d\rho_x(r_x)\,d\vartheta_x\;\; d\gamma(A),$$

where B is a set of pairs of sites, $c_{xy}\ge 0$, ρ_x are finite measures on $[0,\infty)$ with all exponential moments of r^2 finite (point masses at 0, that is killed sites, included), γ is a non-degenerate centred Gaussian measure on $\mathbb{R}^m$, and the ℓ_{xy} are linear functionals. For a real linear functional F of A and $g\ge 0$ on V put $\Theta_{F,g}:=e^{\mathrm{i}F}e^{\sum_x g_x r_x^2}$.

Every state of Definition 2.5 is a BFS measure: $c_{xy}=\varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac12(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$, the diagonal parts $\frac{\varepsilon}{2}r_x^2$ of all bonds at x (including, for boxes, those leading out of X), and the wall; the photon form is positive definite because $\mu_0>0$. The same holds for every measure obtained from such a state by multiplying some c_{xy} by $t\in[0,1]$, multiplying photon gradient terms by $t\in[0,1]$, adding $s|\ell(A)|^2$ ($s\ge 0$) to the action, or multiplying single-site measures by e^{-sr^2} ($s\ge 0$), or killing sites. Curvature sources $F_\Lambda(f)$ are real linear functionals of A, and $e^{:\phi^2:(g)}$ is a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$.

Lemma 2.12 (Brydges–Fröhlich–Seiler, King)**.** *Let $\langle\cdot\rangle$ be a BFS measure, F a real linear functional of A, $g\ge 0$, and $\Theta=\Theta_{F,g}$. Write $\langle X;Y\rangle:=\langle XY\rangle-\langle X\rangle\langle Y\rangle$. Then $\langle\Theta\rangle$ is real, and*

(a) *$\langle\Theta;|\ell(A)|^2\rangle\le 0$ for every complex linear functional ℓ; hence adding $s|\ell(A)|^2$ to the action, $s\ge 0$, does not decrease $\langle\Theta\rangle$, and deleting a photon gradient term does not increase it;*

(b) *$\langle\Theta;r_xr_y\cos(\vartheta_x-\vartheta_y-\ell_{xy}(A))\rangle\ge 0$; hence decreasing c_{xy} does not increase $\langle\Theta\rangle$;*

(c) *$\langle\Theta;r_x^2\rangle\ge 0$; hence multiplying ρ_x by e^{-sr^2}, $s\ge 0$, does not increase $\langle\Theta\rangle$, and killing a site does not increase it;*

(d) *for $\Theta^-:=e^F e^{-\sum_x g_x r_x^2}$ the three inequalities hold with the opposite signs;*

(e) *if the pairs in B and the Gaussian γ do not couple the fields at the sites of $X\subset V$ (and the photon variables attached to them) to the others, the measure is a product, and for Θ depending only on fields in X, $\langle\Theta\rangle$ equals the expectation in the X-factor.*

Proof. (a)–(c) are [BFS, Thms. 6.1, 6.2(1)–(2)] (stated on an arbitrary finite lattice, with an arbitrary Gaussian measure for A and arbitrary single-site measures, [BFS, pp. 275–276]); compare [K2, Prop. 3.1, (3.24)–(3.26)]. (c) is Theorem 6.1 of [BFS] with $P=r_x^2$. For (d), [BFS, Thm. 6.2(3)–(4)] states the photon and the hopping inequality with a sketched proof, and the r^2-inequality, [K2, (3.28)], is stated in [K2] without a proof and is not among the inequalities printed in [BFS]; we prove all three. Let μ be the unnormalised measure and ω' an independent copy of $\omega=(r,\vartheta,A)$; then $(\int d\mu)^2\langle\Theta^-;Y\rangle=\iint\Theta^-(\omega)(Y(\omega)-Y(\omega'))\,d\mu(\omega)\,d\mu(\omega')$. Write $r=u-v$, $r'=u+v$ (so $u_x\ge|v_x|$), $\vartheta=\vartheta_++\vartheta_-$, $\vartheta'=\vartheta_+-\vartheta_-$, $A=a_++a_-$, $A'=a_+-a_-$. Then a_+,a_- are independent with one centred Gaussian law, ϑ_+,ϑ_- are independent and uniform, and the radial law $\prod_x d\rho_x(u_x-v_x)\,d\rho_x(u_x+v_x)$ is invariant under $v_x\mapsto -v_x$ at each site. With $\chi_{xy}:=\vartheta_{+,x}-\vartheta_{+,y}-\ell_{xy}(a_+)$ and $\psi_{xy}:=\vartheta_{-,x}-\vartheta_{-,y}-\ell_{xy}(a_-)$,

$$r_xr_y\cos(\chi+\psi)+r'_xr'_y\cos(\chi-\psi)=2(u_xu_y+v_xv_y)\cos\chi\cos\psi+2(u_xv_y+v_xu_y)\sin\chi\sin\psi,$$
$$r_xr_y\cos(\chi+\psi)-r'_xr'_y\cos(\chi-\psi)=-2(u_xv_y+v_xu_y)\cos\chi\cos\psi-2(u_xu_y+v_xv_y)\sin\chi\sin\psi,$$

$r_x^2-r_x'^2=-4u_xv_x$, $\ell(A)^2-\ell(A')^2=4\ell(a_+)\ell(a_-)$ for real ℓ, and

$$\Theta^-(\omega)=e^{F(a_+)}e^{F(a_-)}e^{-\sum_x g_x(u_x^2+v_x^2)}e^{2\sum_x g_xu_xv_x}.$$

Expanding the combined hopping weight (the first line) and $e^{2\sum g_x u_x v_x}$ in power series, every term of the integrand of $\iint \Theta^-(\omega)(Y(\omega)-Y(\omega'))$ is a sign σ times a monomial in (u,v) with a non-negative coefficient, times the even radial factor $e^{-\sum g_x(u_x^2+v_x^2)}$, times $V(\vartheta_+,a_+)V(\vartheta_-,a_-)$ with one and the same real function V (a product of e^F, of factors $\cos\chi_b$, $\sin\chi_b$, and of ℓ for the photon slot). The angular and photon integral of such a term is the square of the integral of V, and the radial integral vanishes if the degree in v_x is odd at some x and is non-negative otherwise, since $u_x \geq |v_x|$; all series converge absolutely at fixed lattice spacing. The sign is $\sigma=-1$ for $Y=r_x^2$ and for the hopping term (the second line), and $\sigma=+1$ for $Y=\ell(A)^2$ (for complex ℓ apply this to its real and imaginary parts). This is (d). Each monotonicity statement in a parameter follows by differentiating in the parameter, the interpolating measures being BFS measures. The statements on killing need no limit. Let $x\in V$, and let $\langle\cdot\rangle_x$ be the measure with ρ_x replaced by the point mass at 0, in which every term $c_{xy}r_xr_y\cos(\cdot)$ vanishes. Decreasing c_{xy} to 0 for every pair $\{x,y\}\in B$ does not increase $\langle\Theta\rangle$, by (b), and does not decrease $\langle\Theta^-\rangle$, by (d). In the resulting measure the variables (r_x,ϑ_x) are coupled to nothing, so by (e) it is the product of $\rho_x(dr_x)\,d\vartheta_x$ and of the law of the other variables, which is that of $\langle\cdot\rangle_x$; the expectations of Θ and Θ^- in it are therefore $m_x^+\langle\Theta\rangle_x$ and $m_x^-\langle\Theta^-\rangle_x$, with $m_x^\pm := \int e^{\pm g_x r^2}d\rho_x(r)\big/\int d\rho_x$, so that $m_x^-\leq 1\leq m_x^+$. Since $\Theta^->0$, $\langle\Theta^-\rangle_x \geq m_x^-\langle\Theta^-\rangle_x \geq \langle\Theta^-\rangle$. And $\langle\Theta\rangle_x\geq 0$: multiplying the single-site measures by $e^{g_y r_y^2}$ gives a BFS measure, in which the expectation of $e^{\mathrm{i}F}$ is non-negative by Lemma 2.13(a), whose proof does not use the present lemma; hence $\langle\Theta\rangle_x \leq m_x^+\langle\Theta\rangle_x \leq \langle\Theta\rangle$. Several sites are killed one at a time. No condition on ρ_x beyond Definition 2.11 is used, and Lemma 2.7 is not needed here. (e) is clear. □

Lemma 2.13 (current representation and Gaussian domination)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C. Integrating out the matter at fixed A, the matter integral is $\Psi(A)=\sum_q w(q)\cos\langle\kappa(q),A\rangle$, where q runs over integer-valued divergence-free currents on B, $w(q)\geq 0$, $\sum_q w(q)<\infty$, and $\kappa(q)\in\mathbb{R}^m$ is defined by $\langle\kappa(q),A\rangle := \sum_{xy} q_{xy}\ell_{xy}(A)$. Let $\mathbb{E}^\sharp$ be the probability measure on currents with weights proportional to $w(q)e^{-\langle\kappa(q),C\kappa(q)\rangle/2}$. Then for $v\in\mathbb{R}^m$:*

(a) $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cosh\langle Cv,\kappa\rangle$ *and* $\langle e^{s\langle v,A\rangle}\rangle = e^{s^2\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cos(s\langle Cv,\kappa\rangle) \leq e^{s^2\langle v,Cv\rangle/2}$ *for $s\in\mathbb{R}$;*

(b) $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 \in [0,\langle v,Cv\rangle]$.

Proof. Expanding $e^{c\cos t}=\sum_{q\in\mathbb{Z}} I_q(c)e^{\mathrm{i}qt}$ with $I_q(c)\geq 0$ for $c\geq 0$ and integrating over the angles gives Ψ; the weights are even in q because $I_{-q}=I_q$. For a Gaussian γ with covariance C, $\int e^{\mathrm{i}\langle v,A\rangle}\cos\langle\kappa,A\rangle\,d\gamma = e^{-\langle v,Cv\rangle/2-\langle\kappa,C\kappa\rangle/2}\cosh\langle v,C\kappa\rangle$ and, by the Cameron–Martin formula, $\int e^{s\langle v,A\rangle}\cos\langle\kappa,A\rangle\,d\gamma = e^{s^2\langle v,Cv\rangle/2-\langle\kappa,C\kappa\rangle/2}\cos(s\langle v,C\kappa\rangle)$; this gives (a). Replacing v by tv in the first formula of (a) and comparing the terms of order t^2 (all sums converge absolutely at fixed lattice spacing) gives (b). □

The following consequence is the form in which the correlation inequalities are used.

Lemma 2.14 (cutting)**.** *Let $\langle\cdot\rangle_Y$ be a box, torus or walled-torus state on a lattice Λ, possibly with killed sites, and let X be a set of sites of Λ such that X and its nearest neighbours lie in one fundamental cell. Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (wall, killed sites and diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$. Let f,g be sources with $g\geq 0$ such that every plaquette and site at which they do not vanish lies in X. Then, at every lattice spacing,*

$$\langle\Theta(\mathrm{i}f,g)\rangle_X^{\mathrm{ND}} \leq \langle\Theta(\mathrm{i}f,g)\rangle_Y, \qquad \langle\Theta(f,-g)\rangle_X^{\mathrm{ND}} \geq \langle\Theta(f,-g)\rangle_Y.$$

If the wall of Y vanishes on X and no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}} = \langle\cdot\rangle_{\Lambda,X}^{\mathrm{ND}}$ is the state of Definition 2.5(a).

Proof. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2 + r_y^2)$. By Lemma 2.12(a),(b),(d) this does not increase the first expectation and does not decrease the second. The resulting measure is a product (Lemma 2.12(e)); its X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$, since the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms are the Neumann ones. The free Wick constants and δm_ε^2 are the same in both states. □

With $Y = C_{J+1} \cap \Lambda_k$ and $X = C_J \cap \Lambda_k$ this is King's monotonicity in the volume [K2, Thm. 3.2, (3.31)–(3.33)] in the form his proof proves, with one Wick constant for all J. King states it with the Dirichlet Wick constant of [K2, (2.13)], which increases with J; then the c-number factor $\exp[\mp 2\sum_x \varepsilon^3 h(x)(C^\infty - C^J)(x,x)]$ (C^J the Dirichlet covariance of C_J at coinciding points) moves against the monotone factor and the monotonicity does not follow. That form is not used, in either convention for the cube. King's proof [K2, pp. 333–334] is written for the boxes $C_J \cap \Lambda_k$ of Definition 2.5(a), and it begins by replacing the Dirichlet subtraction by the free one: "the advantage of doing this is that $:\phi^2:(h)$ is then independent of J" (p. 333). For the cubes on which King's renormalisation group is run the same monotonicity, with the same J-independent Wick constant, follows from Lemma 2.16 below, with one kill step that King does not print (Corollary 2.17).

2.5. **King's renormalisation-group cubes.**

Definition 2.15 (King's cubes)**.** For $k \geq 0$ and $J \in \mathbb{N}$ put $X_J := C_J \cap \Lambda_k$, and let $T_J := \{x \in X_J : x_\mu = MJ - \varepsilon_k \text{ for some } \mu\}$ be the union of its three top face layers. *King's cube* C_J^K at spacing ε_k is the Neumann–Dirichlet box X_J of Definition 2.5(a) with the matter killed on T_J (Definition 2.5(d)). We write $\langle\cdot\rangle_{k,J}^K$ for its state. Its photon is the Neumann photon of X_J, its scalar field is alive on $X_J \setminus T_J = \{-MJ, \dots, MJ - 2\varepsilon_k\}^3$, and its scalar Dirichlet planes are $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ - \varepsilon_k$, at distance $2MJ$.

The translation by $(MJ + \varepsilon_k)(1,1,1)$ carries X_J onto the box $\{0 < x_\mu \leq 2MJ\}$ of [K2, (3.3)] with $L_\mu = 2MJ$, the Neumann images of the photon onto those of [K2, (3.7)], about the planes half a spacing outside the box, and the scalar Dirichlet condition onto that of [K2, (3.9)], on the planes $x_\mu = 0$ and $x_\mu = L_\mu$, the second of which is a layer of the box; we read (3.9) as setting the scalar field to zero on these planes. The side $2MJ$ is a multiple of every block side, as [K2, (3.4)] requires, and King's block-spin flow [K2, §3.1] is defined on C_J^K with the Dirichlet weight f_D for the block averages of the scalar field, the weight for which the image representation [K2, (3.9)] holds (it does not hold for the weight printed in [K2, (3.8)]), and with the weight 1 and blocks tiling C_J for the photon. For the box X_J itself the scalar Dirichlet planes are at distance $2MJ + \varepsilon_k$, which is a multiple of no block side, and no flow with Dirichlet images is defined on it. The box is the cube of King's proof of the monotonicity in the volume [K2, pp. 333–334], and King's cube C_J^K is the cube of his renormalisation-group construction [K2, §3.1, pp. 328–329]; the two differ by the matter on T_J only.

Lemma 2.16 (interlacing of the two cube families)**.** *Let $k \geq 0$, $J \geq 1$, and let f be a real 2-form and $g \geq 0$, both in C_c^∞, such that every plaquette of Λ_k at which f does not vanish has its four corners in X_J and every site at which g does not vanish lies in $X_J \setminus T_J$; this holds if* $\operatorname{supp} f \cup \operatorname{supp} g$ *has ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$. Then*

$$\langle\Theta(\mathrm{i}f, g)\rangle_{k,J}^K \leq \langle\Theta(\mathrm{i}f, g)\rangle_{k,J}^{\mathrm{ND}} \leq \langle\Theta(\mathrm{i}f, g)\rangle_{k,J+1}^K,$$
$$\langle\Theta(f, -g)\rangle_{k,J}^K \geq \langle\Theta(f, -g)\rangle_{k,J}^{\mathrm{ND}} \geq \langle\Theta(f, -g)\rangle_{k,J+1}^K.$$

Proof. The plaquettes of the sources have their corners in X_J, whose photon variables are present in all three states, and the sites of g are alive in all three; C_ε and δm_ε^2 do not depend on the site. So the observables are the same functions of the shared variables, times the same positive constants. The

state $\langle\cdot\rangle^K_{k,J}$ is obtained from $\langle\cdot\rangle^{\mathrm{ND}}_{k,J}$ by killing the sites of T_J; by the killing clauses of Lemma 2.12(c), (d) this does not increase $\langle\Theta(\mathrm{i}f,g)\rangle$ and does not decrease $\langle\Theta(f,-g)\rangle$, which is the first inequality of each chain. For the second, the sites of T_{J+1} have a coordinate $M(J+1)-\varepsilon_k > MJ-\varepsilon_k$, so no site of X_J is killed in $\langle\cdot\rangle^K_{k,J+1}$, and X_J with its nearest neighbours lies in X_{J+1}. Lemma 2.14, with Y the state of C^K_{J+1} and $X = X_J$, gives the inequalities, its last sentence identifying the cut-out state with $\langle\cdot\rangle^{\mathrm{ND}}_{k,J}$: a bond from a site of X_J to a killed site of T_{J+1} carries its diagonal part at its alive end, which is the Dirichlet term of X_J. A detailed account, with the conventions compared at King's printed pages, is [G]. □

Corollary 2.17 (King's cubes in the volume)**.** *Let f be a real 2-form and $g \geq 0$, both in C^∞_c and supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$.*

(a) *For all $k \geq 0$ and $J \geq J_0$, $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J} \leq \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J+1}$ and $\langle\Theta(f,-g)\rangle^K_{k,J} \geq \langle\Theta(f,-g)\rangle^K_{k,J+1}$: King's monotonicity in the volume* [K2, Thm. 3.2]*, in the form his proof proves, with one J-independent Wick constant, holds on King's cubes.*

(b) *At $g = 0$ and every fixed k, the families $J \mapsto \langle e^{\mathrm{i}F(f)}\rangle^K_{k,J}$ and $J \mapsto \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J}$ are nondecreasing on $J \geq J_0$, bounded by 1, and have the same limit as $J \to \infty$.*

(c) *If $Z^K_J(\mathrm{i}f,0) := \lim_k \langle e^{\mathrm{i}F(f)}\rangle^K_{k,J}$ exists for every $J \geq J_0$, then $\lim_J \liminf_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} = \lim_J \limsup_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} = \lim_J Z^K_J(\mathrm{i}f,0)$; if instead $\lim_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J}$ exists for every $J \geq J_0$, the same holds with the two families exchanged.*

Proof. (a) The hypothesis of Lemma 2.16 holds for every $J \geq J_0$; chain its two inequalities. (b) Both families are nondecreasing, by (a) and by Lemma 2.14 with $Y = C_{J+1} \cap \Lambda_k$, $X = C_J \cap \Lambda_k$; they are real and bounded by 1; and two nondecreasing sequences with $a_J \leq b_J \leq a_{J+1}$ (Lemma 2.16) have the same limit. (c) Lemma 2.16 at every k gives $Z^K_J \leq \liminf_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} \leq \limsup_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} \leq Z^K_{J+1}$, and $J \mapsto Z^K_J(\mathrm{i}f,0)$ is nondecreasing and bounded by (a); for the exchanged statement use $\langle\cdot\rangle^{\mathrm{ND}}_{k,J-1} \leq \langle\cdot\rangle^K_{k,J} \leq \langle\cdot\rangle^{\mathrm{ND}}_{k,J}$, $J \geq J_0 + 1$. □

Remark 2.18 (what the interlacing gives). Lemma 2.16 relates the two families at every spacing and every J, so that they have the same infinite-volume state (Corollary 2.17), but it gives no information on the fixed-J continuum limit of one family from that of the other: the two bounds $\langle\cdot\rangle^K_{k,J}$ and $\langle\cdot\rangle^K_{k,J+1}$ differ macroscopically. The existence statement (13) is therefore posed on the cubes on which King's renormalisation group is run, and every other comparison of cubes in this supplement is transferred to them through Lemma 2.16, at the cost of one index at each end: the comparisons of a cube with walled tori (Theorems 6.4 and 7.5), the monotonicity and the translations (Theorem 3.3), and the reflection (Lemma 6.2). For the reflection $x_3 \mapsto -x_3$ the comparison of C^K_J with $C^K_{J\mp1}$ is also obtained directly, by deletions and kills, when $M \geq 2\varepsilon_k$, which holds at every k for the M of Definition 2.1, since $M \geq 20$ [G]; the proof of Lemma 6.2 uses the comparison with $C^K_{J\mp2}$, which needs no condition.

3. King's state on curvature sources

3.1. **Two elementary properties.** For a state $\langle\cdot\rangle$ of Definition 2.5 and a finite family of real 2-forms f_j and real functions h_j with coefficients $\zeta = (z,w) \in \mathbb{C}^{2N}$ write $Z(\zeta) := \langle\Theta(\sum_j z_j f_j, \sum_j w_j h_j)\rangle$ for the generating functional of Definition 2.8.

Lemma 3.1. (a) *$|Z(\zeta)| \leq Z(\operatorname{Re}\zeta)$, and $\ln Z$ is convex in the real coefficients.*

(b) *$Z(f,h)\,Z(-f,-h) \geq 1$ for real (f,h).*

Proof. (a) $|\Theta(\zeta)| = \Theta(\operatorname{Re}\zeta)$; convexity is Hölder's inequality. (b) $\langle e^X\rangle\langle e^{-X}\rangle \geq 1$ by Cauchy–Schwarz. □

3.2. **The continuity lemma.**

Lemma 3.2 (continuity of convex functionals)**.** *Let E be a Fréchet space and $\Phi : E \to \mathbb{R}$ a convex Borel function. Then Φ is continuous.*

Proof. $E = \bigcup_{j\in\mathbb{N}}\{\Phi \le j\}$, so by the Baire category theorem some $S := \{\Phi \le j\}$ is non-meagre. Being Borel, S has the Baire property: there are a non-empty open set O and a meagre set $\mathcal{M}$ with $O \setminus \mathcal{M} \subset S$. Pick $x_0 \in O$ and a balanced open neighbourhood W of 0 with $x_0 + W + W \subset O$. Let $w \in W$ and $y := x_0 + w$. Then $y + W \subset O$, and the point reflection $\iota(x) := 2y - x$ maps $y + W$ onto itself and preserves meagre sets. The sets $S \cap (y + W)$ and $\iota(S \cap (y + W))$ are comeagre in the non-empty open set $y + W$, so they intersect: there is $u \in S$ with $2y - u \in S$. By convexity $\Phi(y) \le \frac{1}{2}(\Phi(u) + \Phi(2y - u)) \le j$. Hence $\Phi \le j$ on the neighbourhood $x_0 + W$ of x_0. A convex function on a topological vector space that is bounded above on a neighbourhood of one point is continuous. □

3.3. **King's state on curvature sources.** For $\rho > 0$ let $E_\rho := C_c^\infty(\bar{B}_\rho; \Lambda^2)$ be the space of smooth real 2-forms supported in the closed ball $\bar{B}_\rho$, with the C^∞ topology; it is a Fréchet space and it is invariant under $O(3)$. For a finite family of real 2-forms $f_1, \dots, f_N \in C_c^\infty$ and $z \in \mathbb{C}^N$ write

$$Z^K_{k,J}\Big(\sum_j z_j f_j, 0\Big) := \Big\langle \exp\Big(\sum_j z_j F_{\Lambda_k}(f_j)\Big)\Big\rangle^K_{k,J}$$

for the functional of King's cube C^K_J (Definition 2.15) at spacing ε_k, and $Z^{\mathrm{ND}}_{k,J}(\cdot, 0)$ for that of the box $C_J \cap \Lambda_k$ of Definition 2.5(a). If the sources are supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, King's cube limit, item (13) of Section 1.3, gives $Z^K_J(\sum_j z_j f_j, 0) := \lim_k Z^K_{k,J}(\sum_j z_j f_j, 0)$ in $(0, \infty)$ for real z.

Theorem 3.3 (King's state on curvature sources)**.** *Let $f_1, \dots, f_N \in C_c^\infty$ be real 2-forms supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$.*

(i) *For $J \ge J_0$, $Z^K_{k,J}(\sum_j z_j f_j, 0) \to Z^K_J(\sum_j z_j f_j, 0)$ locally uniformly on $\mathbb{C}^N$ as $k \to \infty$, and $z \mapsto Z^K_J(\sum_j z_j f_j, 0)$ is entire.*

(ii) *For a real 2-form f supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $J \mapsto Z^K_J(\mathrm{i}f, 0)$ is nondecreasing and $J \mapsto Z^K_J(f, 0)$ is nonincreasing for $J \ge J_0$.*

(iii) *The limit $Z_\infty(\sum_j z_j f_j, 0) := \lim_{J\to\infty} Z^K_J(\sum_j z_j f_j, 0)$ exists locally uniformly on $\mathbb{C}^N$ and is entire; $Z_\infty(f, 0) > 0$ for every real 2-form f; and $Z^K_J(\mathrm{i}f, 0) \le Z_\infty(\mathrm{i}f, 0)$ for $J \ge J_0$.*

(iv) *For every $\rho > 0$, $\Phi(f) := \ln Z_\infty(f, 0)$ is finite, convex and continuous on E_ρ.*

(v) *$Z_\infty(\sum_j z_j \tau_b f_j, 0) = Z_\infty(\sum_j z_j f_j, 0)$ for all $b \in \mathbb{R}^3$ and $z \in \mathbb{C}^N$.*

Proof. (i) At fixed k the functionals are entire (a finite-dimensional integral with Gaussian decay in A and decay $e^{-\lambda\varepsilon^3 r^4}$ in r). Fix a compact $K \subset \mathbb{C}^N$ and a real box $Q \supset \mathrm{Re}\, K$. By Lemma 3.1(a), $\sup_K |Z^K_{k,J}| \le \sup_Q Z^K_{k,J}$, and by convexity the latter is the maximum over the vertices of Q, at which the sequences converge by item (13). So the family is locally bounded; by Montel's theorem and since $\mathbb{R}^N$ is a set of uniqueness for entire functions, it converges locally uniformly, and the limit is entire (compare [K2, (3.40)–(3.44)]: Lemma 3.1(a) plays the role of (3.40); King's (3.42)–(3.43) are not used).

(ii) Corollary 2.17(a) at fixed k, then $k \to \infty$.

(iii) *Local bound.* For $\zeta \in \mathbb{C}^N$ and $J \ge J_0$, $|Z^K_J(\sum_j \zeta_j f_j, 0)| \le Z^K_J(\sum_j (\mathrm{Re}\,\zeta_j) f_j, 0) \le Z^K_{J_0}(\sum_j (\mathrm{Re}\,\zeta_j) f_j, 0)$, by Lemma 3.1(a) at every k and by (ii), since the real 2-form $\sum_j (\mathrm{Re}\,\zeta_j) f_j$ belongs to the nonincreasing class of (ii); so the family $\{Z^K_J\}_{J\ge J_0}$ is locally bounded on $\mathbb{C}^N$. *Convergence on a set of uniqueness.* On $(\mathrm{i}\mathbb{R})^N$ the source is $\mathrm{i}\sum_j t_j f_j$ with $t \in \mathbb{R}^N$, so $J \mapsto Z^K_J$ is nondecreasing there by (ii) and bounded by 1, hence convergent. $(\mathrm{i}\mathbb{R})^N$ is a totally real subspace of

maximal dimension, hence a set of uniqueness. By Montel's theorem $Z_J^K \to Z_\infty$ locally uniformly on $\mathbb{C}^N$, with Z_∞ entire. Positivity: by Lemma 3.1(b) at every k, $Z_J^K(f,0)\,Z_J^K(-f,0) \geq 1$ for real f, hence $Z_\infty(f,0)\,Z_\infty(-f,0) \geq 1$ in the limit, both factors being limits of positive numbers; so both are positive. The last claim follows from (ii).

(iv) Finiteness and positivity by (iii); convexity as a pointwise limit of convex functions (Lemma 3.1(a)). For J so large that $N_1(\bar{B}_\rho) \subset C_J$, and each k, the function $f \mapsto \ln Z_{k,J}^K(f,0)$ on E_ρ depends continuously on finitely many sampled values of f, hence is continuous; so $\Phi = \lim_J \lim_k$ of continuous functions is Borel. Apply Lemma 3.2.

(v) Let $b \in 5^{-k_0}\mathbb{Z}^3$ with $|b|_\infty \leq M$ and $k \geq k_0$, so that τ_b preserves Λ_k. Write $\langle\cdot\rangle_{k,J}^{\mathrm{ND},b}$ and $\langle\cdot\rangle_{k,J}^{K,b}$ for the states of the box and of King's cube translated by $-b$, which are again states on Λ_k, so that $\langle\Theta(\tau_b\cdot s)\rangle_{k,J}^K = \langle\Theta(s)\rangle_{k,J}^{K,b}$, and let $s = \mathrm{i}\sum_j t_j f_j$ with $t \in \mathbb{R}^N$, the sources being supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$. As sets of sites, $C_{J-1} \cap \Lambda_k \subset (C_J - b) \cap \Lambda_k \subset C_{J+1} \cap \Lambda_k$, so Lemma 2.14 (with $X = C_{J-1} \cap \Lambda_k$ in the state on $(C_J - b) \cap \Lambda_k$, then $X = (C_J - b) \cap \Lambda_k$ in the state on $C_{J+1} \cap \Lambda_k$) gives, for $J \geq J_0 + 1$ (so that, the corners of a plaquette lying within ℓ^∞-distance $\varepsilon/2 \leq \frac{1}{2}$ of its centre, every plaquette at which s does not vanish lies in C_{J-1}), $\langle\Theta(s)\rangle_{k,J-1}^{\mathrm{ND}} \leq \langle\Theta(s)\rangle_{k,J}^{\mathrm{ND},b} \leq \langle\Theta(s)\rangle_{k,J+1}^{\mathrm{ND}}$. Lemma 2.16, applied to the cubes and to their translates, and this sandwich at J and at $J-1$ give for $J \geq J_0 + 3$

$$\langle\Theta(s)\rangle_{k,J-2}^K \leq \langle\Theta(s)\rangle_{k,J-2}^{\mathrm{ND}} \leq \langle\Theta(s)\rangle_{k,J-1}^{\mathrm{ND},b} \leq \langle\Theta(s)\rangle_{k,J}^{K,b} \leq \langle\Theta(s)\rangle_{k,J}^{\mathrm{ND},b} \leq \langle\Theta(s)\rangle_{k,J+1}^{\mathrm{ND}} \leq \langle\Theta(s)\rangle_{k,J+2}^K.$$

Let $k \to \infty$ by item (13) and then $J \to \infty$ by (iii): $Z_\infty(\tau_b \cdot s, 0) = Z_\infty(s,0)$ on $(\mathrm{i}\mathbb{R})^N$. The identity extends to all $z \in \mathbb{C}^N$ by the argument of (iii) (two entire functions that agree on $(\mathrm{i}\mathbb{R})^N$), to all $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration, and to all $b \in \mathbb{R}^3$ at real coefficients by (iv), since $b \mapsto \tau_b f \in E_{\rho'}$ is continuous; then to complex coefficients by the identity theorem. (King states translation invariance as [K2, Thm. 2.3(i)]; his argument is not used.) □

Definition 3.4 (the object of Theorem A: King's printed state)**.** Fix $e > 0$, $\lambda, m, \mu_0 > 0$ in King's range, the block factor 5 and the large-block integer M, of the form $M = 5M'$ with M' divisible by a prime $\mathfrak{p} \neq 5$ and $M' \geq 2\mathfrak{p}$ (Definition 2.1), and let $\Lambda_k = \varepsilon_k\mathbb{Z}^3$, $\varepsilon_k = 5^{-k}$. For $J \in \mathbb{N}$ let $\langle\cdot\rangle_{k,J}^K$ be the state of King's cube C_J^K at spacing ε_k (Definition 2.15): the Neumann photon on $C_J \cap \Lambda_k$, and the scalar field with Dirichlet conditions on the planes $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ - \varepsilon_k$, that is, the cube of [K2, (3.3), (3.7), (3.9), p. 329] with $L_\mu = 2MJ$, where (3.9) is read, after the translation that follows Definition 2.15, as setting the scalar field to zero on the planes $x_\mu = 0$ and $x_\mu = L_\mu$ (a reading of the printed page, marked as such in [G]), for the action of Definition 2.4, whose mass counterterm $\delta m_{\varepsilon_k}^2$ is the free-boundary one, computed on the infinite lattice $\varepsilon_k\mathbb{Z}^3$ and the same at every site (Remark 2.6). For finite families of real 2-forms $f_j \in C_c^\infty$, sampled at plaquette centres in the lattice's own frame (Definition 2.8), and $z \in \mathbb{C}^N$, *King's state on curvature sources* is

$$Z_\infty\Big(\sum_j z_j f_j, 0\Big) := \lim_{J\to\infty} \lim_{k\to\infty} \Big\langle \exp\Big(\sum_j z_j F_{\Lambda_k}(f_j)\Big)\Big\rangle_{k,J}^K.$$

The inner limit is King's cube limit [K2, Thm. 2.1] at $h = 0$ along the spacings 5^{-k}, at fixed μ_0 and fixed J, taken for the J at which the sources have ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; King's proof runs his block-spin flow on C_J^K, whose block averages of the scalar field carry the Dirichlet weight f_D, and the limit is item (13) of Section 1.3, proved in [CA], with the bound at long trees of Proposition 1.2. The outer limit runs along the family $(C_J^K)_J$. Both limits exist, locally uniformly in z, given item (13) (Theorem 3.3), and the same entire function is obtained from the boxes $C_J \cap \Lambda_k$ of Definition 2.5(a) in place of King's cubes, through the iterated limits of Corollary 2.17(c) on imaginary coefficients. The state does not depend on the admissible choice of M: the states $\langle\cdot\rangle_{k,J}^K$ depend on M only through MJ, so if M_1 divides M_2 the cubes of M_2 are the cubes C_{JM_2/M_1}^K of

M_1, and the limits along the two families agree, on imaginary coefficients by Corollary 2.17(a) and then everywhere by Theorem 3.3(iii); two admissible values divide their least common multiple, which is again admissible. This is King's printed state [K2, (2.15)] at $h = 0$, and it is the object of Theorem A. We do not address the regularity of the Schwinger functions (they are distributions), the limit $\mu_0 \to 0$, or the uniqueness of the vacuum. The relation of King's state to infinite-volume limits along periodic boxes, King's expected identity [K2, (2.24)], is never used.

4. The torus comparison

4.1. **The hypothesis.**

Hypothesis 4.1 ((K$_0$): the torus comparison, curvature sector)**.** For every $R \in \mathcal{R}_0$, every walled configuration $\mathfrak{c}$ (of the class (6) of Section 1.3 when (K$_0$) enters Theorem A; Remark 5.5) and every real smooth periodic 2-form G, $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are equal.

Here $R^{-1}\Lambda_k$ carries the frame $(R^{-1}e_\alpha)$ and descends to D_J by Lemma 2.2(b); both lattices have spacing ε_k and carry the same $\delta m^2_{\varepsilon_k}$. Since $\mathcal{R}_0 = \mathcal{R}_0^{-1}$, (K$_0$) for all $R \in \mathcal{R}_0$ is the same as the statement for the lattices $R\Lambda_k$ used in [K2]. Read with a single scalar function on all plaquettes, as printed in [K2, (2.12), Thm. 2.4], the comparison is false already at $e = 0$ (Proposition A.2); the 2-form reading is consistent with King's proof, which treats the source components separately (Remark 2.9).

Lemma 4.2. *Under (K$_0$) the convergence holds locally uniformly in $z \in \mathbb{C}$: the two limits $\lim_k \langle e^{zF(G)} \rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k \langle e^{zF(G)} \rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are the same entire function of z. In particular, for every walled configuration $\mathfrak{c}$ and every real 2-form $f \in C_c^\infty$ supported in one cell, $\lim_k T_{\mathfrak{c},\Lambda_k}(f)$ and $\lim_k T_{\mathfrak{c},R^{-1}\Lambda_k}(f)$ exist and are equal.*

Proof. As in Theorem 3.3(i): local boundedness from $|\langle e^{zF} \rangle| \le \langle e^{\operatorname{Re} z\, F} \rangle$ and convexity, Montel's theorem, and $\mathbb{R}$ as a set of uniqueness. For the second claim take G the periodised f and $z = \mathrm{i}$. □

4.2. **Reduction to one-sided bounds at one lattice spacing.**

King's proof of his Theorem 2.4 compares the lattice of spacing ε with the rotated lattice of spacing 5ε, and obtains the reverse inequality by "switching around the argument" [K2, p. 337]. We compare the two lattices at the *same* spacing, and obtain the reverse inequality from an exact symmetry.

Lemma 4.3 (the involution)**.** *Let $R = R_{\mu\nu}(\pm\theta_0) \in \mathcal{R}_0$, let $\sigma := \sigma_\nu$ be the reflection $x_\nu \mapsto -x_\nu$, and put $S := R^{-1}\sigma$. Then $S^2 = 1$, $S\Lambda_k = R^{-1}\Lambda_k$ and $S(R^{-1}\Lambda_k) = \Lambda_k$, and S maps $D_J = \mathbb{R}^3/P_J$, $P_J = (50MJ\mathbb{Z})^3$, onto the torus $R^{-1}D_J := \mathbb{R}^3/R^{-1}P_J$. Both lattices descend to both tori, and for all real smooth D_J-periodic sources $s = (G, h)$,*

$$\mathcal{Z}_{R^{-1}\Lambda_k, D_J}(s) = \mathcal{Z}_{\Lambda_k, R^{-1}D_J}(S \cdot s), \qquad \mathcal{Z}_{\Lambda_k, D_J}(s) = \mathcal{Z}_{R^{-1}\Lambda_k, R^{-1}D_J}(S \cdot s),$$
$$\mathsf{S}_{R^{-1}\Lambda_k, D_J}(h) = \mathsf{S}_{\Lambda_k, R^{-1}D_J}(S \cdot h).$$

Proof. By Lemma 2.2(a), $\sigma R \sigma = R^{-1}$, so $S^2 = R^{-1}(\sigma R^{-1}\sigma) = R^{-1}R = 1$; $\sigma\Lambda_k = \Lambda_k$ gives $S\Lambda_k = R^{-1}\Lambda_k$, and then $S(R^{-1}\Lambda_k) = S^2\Lambda_k = \Lambda_k$. Also $\sigma P_J = P_J$, so $SP_J = R^{-1}P_J$. The lattice $R^{-1}\Lambda_k$ is invariant under $R^{-1}P_J$ trivially, and Λ_k is invariant under $R^{-1}P_J$ because $R^{-1}p \in 10MJ\mathbb{Z}^3$ for $p \in P_J$ (Lemma 2.2(b)). The map S carries the frame $(R^{-1}e_\alpha)$ to $(SR^{-1}e_\alpha) = (\sigma e_\alpha)$ and the frame (e_α) to $(R^{-1}\sigma e_\alpha)$; these differ from the frames of Λ_k and $R^{-1}\Lambda_k$ only by signs, which do not change the periodic models (Lemma 2.10). Lemma 2.10 with $O = S$, applied to the unnormalised integrals (the bijection of configurations preserves $d\nu$), gives the identities. □

Hypothesis 4.4 ((F) and (F$_0$): one-sided bounds at one spacing)**.** Let $R \in \mathcal{R}_0$ and let $J \in \mathbb{N}$. Write $T_1 := D_J$, $T_2 := R^{-1}D_J$. **(F)**: for $i = 1, 2$ there are real numbers $v_k(T_i)$ and $c_k(T_i)$,

$k \geq k_0(J)$, independent of the fields and the sources, with $|c_k(T_i)| \leq C\varepsilon_k^{-q}$ for some $q < \infty$ and $v_k(T_i)$ bounded, such that for every real smooth T_i-periodic source $s = (G, h)$,

$$\mathcal{Z}_{R^{-1}\Lambda_k,T_i}(s) \leq \big(1 + \delta_k(T_i, s)\big) \exp\big(- v_k(T_i) - c_k(T_i)\, \mathsf{S}_{\Lambda_k,T_i}(h)\big)\, \mathcal{Z}_{\Lambda_k,T_i}(s), \qquad \lim_{k\to\infty} \delta_k(T_i, s) = 0.$$

($\mathbf{F}_0$) is (F) restricted to sources with $h = 0$ and to the sources (G, h_0), where $h_0 = -w \leq 0$ is a wall, in total form ($v_k \equiv 0$), on T_1 for these sources and on T_2 for their images under the involution S of Lemma 4.3, together with **($\mathbf{C}_0$)**: $\limsup_k c_k(T_1) \leq 0$ and $c_k(T_2) - c_k(T_1) \to 0$. The sources $(G, -h_0)$, with a mass-decreasing wall, are not needed.

On each torus, (F) bounds the functional of the rotated lattice from above by that of the axis lattice. The rotated lattice is the one whose flow is King's standard block-spin flow in its own frame, and the axis lattice is the one run through King's hybrid flow; this is the direction of [K2, (4.7)–(4.9)]. Since smooth periodic functions have rapidly convergent Riemann sums, $|\mathsf{S}_{\Lambda,T}(h) - \int_\Gamma h| \leq C_N(h)\varepsilon_k^N$ for every N and every lattice Λ of spacing ε_k descending to T, Γ a fundamental cell; with $N > q$ the error $c_k(\mathsf{S}_{\Lambda,T}(h) - \int h)$ tends to 0.

Theorem 4.5 (closing at one spacing)**.** *Let $R \in \mathcal{R}_0$, $J \in \mathbb{N}$, and assume* (F)*. Write $v = v_k(T_1)$, $v' = v_k(T_2)$, $c = c_k(T_1)$, $c' = c_k(T_2)$, and $\rho_k(s) := \mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s)/\mathcal{Z}_{\Lambda_k,D_J}(s)$.*

(a) *For every real smooth D_J-periodic source $s = (G, h)$, $e^{v'+c'\int h}(1 + o(1)) \leq \rho_k(s) \leq e^{-v-c\int h}(1 + o(1))$ as $k \to \infty$.*

(b) *If (F) holds in total form ($v = v' = 0$) on T_1 for the sources $s_t = (tG, h_0)$, $t \in \mathbb{R}$, with $\int h_0 < 0$, and on T_2 for their S-images, and $\limsup_k(c + c') \leq 0$, then $c + c' \to 0$ and $\rho_k(s_t)e^{c\int h_0} \to 1$ for every real t; in particular, for $s_0 = (0, h_0)$ and $s_1 = (tG, h_0)$,*

$$\frac{\mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s_1)}{\mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s_0)} = \frac{\mathcal{Z}_{\Lambda_k,D_J}(s_1)}{\mathcal{Z}_{\Lambda_k,D_J}(s_0)}\,(1 + o(1)).$$

No convergence of either family is assumed.

Proof. (a) The upper bound is (F) on T_1, with $c\,\mathsf{S}_{\Lambda_k,D_J}(h) = c\int h + o(1)$. Apply (F) on T_2 to the source $S \cdot s$ and use Lemma 4.3: $\mathcal{Z}_{R^{-1}\Lambda_k,T_2}(S \cdot s) = \mathcal{Z}_{\Lambda_k,D_J}(s)$, $\mathcal{Z}_{\Lambda_k,T_2}(S \cdot s) = \mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s)$ and $\mathsf{S}_{\Lambda_k,T_2}(S \cdot h) = \mathsf{S}_{R^{-1}\Lambda_k,D_J}(h) = \int h + O(\varepsilon_k^N)$. This is the lower bound. (b) By (a) at s_t the interval is non-empty, so $(c + c')\int h_0 \leq o(1)$; as $\int h_0 < 0$, $c + c' \geq -o(1)$, and with $\limsup_k(c + c') \leq 0$ this gives $c + c' \to 0$. The two ends of the interval in (a) then differ by a factor $e^{o(1)}$, so $\rho_k(s_t)e^{c\int h_0} \to 1$; dividing the statement for s_1 by the one for s_0 cancels the prefactor. □

The proof uses (F) only at the sources named. In particular, under (F_0) the conclusions (a) and (b) hold for the sources (tG, h_0) (where $\int h_0 < 0$ unless $w = 0$) and, with no condition on c, c', (a) holds in total form for the sources with $h = 0$.

Corollary 4.6 (walled curvature ratios)**.** *Assume* (F_0) *for R and J'. Then for every walled configuration $\mathfrak{c} = (J', w)$ and every real smooth periodic 2-form G, $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 0$ locally uniformly in $z \in \mathbb{C}$. If moreover $\lim_k \langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ exists for real t, then* (K_0) *holds for $\mathfrak{c}$ and G.*

Proof. With $h_0 = -w$, $\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda} = \mathcal{Z}_\Lambda(tG, h_0)/\mathcal{Z}_\Lambda(0, h_0)$ (the Wick constants cancel), a balanced ratio. By Theorem 4.5(b) (at $w = 0$ directly by (a) in total form), $u_k(t) := \langle e^{tF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}/\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 1$ for real t. By Lemma 2.13(a) and the symmetry $(A, \vartheta) \mapsto (-A, -\vartheta)$, $1 \leq \langle e^{tF(G)}\rangle \leq e^{t^2\sigma_\Lambda/2}$ for real t, where $\sigma_\Lambda = \langle J, CJ\rangle$ is the variance of $F(G)$ under the free periodic photon on the lattice; $\sigma_\Lambda \leq \mu_0^{-2}\|\delta_\varepsilon G\|^2$ is bounded uniformly in k, δ_ε the lattice codifferential. Hence the difference of the two functions tends to 0 on $\mathbb{R}$, and it is bounded on every strip $|\operatorname{Re} z| \leq r$ uniformly in k, because $|\langle e^{zF}\rangle| \leq \langle e^{\operatorname{Re} z\, F}\rangle$. Vitali's theorem gives locally uniform convergence to 0 on $\mathbb{C}$. The last statement follows by the same argument. □

Remark 4.7 (what this subsection supplies). Lemma 4.3, Theorem 4.5 and its corollary are a reduction of the torus comparison to the abstract one-sided bound (F): they are proved, and they supply none of the bounds. In the curvature sector the bound (F_0) that Corollary 4.6 consumes, the one-sided forward bound in total form on the mass-increasing family with its source-linear constant on D_J of upper limit at most zero and the constant on the rotated torus $R^{-1}D_J$ equal to it up to $o(1)$, comes from the closing of [CT] itself (Section 4.3), in which King's torus bound [K2, (4.2)] is not used, because at a non-zero wall it is vacuous; its place is taken by the large-field clause of King's upper bound at blocking factor 5, item (1) of Section 1.3, and at a mass-increasing wall the polynomial lower bound (H5) is the wall instance proved in [CT].

4.3. **The corrected torus theorem.** The bound (F_0) and the existence of the axis-lattice limits for walled curvature ratios are the content of the corrected torus theorem of [CT], which we now recall in the form used. Write $X_K := \Lambda_K$ and $Y_K := R\Lambda_K$, $R = R_{12}(\theta_0)$, so that (K_0) for R^{-1} is the statement for the pair (X_K, Y_K). Configurations are taken in the class of item (6) of Section 1.3: $J \in 5^{k_*}\mathbb{N}$ with $J \geq \max(J_0(\varepsilon_0), J_{\mathrm{pos}})$, $\mathfrak{c} = (J, \psi, n)$ with $n > 0$, ψ vanishing on the 2-neighbourhood of $R^{-1}C_L$, the wall condition $n\langle\psi\rangle_{X_K} > \Delta_*|T|/2$, block alignment and the grid offset of the corners of H'.

Theorem 4.8 (the corrected torus theorem, curvature sector; [CT])**.** *Let $e^2, \lambda, m^2, \mu_0^2 > 0$ be in King's range, the block factor 5, and $\mathfrak{c} = (J, \psi, n)$ a walled configuration of the class (6) with wall source $h_0 = -n\psi$. Assume the hypotheses (1)–(3) and (5) of Section 1.3; they supply the hypotheses* (H1)–(H5) *of* [CT] *for the sources (tG, h_0), $t \in \mathbb{R}$, on D_J, for their S^*-images on RD_J, and for their images under the coordinate reflections and permutations on D_J together with the S^*-images of those on RD_J. Then for every real smooth D_J-periodic 2-form G:*

(i) $\langle e^{zF(G)}\rangle_{\mathfrak{c},Y_K} - \langle e^{zF(G)}\rangle_{\mathfrak{c},X_K} \to 0$ *as $K \to \infty$, locally uniformly in $z \in \mathbb{C}$; in particular $\langle F(G)^n\rangle_{\mathfrak{c},Y_K} - \langle F(G)^n\rangle_{\mathfrak{c},X_K} \to 0$ for every n;*

(ii) *if in addition (4) holds (existence of $\lim_K \langle e^{tF(G)}\rangle_{\mathfrak{c},X_K}$ for real t), both families converge, locally uniformly on $\mathbb{C}$, to the same entire function $W_{\mathfrak{c}}(zG)$, and all moments converge to the same limits;*

(iii) *the same holds with R replaced by any $R' \in \mathcal{R}_0$.*

At $e = 0$ the statement holds with no hypothesis.

The chain of [CT] is: exact passive-rotation reductions and the involution of Lemma 4.3; an equal-spacing comparison of the effective actions produced by the two block-spin flows (the standard flow of Y_K and King's hybrid flow of X_K, whose first step averages over rotated blocks) on King's small-field domain, in total form, that is with every field-independent constant included; a one-sided forward bound of the form (F) on each torus, obtained by inserting the comparison into King's upper bound [K1, Thm. 3.1] for Y_K and the hybrid lower bound [K2, Prop. 4.1] for X_K, with the additive large-field term, in the form of the large-field clause of item (1), absorbed on the left by the polynomial lower bound (H5), which at a mass-increasing wall is the wall instance proved in [CT]; and the closing by Theorem 4.5(b) on the mass-increasing family, in which the vacuum constants are absent because the comparison is in total form, and the source-linear constant tends to zero (the two constants of the pair differ by $o(1)$, and the axis-lattice one tends to zero). King's torus bound [K2, (4.2)] is not used anywhere in this chain. The inputs of [CT], in the names used there, and the items of Section 1.3 that supply them, are:

(H1) The two-flow comparison bound of [CT] for the renormalised families of [K1, §3.5] with lines taken from the two flows, in per-flow normal form: the statement (H1*) of item (2) for some $\theta_1 > 0$ (Bałaban's renormalisation bound with one element weakened by a small degree is not an input), proved in [CT] from (3), (5), (2a) and (2b); the one-flow instance is what King asserts in the proof of [K1, Prop. 3.6], p. 665.

(H2) Decay and regularity of the block-spin operators, [K1, Thm. 3.3]: for the standard flow [Ba4], item (5a) (printed with proof, its three asserted steps proved under (1) and (5k)); for the hybrid flow, proved from the hybrid-kernel bounds of item (3).

(H3) King's upper bound [K1, Thm. 3.1, (3.4)] for Y_K, with the additive large-field term in the form of the large-field clause of item (1): proved on RD_J from items (1) and (5k), given (CT), the instance on RD_J being carried from the instance for the axis lattice on D_J by the rotation (Lemma 2.10 with $O = R$), the bridge constant with it [SUP] (the proof of [K1, Thm. 3.1] refers to [Ba1, Ba2, Ba3, Ba4]); on D_J, which is skew in Y's frame, the instance [SKEW-U], proved relative to the same bounds on aligned tori at blocking factor 5 (item (1)), the replacement of the propagators of [Ba2, Prop. 2.6] on the skew torus with the change of domain (P7) of Section 1.4; for the contours of the flow of Y_K, rooted at the block centres, with Proposition 1.3, and at the wall with Proposition 1.4, for $J \geq J_{\mathrm{pos}}$, on D_J under the final-scale cap of the class (6), through the large-field clause of (1) (Section 1.5).

(H4) The hybrid lower bound, the analogue of [K2, Prop. 4.1, (4.5)]: proved from $(\mathrm{BAL})_5$, (CT), (B_Δ) and, on D_J, [SKEW-L] (item (5i)), its own steps (5l)(i)–(iii) included, with Proposition 1.3 for the composite contours of the hybrid flow.

(H5) A polynomial lower bound $\mathcal{Z}_{Y_K}(T; s) \geq e^{-C_1|T|}\varepsilon_K^{q(s)}$: at $h = 0$ Bałaban's lower bound [Ba1, (1.14)] (item (5b), printed as a frame with asserted steps and proved under (1)) on RD_J, and on D_J proved relative to it given $(\mathrm{BAL})_5$ and (CT); at the mass-increasing wall source $h_0 = -n\psi$, the wall instance proved in [CT], at every point of King's range, from the source-free bounds at two other fixed masses, the torus one-point Wick bound (4c), the skew instances under the final-scale cap and exact identities.

(H6) Existence of $\lim_K \langle e^{tF(G)} \rangle_{\mathfrak{c}, X_K}$: item (4), King's Theorem 2.1 of [K1] at a wall, read on the normalised ratio, in which the h-linear c-numbers cancel: proved in [H6], with the residues (4a)–(4d) proved, outside the set E_W of Proposition 1.4 and for $J \geq J_{\mathrm{pos}}$, through the large-field clause of (1) at the wall; King's two statements that it uses are proved (item (4), Section 8.3).

(SK) The skew-torus instances [SKEW-U], [SKEW-L] of King's and Bałaban's bounds, for the rotated lattice on D_J, which is not aligned with its frame: proved relative to the aligned bounds at blocking factor 5 (item (1)), given the skew-torus form of the cumulant bound (5l)(i), and, for [SKEW-U], the change of domain (P7) of Section 1.4.

$(\mathrm{BAL})_5$ Item (1): asserted in print without a complete argument; proved at block factor 5 [AS].

(CT) The bridge between the torus counterterm conventions of [K1] and the free-boundary conventions of [K2, p. 330] (Remark 2.6): (a) exact algebra; (b) the bridge constants in the form consumed, the forward bound for the torus model at the shifted constant source, a single bound Δ_* on $|\Delta_T(s)|$ for $s \leq s_0$ (which is also the constant of the wall condition in (6)) and a bound on the shift $\Delta c_T(s)$ of the Wick constant for $s \leq s_0$ (no bound on the field-independent c-numbers is used: they cancel in every ratio; the literal clause of a bounded factor for the extensive bounds is not what the chain uses), proved, with the uniformity (5f) of King's constant under a constant source shift (King's statement on the inclusion of sources in its consumed form included); its rate clause $|\Delta_T(s) - \Delta_T(0)| \leq Cs$ (item (4c)) enters only through (4); (c) the existence transfer is not an input, (H6) being read on the normalised ratio. On the shell side the same bound Δ_* is the only input of $(\mathrm{CT})_{\mathrm{sh}}$ (item (8f)).

Theorem 4.8 is proved in [CT] from the items (1)–(5). Its closing implication, from (H1)–(H6), (SK), $(\mathrm{BAL})_5$ and (CT) to (i)–(iii), is proved there; its own steps (5l), with (4a) and (F3), its other residues, the items (2b), (4b), (4c), (5i), (5j), and the completions of its steps (F1), (F2) (Section 8.5) are proved in [SUP] ((2b), (5i), (5j) also in [CT]); Bałaban's and King's asserted steps (1) and (5k), with those of (5a), (5b), (5f), (5g) under them, are proved at block factor

5 in [AS], except two statements of King, which are proved in the forms consumed in [SUP] (items (4), (5k); Section 8.3); the reading (B4) of [Ba2, p. 591] behind the constants (5l)(vii) is proved in [SUP] (Section 8.2); on the skew tori the replacement of the propagators of [Ba2, Prop. 2.6] in (1) is proved through the change of domain (P7) of Section 1.4; the repaired cube step in (1) uses an end-functional bound of the re-runs (8e) = (8i) at zero shell mass, which rests on the readings (B3), (B5), (B6) and on the regrouping of the cut families, all proved in [SUP] (Section 8.2); and the forms of (1) for the contour families of its flows and at the walls are proved with Propositions 1.3 and 1.4 of Section 1.5, outside the set E_W there, under the final-scale cap of the class (6), Proposition 1.3 also through (H4) and the comparison of the effective actions. The sector in which the sources are read through a single scalar function on all plaquettes is false (Proposition A.2) and is not what the theorem states. The scalar (Wick-source) sector of [CT], which needs the convergence on the torus of the Wick-source functionals with a renormalised Wick square (Section 10), is outside the theorem of this supplement.

Remark 4.9 (the content of (K_0)). Hypothesis 4.1 is the curvature sector of [K2, Thm. 2.4] with the corrections of Appendix A, for walled tori and with free-boundary counterterms. By Corollary 4.6 its two-lattice content is the one-sided bound (F_0), and its one-lattice content is the existence of the walled limits on the axis lattice. By Theorem 4.8, (K_0) holds for every $R \in \mathcal{R}_0$ and every configuration of the class (6), by the items (1)–(5) of Section 1.3, (1) on the skew tori with the change of domain (P7), with Propositions 1.3 and 1.4 of Section 1.5, outside the set E_W there; they reach the readings (B3), (B5), (B6) only through the end-functional bound of the re-runs (8e) = (8i) that the repaired cube step of (1) uses, and the reading (B4) of Section 8.2 directly; all four are proved in [SUP]. (K_0) is not reduced to any second-order statement, and no part of Section 4 is used to supply it. The rotation enters the whole argument of this supplement only here.

Remark 4.10 (what the torus comparison alone does not give). Given (K_0), the walled values of a configuration $\mathfrak{c}$ on D_J agree on the two lattices, and by passive covariance the walled value of $R\cdot\mathfrak{c}$ on the rotated torus RD_J for the axis lattice equals the walled value of $\mathfrak{c}$ on D_J for the rotated lattice. Rotation invariance of a limit along tori would therefore follow from (K_0) if the limit did not depend on the shape and orientation of the tori, which is the content of King's expected identity [K2, (2.24)] between torus and cube limits. That identity is a uniqueness statement for the infinite-volume limit, and we do not use it. The walls of Section 5 replace it by a one-sided statement about one lattice, (TIGHT_0), whose local form (RES_0) is the hypothesis of Theorem 7.8.

5. Walled tori and the closing hypothesis

Definition 5.1 (admissible walls and walled values)**.** For a bounded set $X \subset \mathbb{R}^3$ let $\mathcal{A}(X)$ be the set of walled configurations $\mathfrak{c} = (J', w)$ with $N_1(X) \subset (-25MJ', 25MJ')^3$ and $w = 0$ on $N_1(X)$; the cell $(-25MJ', 25MJ')^3$ is a fundamental domain of $D_{J'}$ for both lattices, $R^{-1}\Lambda_k$ descending to $D_{J'}$ by Lemma 2.2(b), and sources supported in X are periodised from it. If $X \subset Y$ then $\mathcal{A}(Y) \subset \mathcal{A}(X)$. Assume ($\mathrm{K}_0$). For a real 2-form $f \in C_c^\infty$ supported in X and $\mathfrak{c} \in \mathcal{A}(X)$, the source restricted to the cell and periodised is smooth, and by Lemma 4.2 the *walled value*

$$\mathcal{W}_{\mathfrak{c}}(f,0) := \lim_{k\to\infty} T_{\mathfrak{c},\Lambda_k}(f) = \lim_{k\to\infty} T_{\mathfrak{c},R^{-1}\Lambda_k}(f) \qquad (R \in \mathcal{R}_0)$$

exists. For $b > 0$ and f supported in Q_b let $I_b(f) := \inf_{\mathfrak{c}\in\mathcal{A}(Q_b)} \mathcal{W}_{\mathfrak{c}}(f,0)$, which is nondecreasing in b, and

$$\mathcal{W}^{\mathrm{env}}(f,0) := \lim_{b\to\infty} I_b(f).$$

Lemma 5.2 (the free direction)**.** *Assume (K_0).*

(a) *If f is a real 2-form supported at distance > 1 from $\mathbb{R}^3\backslash C_K$ and $\mathfrak{c} \in \mathcal{A}(\bar{C}_K)$, then $\mathcal{W}_{\mathfrak{c}}(f,0) \geq Z_K^K(\mathrm{i}f,0) := \lim_k \langle e^{\mathrm{i}F(f)}\rangle_{k,K}^K$.*

(b) $\mathcal{W}^{\mathrm{env}}(f,0) \geq Z_\infty(\mathrm{i}f,0) > 0$ *for every real* 2*-form* $f \in C_c^\infty$.

Proof. (a) At each k, Lemma 2.14 with $X = C_K \cap \Lambda_k$ and Lemma 2.16 give $T_{\mathfrak{c},\Lambda_k}(f) \geq \langle e^{\mathrm{i}F(f)}\rangle_{k,K}^{\mathrm{ND}} \geq \langle e^{\mathrm{i}F(f)}\rangle_{k,K}^{K}$; let $k \to \infty$, the limit of the right side existing by item (13) with Theorem 3.3(i). (b) If $\bar{C}_K \subset Q_b$ then $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_K)$, so $I_b \geq Z_K^K(\mathrm{i}f,0)$; let $b \to \infty$ and then $K \to \infty$ (Theorem 3.3(iii)). Positivity: by Lemma 2.13(a), $\langle e^{\mathrm{i}F(f)}\rangle_{k,K}^{K} \geq e^{-\langle v, Cv\rangle/2}$ with v the vector with $\langle v, A\rangle = F(f)$ and C the Neumann photon covariance of the cube, and $\langle v, Cv\rangle \leq \mu_0^{-2}\|d_\varepsilon^* f\|^2$, with $\|d_\varepsilon^* f\|$ the lattice $L^2(\varepsilon_k)$ norm of the lattice coderivative $d_\varepsilon^* f$ of f, is bounded uniformly in k and K. □

Lemma 5.3 (independence of the orientation)**.** *Assume* (K_0)*. For* $R \in \mathcal{R}_0$ *and every real* 2*-form* $f \in C_c^\infty$, $\lim_{L\to\infty} \inf_{\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}_\mathfrak{c}(f,0) = \mathcal{W}^{\mathrm{env}}(f,0)$*, the infimum being defined as soon as* $\operatorname{supp} f \subset Q_{5ML/7}$.

Proof. By Lemma 2.2(c), $Q_{5ML/7} \subset R^{-1}Q_{ML} \subset Q_{7ML/5}$, hence $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML}) \subset \mathcal{A}(Q_{5ML/7})$, and $I_{5ML/7}(f) \leq \inf_{\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}_\mathfrak{c}(f,0) \leq I_{7ML/5}(f)$. Both bounds tend to $\mathcal{W}^{\mathrm{env}}(f,0)$. □

Hypothesis 5.4 ((TIGHT$_0$) and (TIGHT$_2$): the closing hypothesis)**.** **(TIGHT$_0$)**: $\mathcal{W}^{\mathrm{env}}(f,0) \leq Z_\infty(\mathrm{i}f,0)$ for every real 2-form $f \in C_c^\infty$. Equivalently: for every $b > 0$, every real 2-form f supported in Q_b and every $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}_\mathfrak{c}(f,0) \leq Z_\infty(\mathrm{i}f,0) + \eta$; it involves only (K_0) and King's printed state. Where (K_0) is supplied on a sub-class of the walled configurations only, the class (6) of Section 1.3 (Remark 5.5), (TIGHT$_0$) is read with the infimum over that sub-class, whose conditions other than the vanishing of ψ near $R^{-1}C_L$ do not depend on R (J_{pos} being the maximum over the finite set $\mathcal{R}_0$ and the three torus pairs, Proposition 1.4): for every $b > 0$, every real 2-form f supported in Q_b and every $\eta > 0$ there is a configuration $\mathfrak{c}$ of the class (6) in $\mathcal{A}(Q_b)$ with $\mathcal{W}_\mathfrak{c}(f,0) \leq Z_\infty(\mathrm{i}f,0) + \eta$; for $b \geq \frac{7}{5}ML + 2$ such a configuration has $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$ for every $R \in \mathcal{R}_0$ (Lemma 2.2(c)), so it lies in the class (6) for R and L. **(TIGHT$_2$)**: for every real 2-form $f \in C_c^\infty$, $\lim_{b\to\infty} \sup_{\mathfrak{c}\in\mathcal{A}(Q_b)} \sigma_\mathfrak{c}^2(f) \geq S_2(f)$, where $\sigma_\mathfrak{c}^2(f) := \lim_k \langle F_{\Lambda_k}(f)^2\rangle_{\mathfrak{c},\Lambda_k}$ and $S_2(f) := \lim_J \lim_k \langle F_{\Lambda_k}(f)^2\rangle_{k,J}^K$, on King's cubes (Definition 2.15). Where (K_0) is supplied on the class (6) only, the same reading applies to (TIGHT$_2$): the supremum is over the configurations of the class (6) in $\mathcal{A}(Q_b)$.

The equivalence in (TIGHT$_0$) holds because I_b is nondecreasing in b. By Lemma 5.2, (TIGHT$_0$) is equivalent to $\mathcal{W}^{\mathrm{env}}(f,0) = Z_\infty(\mathrm{i}f,0)$ for every real 2-form f: *King's state is the limit of the lower envelopes of walled tori, the continuum limit being taken first.* The limits defining $\sigma_\mathfrak{c}^2$ and S_2 exist under (K_0), respectively by Theorem 3.3, since second derivatives of locally uniformly convergent entire functions converge.

Remark 5.5 (the configuration class). (K_0) is supplied by Theorem 4.8 for configurations of the class (6) of Section 1.3, which is smaller than $\mathcal{A}(X)$: it asks $J \in 5^{k_*}\mathbb{N}$, $J \geq \max(J_0(\varepsilon_0), J_{\mathrm{pos}})$, $n > 0$, the wall condition $n\langle\psi\rangle_{X_K} > \Delta_*|T|/2$, block alignment and the grid offset of the corners of H'; the threshold J_{pos} of Proposition 1.4 belongs to the class because the margin of the positivity condition is proved for $J \geq J_{\mathrm{pos}}$ only. This costs nothing: in the proof of Theorem 6.1 below the infimum over $\mathcal{A}(R^{-1}Q_{ML})$ is used only through an upper bound by the walled value of one configuration supplied by (TIGHT$_0$), so the infimum may be taken over any sub-class that contains that configuration, and (TIGHT$_0$) is then read with the same sub-class. The configuration that Theorem 7.8 produces lies in it: its wall is $(n+K)\psi$ with $n + K \geq n_* > 0$, mass-increasing; the wall condition is linear in the height and holds for K large at fixed T, which Remark 7.9(iii) allows; J (the J' of Theorem 7.10), the block alignment of H', T and the grid offset of the corners of H' are choices of the torus and of the box that Theorem 7.10 makes ($J' \geq \max(J_0, J_{\mathrm{pos}})$ among them), the last moving H' by less than $M'\ell_f$; and Lemma 5.3 uses only shrinking families, so it holds on the intersection. The large-field clause of (1) at the wall on the skew torus takes one step

at the final spacing of Bałaban's fixed-mass bounds; the class carries the cap of Section 1.5 on the coarsest scale, $\varepsilon_0 \le \varepsilon_{\rm sys}(M,R,\rho,\delta_0)$, which does not depend on J and is closed downward, and under it the non-wrapping threshold at that spacing equals 1 for every J of the residue class, so that no further restriction of J is needed.

Lemma 5.6. *Under (K_0), (TIGHT$_0$) implies (TIGHT$_2$), and* $\lim_b \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \le S_2(f)$ *always; so (TIGHT$_2$) is an equality.*

Proof. Fix f, b and $\eta' > 0$. All functions of t below are real and even. Since $\cos x \ge 1 - \frac{x^2}{2}$ and $\cos x \le 1 - \frac{x^2}{2} + \frac{x^4}{24}$ at every lattice level, and derivatives converge in the limits,

$$\mathcal{W}_{\mathfrak{c}}(tf,0) \ge 1 - \tfrac{t^2}{2}\sigma^2_{\mathfrak{c}}(f), \qquad Z_\infty(\mathrm{i}tf,0) \le 1 - \tfrac{t^2}{2}S_2(f) + \tfrac{t^4}{24}M_4(f),$$

with $M_4(f) := \partial_t^4 Z_\infty(tf,0)|_{t=0} < \infty$. By (TIGHT$_0$) applied to tf, for every $t > 0$ and $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}_{\mathfrak{c}}(tf,0) \le Z_\infty(\mathrm{i}tf,0) + \eta$, hence $\sigma^2_{\mathfrak{c}}(f) \ge S_2(f) - \frac{t^2}{12}M_4(f) - \frac{2\eta}{t^2}$. Choosing t with $t^2 M_4(f)/12 < \eta'/2$ and then $\eta = t^2\eta'/4$ gives $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \ge S_2(f) - \eta'$. For the second claim, let $\mathfrak{c} \in \mathcal{A}(\bar{C}_L)$ with f supported well inside C_L. At every k, Lemma 2.16 and Lemma 2.14, as in the proof of Lemma 5.2(a), give $T_{\mathfrak{c},\Lambda_k}(tf) \ge \langle e^{\mathrm{i}tF(f)}\rangle^K_{k,L}$; so $\mathcal{W}_{\mathfrak{c}}(tf,0) \ge Z^K_L(\mathrm{i}tf,0) = \lim_k \langle e^{\mathrm{i}tF(f)}\rangle^K_{k,L}$ for all real t, with equality at $t = 0$, and both sides are even and smooth, so $\sigma^2_{\mathfrak{c}}(f) \le \langle F(f)^2\rangle_{C^K_L} := \lim_k \langle F(f)^2\rangle^K_{k,L}$. Hence $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \le \langle F(f)^2\rangle_{C^K_L}$ whenever $\bar{C}_L \subset Q_b$. By Theorem 3.3(ii), comparing second derivatives at $t = 0$, the cube variances are nonincreasing in L, and they converge to $S_2(f)$. □

6. Rotation invariance

Theorem 6.1 (invariance under the generators)**.** *Assume (K_0) on the class (6) of Section 1.3 and (TIGHT$_0$) read on that class, the sub-class of Remark 5.5 on which (K_0) is proved (Hypothesis 5.4). Then for every $R \in \mathcal{R}_0$ and every real 2-form $f \in C^\infty_c$,*

$$Z_\infty(\mathrm{i}\,R\cdot f,0) = Z_\infty(\mathrm{i}f,0),$$

and consequently the same holds for every $R \in G_0$.

Proof. Fix $R \in \mathcal{R}_0$ and a real 2-form f supported in Q_ρ.

Step 0 (geometry). Take L with $ML \ge \frac{7}{5}\rho + 2$. By Lemma 2.2(c), $R\cdot f$ is supported in $Q_{7\rho/5}$, at distance > 1 from $\mathbb{R}^3 \setminus C_L$, and f is supported in $Q_{5ML/7} \subset R^{-1}Q_{ML}$. Let $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$ be a configuration of the class (6) for R and L. Then $R^{-1}\Lambda_k$ descends to $D_{J'}$ (Lemma 2.2(b)), and $B'_k := R^{-1}(C_L \cap \Lambda_k)$ is a set of sites of $R^{-1}\Lambda_k$ which, with its nearest neighbours, lies in the open cell, and on which $w = 0$. Since $R\cdot f$ has Euclidean distance > 1 from $\mathbb{R}^3 \setminus C_L$, every plaquette of $R^{-1}\Lambda_k$ at which f does not vanish lies in B'_k.

Step 1 (cutting on the rotated lattice). By Lemma 2.14, at every k,

$$\langle e^{\mathrm{i}F(f)}\rangle^{\rm ND}_{R^{-1}\Lambda_k, B'_k} \le T_{\mathfrak{c},R^{-1}\Lambda_k}(f).$$

Step 2 (passive rotation, and King's cube). By Lemma 2.10 with $O = R$, $\langle e^{\mathrm{i}F(f)}\rangle^{\rm ND}_{R^{-1}\Lambda_k,B'_k} = \langle e^{\mathrm{i}F(R\cdot f)}\rangle^{\rm ND}_{k,L}$, and by Lemma 2.16 the functional of King's cube C^K_L for the source $R\cdot f$ is bounded by that of the box: $\langle e^{\mathrm{i}F(R\cdot f)}\rangle^K_{k,L} \le \langle e^{\mathrm{i}F(R\cdot f)}\rangle^{\rm ND}_{k,L}$.

Step 3 (continuum limits). As $k \to \infty$ the left side converges by item (13) (Theorem 3.3(i)) and the right side by (K_0) (Definition 5.1):

$$Z^K_L(\mathrm{i}\,R\cdot f,0) \le \mathcal{W}_{\mathfrak{c}}(f,0) \qquad \text{for every configuration } \mathfrak{c} \text{ of the class (6) in } \mathcal{A}(R^{-1}Q_{ML}).$$

Step 4 (infimum, volume limit, closing hypothesis). Take the infimum over these $\mathfrak{c}$ and let $L \to \infty$. The infimum is used only through an upper bound by the walled value of one configuration

supplied by (TIGHT$_0$), so it may be taken over the sub-class (Remark 5.5): the configuration that (TIGHT$_0$) supplies for a source supported in Q_b with $b \geq \frac{7}{5}ML + 2$ has $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$ (Lemma 2.2(c)), so it is admissible in Step 0. By Theorem 3.3(iii), Lemma 5.3 (on the sub-class, Remark 5.5) and (TIGHT$_0$),

$$Z_\infty(\mathrm{i}\, R\cdot f,0) = \lim_{L\to\infty} Z_L^K(\mathrm{i}\, R\cdot f,0) \leq \mathcal{W}^{\mathrm{env}}(f,0) \leq Z_\infty(\mathrm{i}f,0).$$

Step 5 (the inverse). $R^{-1} \in \mathcal{R}_0$ and $R\cdot f$ is a real 2-form, so Steps 0–4 for $(R^{-1}, R\cdot f)$ give $Z_\infty(\mathrm{i}f,0) \leq Z_\infty(\mathrm{i}\, R\cdot f,0)$. Since the real 2-forms are G_0-invariant, the claim extends to G_0. □

The order of limits is: at fixed R, f, L and $\mathfrak{c}$, first $k\to\infty$; then the infimum over $\mathfrak{c}$; then $L\to\infty$. The continuum limit is always taken first, at fixed volume and fixed wall, as in King's construction [K2, p. 336], and (K$_0$) is used only as an equality of continuum limits, not at fixed lattice spacing.

Lemma 6.2 (reflection)**.** *Let* $\sigma_3 = \mathrm{diag}(1,1,-1)$. *Then* $Z_\infty(\mathrm{i}\,\sigma_3\cdot f,0) = Z_\infty(\mathrm{i}f,0)$ *for every real* 2*-form* $f \in C_c^\infty$.

Proof. σ_3 preserves Λ_k, and maps $C_J\cap\Lambda_k$ onto $X_+ := (C_J + \varepsilon_k e_3)\cap\Lambda_k$. By Lemma 2.10, $\langle e^{\mathrm{i}F(\sigma_3\cdot f)}\rangle_{k,J}^{\mathrm{ND}}$ equals the corresponding expectation in the Neumann–Dirichlet state on X_+ defined with the frame $(e_1,e_2,-e_3)$. Written in the frame (e_α) (with $A_3'(x) = -A_3(x-\varepsilon e_3)$), this is the BFS measure in which ϕ, A_1, A_2 live on X_+ and A_3 on $X_+ - \varepsilon e_3 = C_J\cap\Lambda_k$, with Dirichlet conditions for ϕ on X_+ and photon gradient terms within these sets. It is obtained from the Neumann–Dirichlet state on $C_{J+1}\cap\Lambda_k$ by deleting photon gradient terms and cosine parts of hopping terms, since $X_+\cup(X_+-\varepsilon e_3)\subset C_{J+1}$; and the state on $C_{J-1}\cap\Lambda_k$ is obtained from it in the same way, since $C_{J-1}\subset C_J\cap(C_J+\varepsilon e_3)$. As in Lemma 2.14, for J large and every k, $\langle e^{\mathrm{i}F(f)}\rangle_{k,J-1}^{\mathrm{ND}} \leq \langle e^{\mathrm{i}F(\sigma_3\cdot f)}\rangle_{k,J}^{\mathrm{ND}} \leq \langle e^{\mathrm{i}F(f)}\rangle_{k,J+1}^{\mathrm{ND}}$.

On King's cubes: this sandwich at J and at $J-1$, and Lemma 2.16, applied at its two ends and twice to the source $\sigma_3\cdot f$ in the middle, give for J large and every $k\geq 0$

$$\begin{aligned}\langle e^{\mathrm{i}F(f)}\rangle_{k,J-2}^{K} \leq \langle e^{\mathrm{i}F(f)}\rangle_{k,J-2}^{\mathrm{ND}} \leq \langle e^{\mathrm{i}F(\sigma_3\cdot f)}\rangle_{k,J-1}^{\mathrm{ND}} \leq \langle e^{\mathrm{i}F(\sigma_3\cdot f)}\rangle_{k,J}^{K}\\ \leq \langle e^{\mathrm{i}F(\sigma_3\cdot f)}\rangle_{k,J}^{\mathrm{ND}} \leq \langle e^{\mathrm{i}F(f)}\rangle_{k,J+1}^{\mathrm{ND}} \leq \langle e^{\mathrm{i}F(f)}\rangle_{k,J+2}^{K};\end{aligned}$$

let $k\to\infty$ by item (13) and then $J\to\infty$ (Theorem 3.3(iii)). The comparison of C_J^K with $C_{J\mp1}^K$ is also obtained by deletions and kills when $M\geq 2\varepsilon_k$, which holds at every k for the M of Definition 2.1 [G]; the two-step form used here needs no such condition. □

Theorem 6.3 (closure: from the generators to $O(3)$)**.** *Suppose* $Z_\infty(\mathrm{i}\, R\cdot f,0) = Z_\infty(\mathrm{i}f,0)$ *for all real* 2*-forms* $f\in C_c^\infty$ *and* $R\in\mathcal{R}_0$. *Then for every* $R\in O(3)$, *every* N, *all real* 2*-forms* $f_j\in C_c^\infty$ *and all* $z\in\mathbb{C}^N$,

$$Z_\infty\Big(\sum_j z_j\, R\cdot f_j, 0\Big) = Z_\infty\Big(\sum_j z_j f_j, 0\Big).$$

Proof. Complex coefficients. The hypothesis passes to G_0, since the real 2-forms are invariant under $O(3)$ and $\mathcal{R}_0 = \mathcal{R}_0^{-1}$. For $R\in G_0$ and a finite family, the difference of the two sides, as a function of $z\in\mathbb{C}^N$, is entire (Theorem 3.3(iii)) and vanishes on $(\mathrm{i}\mathbb{R})^N$, where $\sum_j z_j f_j = \mathrm{i}\sum_j t_j f_j$ with $t\in\mathbb{R}^N$ and the hypothesis applies to the real 2-form $\sum_j t_j f_j$; hence it vanishes identically.

Closure in $SO(3)$. At real coefficients, $\Phi(R\cdot f) = \Phi(f)$ for $f\in E_\rho$ and $R\in G_0$. The map $R\mapsto R\cdot f\in E_\rho$ is continuous on $O(3)$, and Φ is continuous on E_ρ by Theorem 3.3(iv). Hence $G := \{R\in O(3) : \Phi(R\cdot f) = \Phi(f) \text{ for all } \rho \text{ and } f\in E_\rho\}$ is a closed subgroup of $O(3)$ containing G_0, so $G\supset SO(3)$ by Lemma 2.2(d).

Reflection. Lemma 6.2 and the first step (with σ_3 in place of R) give $\sigma_3\in G$, so $G = O(3) = SO(3)\cup\sigma_3 SO(3)$.

Conclusion. For $R \in O(3)$ both sides of the claim are entire in $z \in \mathbb{C}^N$ and agree on $\mathbb{R}^N$. □

Theorem 6.4 (the curvature sector)**.** *Assume (K_0) on the class (6) of Section 1.3 and (TIGHT$_0$) read on that class, the sub-class of Remark 5.5 on which (K_0) is proved and whose conditions do not depend on R (Hypothesis 5.4). Then King's printed state on curvature sources is $O(3)$-invariant: for every $R \in O(3)$, all real 2-forms $f_j \in C_c^\infty$ and all $z \in \mathbb{C}^N$, $Z_\infty(\sum_j z_j\, R{\cdot}f_j, 0) = Z_\infty(\sum_j z_j f_j, 0)$, and all curvature Schwinger functions are $O(3)$-covariant. Moreover (item (14) of Section 1.3, proved from (13)), for every Euclidean motion $x \mapsto Rx+b$, $Z_\infty(\sum_j z_j\, \tau_b R{\cdot}f_j, 0) = Z_\infty(\sum_j z_j f_j, 0)$.*

Proof. Theorem 6.1 gives invariance under $\mathcal{R}_0$ on imaginary coefficients; Theorem 6.3 gives $O(3)$ and the entire extension. Translations are Theorem 3.3(v), proved there from (13), Lemma 2.16 and the cut sandwich; King's printed argument [K2, Thm. 2.3(i)] is not used. □

Corollary 6.5 (the curvature two-point function)**.** *Assume (K_0) and (TIGHT$_2$). Then $S_2(f_1, f_2) := \lim_J \lim_k \langle F_{\Lambda_k}(f_1) F_{\Lambda_k}(f_2)\rangle^K_{k,J}$ satisfies $S_2(R\cdot f_1, R\cdot f_2) = S_2(f_1, f_2)$ for all $R \in O(3)$ and real 2-forms $f_1, f_2 \in C_c^\infty$.*

Proof. Let $R \in \mathcal{R}_0$. Steps 0–3 of the proof of Theorem 6.1, with source tf, give $Z^K_L(\mathrm{i}t\, R\cdot f, 0) \le \mathcal{W}_{\mathfrak{c}}(tf, 0)$ for all real t and every configuration $\mathfrak{c}$ of the class (6) in $\mathcal{A}(R^{-1}Q_{ML})$. Both sides are even and smooth in t and equal 1 at $t = 0$, so $\langle F(R\cdot f)^2\rangle_{C^K_L} \ge \sigma^2_{\mathfrak{c}}(f)$. Since every configuration of the class (6) in $\mathcal{A}(Q_{7ML/5+2})$ lies in $\mathcal{A}(R^{-1}Q_{ML})$ and has $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$ (Lemma 2.2(c)), and $b \mapsto \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f)$, the supremum over the class (6) in $\mathcal{A}(Q_b)$ (Hypothesis 5.4), is nonincreasing,

$$\langle F(R\cdot f)^2\rangle_{C^K_L} \ge \sup_{\mathcal{A}(Q_{7ML/5+2})} \sigma^2_{\mathfrak{c}}(f) \ge \lim_{b\to\infty} \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \ge S_2(f)$$

by (TIGHT$_2$). The cube variances converge to S_2 (proof of Lemma 5.6), so $S_2(R\cdot f) \ge S_2(f)$, and the inverse generator gives equality; hence S_2 is G_0-invariant. By Lemma 2.13, $\langle F(f)^2\rangle^K_{k,J} \le \langle v, Cv\rangle_\varepsilon$ with $v = d^*_\varepsilon f$, where C is the photon covariance of the cube, the Neumann photon of $C_J \cap \Lambda_k$; the photon form is bounded below by μ_0^2, so $0 \le S_2(f) \le \mu_0^{-2}\|d^*f\|^2_{L^2}$. Since S_2 is a positive semidefinite quadratic form, $|S_2(f) - S_2(f')| \le S_2(f-f')^{1/2} S_2(f+f')^{1/2}$, and $R \mapsto S_2(R\cdot f)$ is continuous on $O(3)$. Density of G_0 gives $SO(3)$, Lemma 6.2 (expanded to second order) gives σ_3, and polarisation gives the bilinear form. □

7. The heavy-shell hypothesis

Throughout this section $\Lambda = \Lambda_k$ is the axis lattice, $\varepsilon = \varepsilon_k \le 1/\mu_0$, and $m_\varepsilon := \frac{2}{\varepsilon}\operatorname{arcsinh}(\varepsilon\mu_0/2)$; since $t \mapsto \operatorname{arcsinh}(t)/t$ is decreasing, $m_\varepsilon \ge 0.96\,\mu_0$. Let $G_{\mathbb{Z}^3}$ be the Green's function of simple random walk on $\mathbb{Z}^3$ and

$$\mathsf{a} := \sup_{z\in\mathbb{Z}^3\setminus\{0\}} |z|_\infty\, G_{\mathbb{Z}^3}(0, z),$$

which is finite by the Green's function asymptotics of simple random walk in three dimensions [LL, Chap. 4]. For a set $X \subset \mathbb{R}^3$ put $\mathrm{vol}_\varepsilon(X) := \varepsilon^3 \#(X \cap \Lambda)$.

7.1. **Photon boundary conditions across a matter-free collar.** The photon in Feynman form consists of three independent scalar Gaussian fields A_α, $\alpha = 1, 2, 3$, and on a finite graph $\mathcal{G}$ of sites of Λ each has the precision $K_{\mathcal{G}} = \varepsilon L_{\mathcal{G}} + \mu_0^2\varepsilon^3$, with $L_{\mathcal{G}}$ the graph Laplacian of $\mathcal{G}$ with unit conductances and $\langle u, v\rangle = \sum_x u(x)v(x)$; King's Neumann photon on a cube is the case of the induced subgraph [K2, (2.5), p. 333], and the periodic photon the case of the torus graph.

Lemma 7.1 (photon boundary switch)**.** *Let $H = Q_{b'}$ be a closed cube and C_L a King cube with $\mathrm{dist}_\infty(H, \mathbb{R}^3 \setminus C_L) = D + 3$, where $D \ge \max(1, 3/\mu_0)$ and $\varepsilon \le \min(1/\mu_0, 1/4)$. Let $\langle\cdot\rangle_\natural$ and $\langle\cdot\rangle_N$ be BFS measures with the same matter part: the sites of $H \cap \Lambda$, BFS single-site measures (arbitrary;*

wall and killed sites included), and hopping terms only between neighbouring sites of H, coupled to the photon variables of the bonds of H as in King's action. Their photon parts are:

- (♮) *the variables $A_\alpha(x)$ with $x, x+\varepsilon e_\alpha \in C_L$, with the law of the periodic photon of a torus whose cell contains $N_1(C_L)$, conditioned on $A_\beta(y)=0$ for all other (β,y);*
- (N) *King's Neumann photon on $C_L \cap \Lambda$.*

Let $\Theta = e^{\mathrm{i}F(f)} e^{\sum_x \varepsilon^3 g(x) r_x^2 - c}$ with f supported on plaquettes of H, $g \ge 0$ supported in H, and $c \in \mathbb{R}$. Then

$$0 \le \langle\Theta\rangle_\natural - \langle\Theta\rangle_N \le \langle|\Theta|^2\rangle_N^{1/2} \big(e^\tau - 1\big)^{1/2}, \qquad \tau \le C_{\mathrm{GB}}\, \mathsf{a}\, \mathrm{vol}_\varepsilon(N_2(H))\, D^{-1} e^{-\frac{3}{2} m_\varepsilon D},$$

with an absolute constant C_{GB}.

Proof. Step 1 (marginals and order). Fix a component α. Let $X_\alpha := \{x \in C_L \cap \Lambda : x + \varepsilon e_\alpha \in C_L\}$; conditioning a Gaussian on the vanishing of some coordinates leaves the principal submatrix of the precision on the others, so in (♮) the variables $A_\alpha(x)$, $x \in X_\alpha$, have precision $K_\natural = \varepsilon(6 - \mathrm{Adj}_{X_\alpha}) + \mu_0^2 \varepsilon^3$, where Adj_{X_α} is the adjacency matrix of X_α. In (N) the variables are $A_\alpha(x)$, $x \in C_L \cap \Lambda$, with precision $K_N = \varepsilon L_{C_L} + \mu_0^2 \varepsilon^3$. The principal submatrix of K_N on X_α is $\varepsilon(\deg_{C_L} - \mathrm{Adj}_{X_\alpha}) + \mu_0^2\varepsilon^3 \le K_\natural$, and the Schur complement of K_N on X_α is at most its principal submatrix; hence $C_\natural := K_\natural^{-1} \le C_N|_{X_\alpha}$, $C_N := K_N^{-1}$. Let $\hat X := \{(\alpha, x) : x, x + \varepsilon e_\alpha \in H\}$. The integrands depend on A only through $A_{\hat X}$, whose laws are $N(0, C_\natural|_{\hat X})$ and $N(0, C_N|_{\hat X})$, with $C_\natural|_{\hat X} \le C_N|_{\hat X}$. Put $M := (C_\natural|_{\hat X})^{-1} - (C_N|_{\hat X})^{-1} \ge 0$. The ♮-measure is obtained from the N-measure, after integrating out the photon variables off $\hat X$, by adding $\frac12\langle A, M\rangle A$ to the action; by Lemma 2.12(a) along $s \mapsto \frac{s}{2}\langle A, M\rangle A$, $\langle\Theta\rangle_\natural \ge \langle\Theta\rangle_N$.

Step 2 (Cauchy–Schwarz). With $Q := \langle A, M\rangle A$ and $\varsigma := e^{-Q/2}$, $\langle\Theta\rangle_\natural = \langle\Theta\varsigma\rangle_N / \langle\varsigma\rangle_N$, so

$$\langle\Theta\rangle_\natural - \langle\Theta\rangle_N = \frac{\langle(\Theta - \langle\Theta\rangle_N)(\varsigma - \langle\varsigma\rangle_N)\rangle_N}{\langle\varsigma\rangle_N} \le \langle|\Theta|^2\rangle_N^{1/2} \Big(\frac{\langle e^{-Q}\rangle_N}{\langle e^{-Q/2}\rangle_N^2} - 1\Big)^{1/2} \le \langle|\Theta|^2\rangle_N^{1/2} \big(e^{\langle Q\rangle_N} - 1\big)^{1/2},$$

by $\langle e^{-Q}\rangle_N \le 1$ and Jensen's inequality. Writing $M = \sum_j \lambda_j v_j v_j^T$, Lemma 2.13(b) in the N-measure gives $\langle Q\rangle_N \le \sum_j \lambda_j \langle v_j, C_N v_j\rangle = \mathrm{tr}(M C_N|_{\hat X}) =: \tau$.

Step 3 (the trace). Let $\Delta := C_N|_{X_\alpha} - C_\natural \ge 0$, for each α. Then $M C_N|_{\hat X} = (C_\natural|_{\hat X})^{-1} \Delta_{\hat X}$, so $\tau = \mathbb{E}\langle Y, (C_\natural|_{\hat X})^{-1} Y\rangle$ with $Y \sim N(0, \Delta_{\hat X})$. Since the inverse of a principal block of the covariance is the minimum of the precision form over extensions, $\langle a, (C_\natural|_{\hat X})^{-1} a\rangle = \min\{\langle u, K_\natural u\rangle : u|_{\hat X} = a\}$. Let $\tilde Y \sim N(0, \Delta)$ on X_α and $\eta(x) := (1 - \mathrm{dist}_\infty(x, H))_+$; then $\eta\tilde Y$ is an extension of $\tilde Y|_{\hat X}$, so $\tau \le \sum_\alpha \mathbb{E}\langle \eta\tilde Y, K_\natural \eta\tilde Y\rangle$. Let int be the set of $x \in X_\alpha$ whose six neighbours lie in X_α; it contains $N_2(H) \cap \Lambda$. At $x \in \mathrm{int}$ the rows of $K_\natural$, K_N and of the infinite-lattice precision K_∞ coincide, so $(K_\infty \Delta)(x, w) = (K_N C_N)(x, w) - (K_\natural C_\natural)(x, w) = 0$ for all $w \in X_\alpha$, hence $\mathrm{Var}((K_\infty \tilde Y)(x)) = 0$ and $K_\infty \tilde Y = 0$ on int almost surely. For u with $K_\infty u = 0$ on $\mathrm{int} \supset \mathrm{supp}\,\eta$, summation by parts and the identity $(\eta_x u_x - \eta_y u_y)^2 - (u_x - u_y)(\eta_x^2 u_x - \eta_y^2 u_y) = (\eta_x - \eta_y)^2 u_x u_y$ give $\langle \eta u, K_\natural \eta u\rangle = \langle \eta u, K_\infty \eta u\rangle = \varepsilon \sum_{\langle xy\rangle} (\eta_x - \eta_y)^2 u_x u_y$. Since $|\eta_x - \eta_y| \le \varepsilon$ and $|\Delta(x,y)| \le \frac12(\Delta(x,x) + \Delta(y,y))$,

$$\tau \le \sum_\alpha \varepsilon^3 \sum_{\langle xy\rangle \subset N_2(H)} |\Delta(x,y)| \le 9\, \mathrm{vol}_\varepsilon(N_2(H)) \sup_{\alpha,\ x \in N_2(H)} \Delta(x,x).$$

Step 4 (decay of Δ). For $K = \mathcal{D} - \varepsilon \mathrm{Adj}_{\mathcal{G}}$ with $\mathcal{D}$ diagonal, $K^{-1}(x,y) = \sum_\omega \prod_{(u\to v)\in\omega} (\varepsilon/\mathcal{D}(u)) \cdot \mathcal{D}(y)^{-1}$, the sum over walks ω from x to y in $\mathcal{G}$. For $K_\natural$ and K_N the weights of steps from sites of int coincide. Decomposing walks from $x \in N_2(H)$ at their first visit to $\Sigma := X_\alpha \setminus \mathrm{int}$, $K^{-1}(x,y) = G^{\mathrm{int}}(x,y) + \sum_{z\in\Sigma} \mathsf{H}(x,z) K^{-1}(z,y)$ with G^{int} and H the same for both precisions. Hence

$$\Delta(x,x) = \sum_{z\in\Sigma} \mathsf{H}(x,z)\big(K_N^{-1}(z,x) - K_\natural^{-1}(z,x)\big) \le \Big(\sum_{z\in\Sigma} \mathsf{H}(x,z)\Big) \sup_{z\in\Sigma} K_N^{-1}(z,x).$$

The first factor is the probability that simple random walk from x, killed with probability $\mu_0^2\varepsilon^2/(6+\mu_0^2\varepsilon^2)$ per step, reaches Σ, which lies at ℓ^∞-distance at least D from x. With $\cosh\lambda = 1+\mu_0^2\varepsilon^2/2$, that is $\lambda = \varepsilon m_\varepsilon$, $e^{\pm\lambda S_n^\beta}\mathbf{1}_{\{\text{alive at } n\}}$ is a martingale for each coordinate S^β; optional stopping and a union over the six half-spaces give $\sum_z \mathsf{H}(x,z) \le 6e^{-m_\varepsilon D}$. The same argument, with the decomposition at the first exit from the ℓ^∞-ball of radius $|y|_\infty/2$ and the bound $K^{-1}_{\varepsilon\mathbb{Z}^3,\mu_0} \le K^{-1}_{\varepsilon\mathbb{Z}^3,0} = (6\varepsilon)^{-1}G_{\mathbb{Z}^3}(\cdot/\varepsilon)$ entrywise, gives the free bound

$$G_\varepsilon(y) := K^{-1}_{\varepsilon\mathbb{Z}^3}(0,y) \le 4\mathsf{a}\,|y|_\infty^{-1}e^{-m_\varepsilon|y|_\infty/2} \qquad (|y|_\infty \ge 4\varepsilon).$$

King's Neumann precision on the box $C_L\cap\Lambda$ has the Green's function $K_N^{-1}(z,x) = \sum_{x'} G_\varepsilon(z-x')$, the sum over the images x' of x under the reflections of $\varepsilon\mathbb{Z}^3$ in the planes half a lattice spacing outside the faces of the box (method of images; the mass term is uniform). Every image other than x lies outside C_L at ℓ^∞-distance at least D from every site of C_L, and so does x from $z\in\Sigma$. If $s\ge 2D$ is the side of C_L, at most $8(j+3)^3$ images satisfy $|z-x'|_\infty < D+(j+1)s$; since $m_\varepsilon D \ge 2$,

$$K_N^{-1}(z,x) \le \sum_{j\ge 0} 8(j+3)^3\,\frac{4\mathsf{a}\,e^{-m_\varepsilon(D+js)/2}}{D+js} \le C_1\,\mathsf{a}\,D^{-1}e^{-m_\varepsilon D/2}$$

with an absolute constant C_1. Combining, $\Delta(x,x)\le 6C_1\mathsf{a}D^{-1}e^{-\frac{3}{2}m_\varepsilon D}$, and Step 3 gives the claim with $C_{\mathrm{GB}} = 54C_1$. □

The Gaussian measures in Lemma 7.1 are compared only on the bonds of H, where no live matter crosses the region in which the two photon boundary conditions differ; this is what makes the bound uniform in ε. A comparison in which the photon is pinned or cut on a surface that live matter crosses is not of this type: a one-loop computation shows that the order-e^2 difference of such states diverges logarithmically as $\varepsilon\to 0$, because the free-boundary mass counterterm compensates the photon self-energy of the matter only where the photon has its bulk covariance.

7.2. **Shell configurations, two-level walls, and the dead-shell bound.**

Definition 7.2 (shell configurations and shell profiles)**.** Let $b>0$. A *shell configuration* for Q_b is $\mathfrak{s} = (\mathfrak{c}, b', L)$, where $\mathfrak{c} = (J',\psi,n)$ with $\psi\in C^\infty(D_{J'};[0,\infty))$ and $n\ge 0$ is a walled configuration in $\mathcal{A}(Q_b)$ (so $\psi = 0$ on $N_1(Q_b)$), $H := Q_{b'}$ with $b'\ge b+3$, and C_L is a King cube (the shell's cube, with its own index L; not the source cube C_L of the class (6) and of Theorems 6.1 and 6.4) with $\mathrm{dist}_\infty(H,\mathbb{R}^3\setminus C_L) = D(\mathfrak{s})+3$, $D(\mathfrak{s})\ge\max(1,3/\mu_0)$, and $N_1(\bar C_L)\subset(-25MJ',25MJ')^3$. Its *shell* is $U := C_L\setminus H$, and we require the *plateau condition* $\psi\ge 1$ on $N_1(U)$ (compatible with $\psi=0$ on $N_1(Q_b)$, since $\mathrm{dist}_\infty(U,Q_b)\ge 3$). Put

$$\tau(\mathfrak{s}) := C_{\mathrm{GB}}\,\mathsf{a}\,\mathrm{vol}(N_3(H))\,D(\mathfrak{s})^{-1}e^{-1.44\,\mu_0 D(\mathfrak{s})}.$$

A *shell profile* for $\mathfrak{s}$ is $\chi\in C^\infty(D_{J'};[0,1])$ with $\chi = 1$ on U and $\mathrm{supp}\,\chi\subset N_1(U)$; then $\chi\le\psi$ automatically, and $\chi = 0$ on $N_1(Q_b)$. Its *dead set* is $D_\chi := \{\chi>0\}\subset N_1(U)$; every site of C_L within ε of $\mathbb{R}^3\setminus C_L$, and every site of U, lies in D_χ. We call $(\mathfrak{s},\chi)$ a *two-level configuration* for Q_b.

Definition 7.3 (two-level states and the tail)**.** Let $(\mathfrak{s},\chi)$ be a two-level configuration for Q_b, f a real 2-form in C_c^∞ supported in Q_b, and k with $\varepsilon_k\le\min(1/\mu_0,1/4)$. For $K\in[0,\infty)$ let

$$\Psi_k(n,K) := T_{(J',n\psi+K\chi),\Lambda_k}(f)$$

be the walled expectation of the configuration with the two-level wall $n\psi+K\chi$, which lies in $\mathcal{A}(Q_b)$; and let $\Psi_k(n,\infty) := \langle e^{\mathrm{i}F(f)}\rangle$ in the walled torus state of $\mathfrak{c} = (J',n\psi)$ with the matter killed on $D_\chi\cap\Lambda_k$ (Definition 2.5(d)). Put $\mathrm{tail}_k(K) := \Psi_k(n,K)-\Psi_k(n,\infty)$.

Lemma 7.4 (elementary properties)**.** (a) *$K\mapsto\Psi_k(n,K)$ is nonincreasing on $[0,\infty)$, $\Psi_k(n,K)\ge\Psi_k(n,\infty)$, and $\Psi_k(n,K)\to\Psi_k(n,\infty)$ as $K\to\infty$. Hence $\mathrm{tail}_k(K)\ge 0$ is nonincreasing in K and tends to 0 as $K\to\infty$, at every fixed k.*

(b) *(two-level sandwich) For $n, K \geq 0$:* $\Psi_k(n+K,0) \leq \Psi_k(n,K) \leq \Psi_k(n,0)$.

Proof. (a) Increasing K multiplies the single-site measures on D_χ by $e^{-(K'-K)\varepsilon^3\chi(x)r_x^2}$; Lemma 2.12(c). Killing D_χ is the limit (Lemma 2.7 with the rates $\chi > 0$ on D_χ), and it does not increase the expectation. (b) The wall of $\Psi_k(n+K,0)$ exceeds that of $\Psi_k(n,K)$ by $K(\psi-\chi) \geq 0$, since $\chi \leq \psi$; the wall of $\Psi_k(n,K)$ exceeds that of $\Psi_k(n,0)$ by $K\chi \geq 0$. Apply Lemma 2.12(c). □

Part (b) is the two-level sandwich (Lemma 7.6 below in the form used); it is the only place where $\chi \leq \psi$ is needed. In words: raising the matter mass on the shell alone, by any amount, gives a state above the one in which the whole plateau is raised by the same amount, and below the one with the shell untouched.

Theorem 7.5 (the dead-shell bound)**.** *For every two-level configuration $(\mathfrak{s}, \chi)$ for Q_b, every real 2-form $f \in C_c^\infty$ supported in Q_b and every k with ε_k small,*

$$\Psi_k(n,\infty) \leq \langle e^{\mathrm{i}F(f)}\rangle_{k,L}^{\mathrm{ND}} + \big(e^{\tau(\mathfrak{s})}-1\big)^{1/2} \leq \langle e^{\mathrm{i}F(f)}\rangle_{k,L+1}^{K} + \big(e^{\tau(\mathfrak{s})}-1\big)^{1/2}, \tag{7.1}$$

C_L the shell's cube, and hence $\limsup_k \Psi_k(n,\infty) \leq Z_\infty(\mathrm{i}f,0) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$.

Proof. Fix k with $\varepsilon = \varepsilon_k$ small, and write $\Theta = e^{\mathrm{i}F(f)}$. Let E be the set of sites of the torus outside C_L.

Step 1 (pinning). In the state with the matter killed on $D_\chi \cap \Lambda_k$, add $s|A_\beta(y)|^2$ for every photon variable on a bond with an endpoint in E, and let $s \to \infty$. By Lemma 2.12(a) the expectation of Θ does not decrease, and in the limit the photon is the conditioned law ($\natural$) of Lemma 7.1. In this state the hopping terms within E do not involve A, and there are no hopping terms between E and live sites: every site of C_L adjacent to E lies in D_χ and is dead. So the matter on E is a factor independent of A and cancels (Lemma 2.12(e)). What remains is the $\natural$-measure of Lemma 7.1, with the matter on H (dead on $D_\chi \cap H$), the wall, and Dirichlet conditions at ∂H, since $U \subset D_\chi$ is dead. Hence $\Psi_k(n,\infty) \leq \langle\Theta\rangle_\natural$.

Step 2 (photon switch). By Lemma 7.1, $\langle\Theta\rangle_\natural \leq \langle\Theta\rangle_N + \langle|\Theta|^2\rangle_N^{1/2}(e^{\tau(\mathfrak{s})}-1)^{1/2}$, and $|\Theta|^2 = 1$; here $m_\varepsilon \geq 0.96\mu_0$ and $\mathrm{vol}_\varepsilon(N_2(H)) \leq \mathrm{vol}(N_3(H))$ for small ε give $\tau \leq \tau(\mathfrak{s})$.

Step 3 (to the box C_L, and to King's cube). The N-measure is the limit $s \to \infty$ of the walled Neumann–Dirichlet state on $C_L \cap \Lambda_k$ with the single-site measures on D_χ multiplied by $e^{-s\varepsilon^3\chi(x)r^2}$ (Lemma 2.7). By Lemma 2.12(c) the expectation of Θ does not decrease as s decreases to 0, nor when the wall is then removed. The result is the box state $\langle\cdot\rangle_{k,L}^{\mathrm{ND}}$ of Definition 2.5(a): this is the first inequality of (7.1). The second is Lemma 2.16, since f is supported in H, at distance > 1 from $\mathbb{R}^3 \setminus C_L$.

Step 4 (limits). The limit of $\langle e^{\mathrm{i}F(f)}\rangle_{k,L+1}^{K}$ as $k \to \infty$ exists by Theorem 3.3(i) (item (13), Lemma 3.1(a), Montel) and is at most $Z_\infty(\mathrm{i}f,0)$ (Theorem 3.3(iii)). This gives the lim sup bound. □

7.3. **The main hypothesis.**

Lemma 7.6 (two-level sandwich)**.** *For every two-level configuration, every real 2-form $f \in C_c^\infty$ supported in Q_b, every k and $n, K \geq 0$,*

$$\Psi_k(n+K,0) \;\leq\; \Psi_k(n,K) \;=\; \Psi_k(n,\infty) + \mathrm{tail}_k(K) \;\leq\; \Psi_k(n,0), \qquad \mathrm{tail}_k(K) \geq 0.$$

Proof. Lemma 7.4. □

Hypothesis 7.7 ((RES$_0$): heavy-shell decoupling at one configuration)**.** For every $b > 0$, every real 2-form $f \in C_c^\infty$ supported in Q_b and every $\eta > 0$ there are a two-level configuration $(\mathfrak{s}, \chi)$ for Q_b with $\tau(\mathfrak{s}) \leq \eta$ and a shell mass $K \in [0,\infty)$ such that

$$\liminf_{k\to\infty} \mathrm{tail}_k(K) = \liminf_{k\to\infty} \big[\Psi_k(n,K) - \Psi_k(n,\infty)\big] \leq \eta.$$

In words: in a walled torus, once the scalar mass squared on a thick shell of the heavy plateau has been raised by $2K$, removing the matter on the shell altogether changes the expectation of the observable by at most η in the continuum limit, for one geometry and one K, with no rate in K and no uniformity in anything. Since $\tau(\mathfrak{s})$ depends only on H and the width $D(\mathfrak{s})$, the condition $\tau(\mathfrak{s}) \le \eta$ only asks the shell to be wide enough, logarithmically in b' and $1/\eta$. By Lemma 7.4(a) the bracket tends to 0 as $K \to \infty$ at every fixed k; the hypothesis says that the heavy-shell limit and the continuum limit may be interchanged at one configuration. The quantifiers are the weakest that Theorem 7.8 accepts: one configuration, one K, the lim inf along the spacings, and no uniformity in n, L, D or the torus.

Theorem 7.8 ((RES$_0$) implies (TIGHT$_0$))**.** *Under* (K$_0$)*,* (RES$_0$) *implies* (TIGHT$_0$)*. More precisely, if* $(\mathfrak{s}, \chi)$ *and* K *are as in* (RES$_0$) *for* f *and* η*, then the single-level configuration* $\mathfrak{c}' := (J', \psi, n + K) \in \mathcal{A}(Q_b)$ *satisfies*

$$\mathcal{W}_{\mathfrak{c}'}(f, 0) \le Z_\infty(\mathrm{i}f, 0) + \big(e^{\eta} - 1\big)^{1/2} + \eta.$$

Proof. By Lemma 7.6, $T_{\mathfrak{c}',\Lambda_k}(f) = \Psi_k(n + K, 0) \le \Psi_k(n, \infty) + \mathrm{tail}_k(K)$ at every k. The left side converges to $\mathcal{W}_{\mathfrak{c}'}(f, 0)$ by (K$_0$), since $\mathfrak{c}'$ is a walled configuration (Lemma 4.2); hence

$$\begin{aligned}\mathcal{W}_{\mathfrak{c}'}(f, 0) = \liminf_k \Psi_k(n + K, 0) &\le \limsup_k \Psi_k(n, \infty) + \liminf_k \mathrm{tail}_k(K) \\ &\le Z_\infty(\mathrm{i}f, 0) + \big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2} + \eta\end{aligned}$$

by Theorem 7.5 and (RES$_0$), using $\liminf(x_k + y_k) \le \limsup x_k + \liminf y_k$. With $\tau(\mathfrak{s}) \le \eta$ this is the displayed bound; as η is arbitrary and $\mathfrak{c}' \in \mathcal{A}(Q_b)$, $I_b(f) \le Z_\infty(\mathrm{i}f, 0)$ for every b, which is (TIGHT$_0$). □

Remark 7.9 (on the proof). (i) The two-level state is only an intermediate inequality; the configuration that enters (TIGHT$_0$) is the single-level wall $(n + K)\psi$, and the only limit input is the existence of its walled value, which is part of (K$_0$) (the hypothesis (H6) of Theorem 4.8: King's Theorem 2.1 of [K1] with the wall as King's scalar source, read on the normalised ratio). No existence statement is needed for the two-level state or for the dead-shell state. (ii) The lattice statements of this subsection hold verbatim for the sharp profile $\chi = \mathbf{1}_U$ (then $D_\chi = U$ and $\chi \le \psi$ needs only $\psi \ge 1$ on U); smoothness of χ would be needed only where the two-level walled value must have a continuum limit, which, by (i), the chain never needs. (iii) Since $\mathrm{tail}_k(K)$ is nonincreasing in K, (RES$_0$) for one K implies it for every larger K; so K may be taken as large as one likes, which is used to place $n + K$ in any prescribed window of wall heights.

7.4. **The heavy-shell estimate from the shell bounds.** In the curvature sector (RES$_0$) is not left as a hypothesis. The shell is now taken to be $U := T \setminus H'$, the whole exterior of the source box $H' := H$ in the torus T of the configuration, so that $\partial U = \partial H'$ is a single convex surface with no re-entrant edge (on a cubic shell $C_L \setminus H$ the derivative clause of King's bound on the block propagators, [K1, Thm. 3.3], fails at the re-entrant edges); the plateau condition of Definition 7.2 is read for this shell, $\psi \ge 1$ on $N_1(U)$ with $U = T \setminus H'$, that is, $\psi \ge 1$ except at the points of H' at distance more than one from $\partial H'$, which is compatible with $\psi = 0$ on $N_1(Q_b)$ since $H' = Q_{b'}$ with $b' \ge b + 3$, and the shell profile χ and the dead set D_χ are those of Definition 7.2 for this U; the steps of the dead-shell chain of Theorem 7.5 in this geometry are item (10c) of Section 1.3. For $\kappa \ge 0$ let $\Psi_k(n, \kappa)$ be the two-level expectation of Definition 7.3 with the shell mass $\kappa \mathbf{1}_U$ added to the wall (so that the scalar mass squared is raised by 2κ on U, Definition 2.5(c); in the normalisation of [SW1, SW3], whose profile V enters $\frac{1}{2}\langle\phi, V\phi\rangle$, the wall $n\psi + \kappa\mathbf{1}_U$ is their $2n\psi + 2\kappa\mathbf{1}_U$), so that $\Psi_k(n, K)$ and $\Psi_k(n, \infty)$ in (RES$_0$) are the cases $\kappa = K$ and $\kappa = \infty$, and let $Z_\kappa(zf, g')$ be the corresponding partition function with sources. King's block-spin flow [K1, §3] is run for the model in which $\kappa\mathbf{1}_U$ is part of the matter covariance at

every scale; $\eta_0 \in 5^{-\mathbb{N}}$ is the stopping scale, $\kappa_0 := \max(c_0, 1)/\eta_0^2$ the threshold above which the shell is heavy in block units at the stopping scale and both (8) (stated for $\kappa \geq c_0/\eta_0^2$) and (10) hold, I_κ^{cap} the stopping-scale integral over the capped test set (a set of block fields that does not depend on κ), and $\Delta_z(b)$ the difference of the reduced small-field actions at shell masses κ' and κ at the block field b and source coefficient z; $n_* := 1$, so that for $n \geq n_*$ the plateau height $2n$ of [SW1, SW3] satisfies the condition $2n \geq 1$ under which the estimates (8), (10) and (11) are stated there (both assume a plateau height ≥ 1); the estimates (8)–(11) are read through this dictionary, which changes the threshold κ_0 by a factor 2 and nothing else, since they hold uniformly in $\kappa \geq \kappa_0$ at every fixed plateau height. The dictionary carries the profile shape as well as n and κ: the estimates (8)–(11) are stated in [SW1, SW3] for King's wall profile only, $\psi\colon T \to [0,1]$ with $\psi = 0$ on the source region and $\psi = 1$ off its 1-neighbourhood, so that $\psi \equiv 1$ on U and on its η_0-neighbourhood, at a fixed plateau height $n \geq 1$ in their normalisation, and the two-level configurations to which Theorem 7.10 is applied are taken with this profile (one configuration suffices for (RES_0), Theorem 7.8); the notation is that of [SW1, SW3].

Theorem 7.10 (the heavy-shell estimate from the shell bounds)**.** *Let $g = 0$ and let $(\mathfrak{s}, \chi)$ be a two-level configuration with the shell $U = T \setminus H'$, the torus $T = D_{J'}$ with $J' \in 5^{k_*}\mathbb{N}$ and $J' \geq \max(J_0, J_{\mathrm{pos}})$, $J_0 = J_0(\varepsilon_0)$ the non-wrapping threshold of the class (6), King's wall profile ($\psi \in [0,1]$, $\psi = 0$ on $N_1(Q_b)$, $\psi = 1$ off $N_2(Q_b)$), plateau height $n \geq n_*$, H', T unions of blocks of the final scale and the corners of H' in the residue class of (6). Assume $(\mathrm{SW1}^\circ)_{\mathrm{int}}$, $(\mathrm{SW2}^{\min})$, $(\mathrm{SW3}^\circ)_{\mathrm{int}}$ and (SWT) in the forms (8)–(11) of Section 1.3. Then for every $\eta > 0$ there are a stopping scale η_0 and a shell mass $K < \infty$ with $\limsup_k \mathrm{tail}_k(K) \leq \eta$. The quantifiers of (RES_0) are met in this order: given b, a source f supported in Q_b and $\eta > 0$, the configuration $(\mathfrak{s}, \chi)$ is chosen first, with $D(\mathfrak{s})$ so large that $\tau(\mathfrak{s}) \leq \eta$, and then η_0 and K as above; so (RES_0) holds at $(\mathfrak{s}, \chi)$ with this K. Consequently, under (K_0), (TIGHT_0) holds (Theorem 7.8), with the single-level configuration $(J', \psi, n+K)$ in the class (6), K being taken large enough for its wall condition (Remark 7.9(iii)) and the vanishing of ψ near $R^{-1}C_L$, C_L the source cube of the class (6), being the choice of b in Hypothesis 5.4.*

Proof. Fix $(\mathfrak{s}, \chi)$, the source f and $g = 0$. For $\kappa \in [0, \infty]$ and $z \in \mathbb{C}$ put $\Psi_\kappa(z) := Z_\kappa(zf, 0)/Z_\kappa(0, 0)$, where Z_∞ is the partition function of the state with the matter killed on D_χ; the shell mass $\kappa \mathbf{1}_U$ is the second level $\kappa\chi$ of the wall, so that by Definition 7.3, at $g = 0$, $\Psi_K(\mathrm{i}) = \Psi_k(n, K)$, $\Psi_\infty(\mathrm{i}) = \Psi_k(n, \infty)$ and $\mathrm{tail}_k(K) = \Psi_K(\mathrm{i}) - \Psi_\infty(\mathrm{i})$. For $\kappa \geq \kappa_0$ and real z put $J_\kappa(z) := I_\kappa^{\mathrm{cap}}(zf, 0)/I_\kappa^{\mathrm{cap}}(0, 0)$. Every constant below is uniform in $\varepsilon \leq \varepsilon_0(\eta_0)$.

Step 0: the disc bound. At fixed spacing $z \mapsto \Psi_\kappa(z)$ is entire, a finite-dimensional integral of $e^{zF(f)}$ against a measure with Gaussian photon tails. For $|z| \leq 2$, $|e^{zF(f)}| \leq e^{2F(f)} + e^{-2F(f)}$, and Lemma 2.13(a) with $s = \pm 2$ and v the vector with $\langle v, A\rangle = F(f)$ gives $\langle e^{\pm 2F(f)}\rangle_\kappa \leq e^{2\langle v, Cv\rangle}$ in every state with the photon covariance C, the shell mass and the killing acting on the matter only. Hence $|\Psi_\kappa(z)| \leq M := 2e^{2\langle v, Cv\rangle}$ for all $\kappa \in [0, \infty]$, and $\langle v, Cv\rangle$, the variance of $F(f)$ under the free massive photon, is bounded uniformly in ε for smooth f. This is item (9): at $g = 0$ the matter moment factor of the disc bound is 1. For real z, $\Psi_\kappa(z) \to \Psi_\infty(z)$ as $\kappa \to \infty$ at fixed ε, by the substitution and dominated convergence of the proof of Lemma 2.7, the factor $e^{zF(f)}$ being integrable against the photon Gaussian by the same bound.

Step 1: the sandwich. Let $\eta_0 \leq \min(\eta_1, \eta_2)$ be so small that $b(\eta_0) \leq \frac{1}{2}$, and let $\kappa \geq \kappa_0$, $z \in [-1, 1]$. By (8) at $(0, 0)$, $e^{-a}(1-b)Z_\kappa(0,0) \leq I_\kappa^{\mathrm{cap}}(0,0) \leq e^a Z_\kappa(0,0)$; by (8) at $(zf, 0)$ and these two bounds, $e^{-2a}(1-b)J_\kappa(z) \leq \Psi_\kappa(z) \leq e^{2a}J_\kappa(z) + b$. Hence $J_\kappa(z) \leq 2e^{2a}M =: \bar{J}$ and $|\Psi_\kappa(z) - J_\kappa(z)| \leq \bar{J}(e^{2a} - e^{-2a} + b) + b =: m_1(\eta_0)$, and $m_1(\eta_0) \to 0$ as $\eta_0 \to 0$ because $a, b \to 0$.

Step 2: the κ-Cauchy property. Let $\kappa_0 \leq \kappa \leq \kappa' < \infty$ and $z \in [-1, 1]$. Write $I_\kappa^{\mathrm{cap}}(zf, 0) = \int_{\tilde{\chi}} e^{-\tilde{S}_\kappa(b; z)}\, db$ with $\tilde{S}_\kappa(\cdot; z)$ the reduced small-field action at shell mass κ and $\tilde{\chi}$ the capped test set, which is the same for all $\kappa \geq \kappa_0$; this is where (11) enters: no threshold term arises between

κ and κ'. Then

$$\frac{J_\kappa(z)}{J_{\kappa'}(z)} = \frac{\iint_{\tilde\chi\times\tilde\chi} e^{-\tilde S_\kappa(b;z)-\tilde S_{\kappa'}(b';0)}\,db\,db'}{\iint_{\tilde\chi\times\tilde\chi} e^{-\tilde S_{\kappa'}(b;z)-\tilde S_\kappa(b';0)}\,db\,db'},$$

and the two integrands differ pointwise by the factor $e^{\Delta_z(b)-\Delta_0(b')}$, which lies in $[e^{-2r}, e^{2r}]$ with $r := r(\kappa;\eta_0)$ by (10). So $J_\kappa(z)/J_{\kappa'}(z) \in [e^{-2r}, e^{2r}]$ and $|J_\kappa(z) - J_{\kappa'}(z)| \le \bar J(e^{2r}-1)$. Put $\bar r(K;\eta_0) := \sup_{\kappa\ge K} r(\kappa;\eta_0)$, which tends to 0 as $K\to\infty$ at fixed η_0. With Step 1, for all $\kappa' \ge \kappa \ge K \ge \kappa_0$,

$$|\Psi_\kappa(z) - \Psi_{\kappa'}(z)| \le m(K,\eta_0) := 2m_1(\eta_0) + \bar J\big(e^{2\bar r(K;\eta_0)}-1\big), \qquad z\in[-1,1],$$

uniformly in $\varepsilon \le \varepsilon_0(\eta_0)$.

Step 3. Letting $\kappa'\to\infty$ at fixed ε (Step 0): $|\Psi_K(z) - \Psi_\infty(z)| \le m(K,\eta_0)$ for $z\in[-1,1]$.

Step 4: two constants. $D := \Psi_K - \Psi_\infty$ is entire, $|D| \le 2M$ on $\{|z|\le 2\}$ and $|D| \le m(K,\eta_0)$ on $[-1,1]$. Since $\log|D|$ is subharmonic on $\Omega := \{|z|<2\}\setminus[-1,1]$ and is dominated there by the harmonic function with boundary values $\log(2M)$ on $|z|=2$ and $\log m(K,\eta_0)$ on the slit, $|D(\mathrm{i})| \le (2M)^{1-\omega} m(K,\eta_0)^\omega$, where $\omega\in(0,1)$ is the harmonic measure of the slit $[-1,1]$ at the point i in Ω, an absolute constant.

Step 5: the real point. The map $(A,\phi)\mapsto(-A,\bar\phi)$ preserves the action, the wall, the shell mass, the gauge fixing and the measure, and reverses the sign of $F(f)$; so $\Psi_\kappa(-z) = \Psi_\kappa(z)$, and since $\Psi_\kappa(\bar z) = \overline{\Psi_\kappa(z)}$, the value $\Psi_\kappa(\mathrm{i})$ is real. Thus $D(\mathrm{i}) = \mathrm{tail}_k(K)$ and

$$0 \le \mathrm{tail}_k(K) \le (2M)^{1-\omega}\, m(K,\eta_0)^\omega \qquad \text{for every } \varepsilon_k \le \varepsilon_0(\eta_0).$$

Choice of the parameters. Given $\eta>0$: take $\delta>0$ with $(2M)^{1-\omega}\delta^\omega \le \eta$; then $\eta_0 \le \min(\eta_1,\eta_2)$ with $b(\eta_0) \le \frac12$ and $2m_1(\eta_0) \le \delta/2$, which is possible because $a, b \to 0$; then $K \ge \kappa_0$ with $\bar J(e^{2\bar r(K;\eta_0)}-1) \le \delta/2$, which is possible because $r(K;\eta_0)\to 0$ as $K\to\infty$ at fixed η_0; and only then $\varepsilon \le \varepsilon_0(\eta_0)$. Then $m(K,\eta_0)\le\delta$ and $\mathrm{tail}_k(K)\le\eta$ for all $\varepsilon_k \le \varepsilon_0(\eta_0)$, so $\limsup_k \mathrm{tail}_k(K) \le \eta$: this is ($\mathrm{RES}_0$) at $(\mathfrak s,\chi)$ with this K, in its quantifier order: one configuration, one K, no rate, the limit along the spacings last. Since tail_k is nonincreasing in K (Lemma 7.4(a)), K may be enlarged at will (Remark 7.9(iii)), which places $(J',\psi,n+K)$ in the class (6) (Remark 5.5); Theorem 7.8 then gives (TIGHT_0) under (K_0). □

Remark 7.11 (the shell bounds). (8) and (10) are proved in [SW1] and [SW3], each from the residues inside its proof, the items (8a)–(8k) and (10a), (10b), (10d)–(10g), with the steps (F2), (F3), (F5), respectively (F6), of Section 8.5 in their completed forms; (9) and (11) are proved here; the literature inputs (12) are listed with their sources. Theorem 7.10 uses (10a) only through $\tau(\kappa)\to 0$, so a change of the exponent of τ or a logarithmic factor leaves the theorem unchanged. The residues (8k) and (10g), with (8e) = (8i), are proved in [SUP], in the order stated at the end of Section 1.4, the determinant clause of (8k)(ii) with the bound (DOM) of that section; on the histories with proper propagator domains the removal of the tail of the background in (8e) = (8i) uses the sub-domain form (P9) at order fourteen, which enters the upper bound of (8) only; (8e) = (8i) rests on the norm form of (8k)(ii) and on (8k)(i) through the bound for the mixed classes, and directly on the walled renormalisation bound (P1). They enter (8) directly and (10) through the statements of [SW1] that it cites. A photon Ward identity for each fixed ordering of King's graphs would not supply (10d), since that identity is false (Appendix A); the proof of (10d) uses the identities that hold, those for full lines, and Bałaban's resummed organisation keeps a convergent remainder, in whose place (10d) uses the bound (10g)(vi). The residues (8b)–(8f), (8k), (10e)–(10g) are internal to the proofs of (8) and (10), and (10c) to the use of Theorem 7.5 inside Theorem 7.8 at the configuration of Theorem 7.10: none of them changes the statement of (RES_0) that Theorem 7.8 consumes.

8. The inputs of the theorem, in plain words

This section states, in plain words and grouped by kind, every input of Theorem A that is not proved here: the steps proved in the main paper, among them the inputs stated in Section 1.4, and the statements of the literature, cited as printed; the numbers are those of Section 1.3. Every step of a printed proof that is asserted there without a justification and is used on the chain is named below, with the part of the main paper in which it is proved.

8.1. **The steps completed in the main paper.** Three statements inside the proofs of the torus comparison, of King's cube limit and of the shell estimates are proved in the main paper, each by more than one argument, (P7)–(P9) of Section 1.4. In plain words: (P7) says that on the skew tori of the corrected torus theorem, as on the cubic tori, the difference of Bałaban's propagators on two nested domains decays with the depth of its two points in the smaller domain; this is what makes his replacement of the propagators of the interaction terms at a non-zero background small. Its proof on the cubic tori goes over to the skew tori, since it uses no symmetry of the cubic lattice and its inputs hold there, and a second proof obtains it from an aligned cubic torus that covers the skew torus; only the torus comparison uses it. (P8) says that in King's cube limit the graphs whose trees are long, at the faces of the cube and away from them, are bounded at each of two cutoffs by the cut of the long lines, so that their change between the two cutoffs needs no rate; it is used by the cube limit, and so by the definition of King's state. (P9) says that the transfer of the perturbative bound to Bałaban's propagator domains holds for the families of order fourteen that the removal of the tail of the background produces, as it does up to order thirteen; only the upper bound of the shell estimates uses it.

(DOM) of Section 1.4 is proved in [SUP], by two independent arguments. In plain words: the shell estimates expand the determinant of each renormalisation step in the photon fluctuation field, on the torus and, in the upper bound, on the domains on which Bałaban defines his propagators; (DOM) says that the first-order term of that expansion, taken block by block, is bounded by the first differences of the background and of the fluctuation field, uniformly in the shell mass and with no bound on the size of the background, up to a term smaller than every power of the block size, at every block of the domain, its boundary included. The full determinant is invariant under every gauge transformation on the sites of the domain. The first argument uses this invariance at each point, together with a large gauge transformation of the background, and controls the resolvent at every point by an energy estimate; the second compares the coefficient at the blocks that carry the vertices, which lie deep inside the domain, with a coefficient on the torus at a regular field built from the background near the block. The torus comparison and King's cube limit are statements at zero shell mass, where the shell-mass residual of this term vanishes. The renormalisation bound for the operator with the scalar wall, the vacuum families with the mismatch vertex at King's backgrounds, Bałaban's enumeration read block by block at higher orders and King's cut of the long lines on his cubes are proved, in the forms (P1)–(P3) and (P5) of Section 1.4.

The four statements of Section 1.5 are proved in [SUP]. In plain words: Proposition 1.2 says that on King's cube the graphs with block vertices, in the families that carry one inserted external field, are bounded at long trees, at each of two cutoffs, by the cut of the long lines, as the same graphs without the inserted field are by (P8); it is used by the ultraviolet construction of King's cube limit, and so by the definition of King's state. Proposition 1.3 says that Bałaban's and King's estimates of the fluctuation integrals, printed for King's in-block contours rooted at a corner of each block, hold for the contours rooted at the block centres and for the composite contours of the hybrid flow, which the corrected torus theorem uses. Proposition 1.4 says that where the change of counterterm convention lowers the bare scalar mass on a walled torus, there is a mass in King's range whose bare mass squared lies below that of King's mass by more than this lowering, uniformly in the spacing, at every point of King's range except those at which the scalar mass minimises the bridge constants' function of the mass; it is used by the large-field bound at the wall,

and at the exceptional points the invariance follows by right-continuity in the scalar mass (item (15)), proved in the main paper from the monotonicity of the state in the bare mass coefficient and in the cube, the real-analyticity in the mass of the limiting difference of the bare mass coefficients, and a comparison of King's cube with the dead-shell torus at a fixed stopping scale, the endpoint step at $\kappa = \infty$ of [SW1] included. The final-scale cap says that the coarsest scale of the flow lies below a threshold fixed by the geometry, under which this large-field bound holds on every skew torus of the configurations used. The first enters King's cube limit; the other three enter the torus comparison, and the positivity condition also the walled limit (4). The odd-block clause of the degree census of the shell model is proved there as well and enters the κ-Cauchy estimate (10) only.

(10d) is proved in [SW3]. In the κ-Cauchy estimate of [SW3] the κ-derivative of King's effective action is King's sum of graphs with one insertion of the shell mass. Consider an insertion on a line used as a leg, deep inside the shell, and the cluster of short lines around it. If the cluster is joined to the rest of the graph through a scalar leg or a scalar line, the heavy matter pays for it (the deep payment). If all its connections are photon lines or photon legs, the photon factors do not see the shell mass and the deep payment fails. The bound then treats the photon subfamily containing the insertion as a whole, with the insertion unsplit: the members whose matter component through the insertion is closed are grouped into classes with the same full lines, one gauge identity is applied to each class, and only then is each class expanded into King's members, each with its own background; its base layer is the first-moment bound (10g)(vi). The route through a photon Ward identity for each fixed ordering, or each fixed scale assignment, of King's graphs is closed: that identity is false, the violation being of order one in exact computations. The Ward identities that hold are those for full propagators and for Bałaban's resummed organisation, which sums over the scales first and keeps a convergent remainder. King states the per-ordering cancellation [K1, p. 666] by citation only.

(8g) is proved by two independent arguments, which [SW1] cites. The perturbative bound of [SW1] expands the effective action around the background and Taylor-expands the photon legs; the remainder of that step is controlled by the bound $|\Pi_j^\kappa(x,y)| \le C|x-y|^{-4}$, uniform in the shell mass, for the current two-point function of the step-j lines. That bound is a pointwise statement about the line kernels: it needs, besides the decay of the kernel itself, a first difference at one end with one more power of decay, and a mixed second difference, one difference at each end of one line, with two more powers, on both sides of the shell boundary and uniformly in the shell mass. These are interior kernel estimates, which the estimates of [SW1] for bounded sources do not imply; they are proved by two independent arguments on the torus, the second of which also gives the restricted form used on the Neumann sub-domains, with both points at distance at least one block from the complement of the sub-domain.

(8h) is not an input. On a sub-domain with a Neumann face, the interior estimates of [SW1] are extended up to the face by reflecting the solution, and with it the gauge field, across the face. The reflected field has zero curvature on the plaquettes straddling the face, but the regularity condition of [K1, (3.5)] bounds every first difference of the field, and the odd reflection of the normal component, set to 0 on the bonds crossing the face, has a first difference of size $|A_m|/\eta$ there; for a constant normal component the reflected field is therefore not regular. What is used is regularity after a gauge transformation on the face layer, which a smoothed gauge supplies, and the coercivity of the doubled operator, which holds for every field.

(8j), the light-side estimates on sub-domains, holds at the deep points near which every cube of Bałaban's decomposition that meets ∂U is a flat-face cube. Near the edges and corners of H' that condition fails, and the light-side estimates are not used there: the localisation lemma of [LT] reduces Bałaban's local inverse on any Neumann union of blocks, at a point of depth at least 4ρ,

to the torus inverse for the data near the point, with an error exponentially small in ρ, and this supplies the forms that the consumers use, the payment of the far blocks included, (8k)(vi).

The steps of the parts [CT, H6, SW1, SW3] of the main paper that are completed in [SUP], the determinant clause of (8k)(ii) with (DOM), and the removal of the tail of the background on the histories with proper propagator domains with (P9), grouped by content, with the labels of Section 1.3, are:

- *The shell mass inside Bałaban's estimates.* The transfer of Bałaban's inputs at block factor 5 to the model with the shell mass inside the covariance, (8e) = (8i); the Hölder bound for the external fields required by [Ba3, (1.33)], (F3); the constants of Bałaban's completions at block factor 5 and the uniformity of his unqualified bounds in the scale index, (5l)(vii); the cumulant bound that replaces the cumulant lemma of Benfatto et al., on cubic and on skew tori, (5l)(i); and the reading of the wall as constant on each cube in the light-side estimates, (8k)(iv).
- *The perturbative bound on sub-domains and its cut members.* The transfer of its torus form to the propagator domains, with its norm form and the amplitude bounds there and with the regrouping of the cut families of Bałaban's classes by cut datum, into products of full classes, (8k)(i); the bound on the cut members of the resummed photon two-leg kernel, the mismatch clause at King's backgrounds and the determinant clause on the torus and on the propagator domains, with the bound (DOM) for its first-order term, (8k)(ii).
- *The random walk and the edges of the source box in the first shell estimate.* The gradient bound for the simple random walk, (8k)(iii); and the far-block payment, with a radius free of the large-block integer and near the edges of H', (8k)(vi).
- *The restricted renormalisation theorem and the one-loop term in the κ-Cauchy estimate.* The degree counts for the classes with block vertices, in which King's credit for photon fields contracted outside a subgraph is replaced by a split count, (5l)(x); the restricted form of [K1, (3.38)] for the shell model, the interpolated members included, (10g)(i); and the one-loop z^2-term with the block fluctuation line, in both regimes and with the tadpole in g', (10g)(ii).
- *The Ward structure in the κ-Cauchy estimate.* The first-moment bound for the two-photon kernel of full lines that replaces the regrouped Ward remainder in the shell model, (10g)(vi); the photon-leg classes with an insertion, (10g)(iv); the members linear in the source at the light local vertex, (10g)(vii); and the field-dependent members that this vertex may bring into the closed class of (10d), (10g)(viii).
- *Scale and position sums in the κ-Cauchy estimate.* The Hölder remainder for lines used as legs with an open matter component, (10g)(iii); the lattice regime of the scale sum of the Hölder-leg remainders, (10g)(v); the bookkeeping of the crossing and winding loops in (10a); and the dead-shell chain (10c).
- *The own steps of the corrected torus theorem.* The absorption in the hybrid lower bound, (5l)(ii); the hybrid-flow form of [Ba3, Prop. 1], (5l)(iii); the two repair steps, the determinants included, (5l)(v), (vi); the transfer of Bałaban's Green's-function bounds to the tori of [CT], read at the printed page, (5l)(iv); the two statements of (3), (5l)(viii), and the four steps of lower weight, (5l)(ix); and the Taylor step (4a), which both (K_0) and (TIGHT_0) use through (4).

The statements of the literature that the chain uses and that are asserted in print without a printed argument, or read off a printed construction, are all proved in the forms consumed. Four of them are proved at block factor 5 in [AS] and [CA]: Bałaban's same-scale estimates (item (1), the form for the operator with the scalar wall, (P1), included, except the six readings below, and on the skew tori with the change of domain (P7), for the contour families of the flows of [CT] and at the walls with Propositions 1.3 and 1.4 of Section 1.5), King's cube theorems that define King's state (item

(13), with the long trees (P8) and Proposition 1.2), Bałaban's all-orders enumeration with one inserted vertex (item (10f)), and King's statement that the remaining bounds of his graph analysis follow in the same way [K1, pp. 675–676] (item (5k)). The others are proved in [SUP], in the forms consumed: two readings of Bałaban's papers, (B1) and (B4) of Section 8.2, the bounds for the vector-field propagators [Ba3, p. 426] on the torus ((10g)(ii)) and the vector half of the completion of [Ba2, §3C] on p. 591 there ((5l)(vii), hence (K_0) and (8)); two statements of King, the source-free strict degree property of his renormalised graphs as rewritten by Bałaban [K1, p. 668], in the componentwise form of [Ba3, (3.5)] (items (4), (5g), (13)), and his treatment of the sources [K1, pp. 664–665, 677], as the small-field half of his upper bound at fixed sources (items (4), (5f), (5k), and the large-field clause of (1) at the walls); the photon legs of the per-step constants of the field-independent part of the perturbative bound of the shell model, as Bałaban's external fields in his norm ((12e), hence (8)); and four readings of Bałaban's papers, (B2), (B3), (B5), (B6) of Section 8.2: the pointwise bounds for the massive lattice Green's function and its differences [Ba3, p. 437] ((13)), the local coefficient of the classes with two scalar legs and one vector leg at zero shell mass [Ba3, (3.33)–(3.38)] ((8k)(ii), (10g)(iii), (iv), (8e), (K_0)) with the first moment of the vacuum-polarisation bubble at zero shell mass with the wall [Ba3, (3.26)–(3.28), p. 441] ((10)), and two readings of the construction of [Ba2], the regularity of the uncut background next to each propagator domain and the way a bound on the scalar block field near the shell is carried to the next step ((8k)(i), (13), (8e), (K_0)); with them the closure of his renormalised classes under his completion moves [Ba3, pp. 417–432], behind the regrouping of the cut families ((8k)(i), (13), (8e), (K_0)). The steps of the main paper named in Section 8.5 are used in their completed forms.

8.2. **Bałaban's asserted estimates.** Item (1) is the reading that every estimate of [Ba1, Ba2, Ba3, Ba4] asserted without a printed argument holds as asserted, at blocking factor 5, with per-step constants that depend on the step only through the running scale (scale-local), hence uniformly in the initial spacing. The asserted steps on the chain are, in the order of the papers: in [Ba1], the identities (2.39)–(2.42) ("after easy calculations", p. 612), the lower bound (2.33), the deferred statements (3.20)–(3.21) and (3.25), the perturbative formula (3.34)–(3.36) ("close inspection of the procedures", p. 618), the bound $O(\varepsilon^{\kappa})|T_1|$ on p. 615, which needs the order of the expansion to be at least 6, and the use of the cumulant lemma of Benfatto et al. (pp. 616, 623); the frame proof of (1.14); in [Ba2], the replacement of A'' by A' (pp. 561–562), the estimates of unnecessary terms through [Ba1, Props. 2.1–2.3] (p. 568; Proposition 2.2 of [Ba1] has no proof), the analysis of $W^{(j)}$ (p. 578), the transfer of [Ba1, Prop. 3.1] (p. 579) and Proposition 2.6 (p. 580, stated without proof), the inductive hypotheses (3.5), (3.7)–(3.8), the error arithmetic of (2.78), and Lemma 2.7 with (2.113), whose proof is omitted; in [Ba3], Proposition 1 with (1.33) and its consequences on pp. 421–422 ("rather easy consequences", "completely obvious"), and the Hölder clause on p. 412 ("similar considerations lead to the desired bounds"); in [Ba4], the scale-uniform analyticity neighbourhood of Lemma 2.4 (p. 586), the transfer of Corollary 2.3 to the mixed operators and the bounds (1.15), (1.16), (1.20) as its consequences (p. 588; the domain-difference clause (1.20) is not argued), the step from (3.10) to (1.14) (p. 589), the change of domain (5.5) (p. 594), and (1.8) on general domains, whose proof is not printed and is completed in [CT, SW1]. Behind King's theorems stand the same estimates: [K1, Thm. 3.1] is proved there by a reference to [Ba1, Ba2, Ba3, Ba4] and to its Appendix, and [K2, (4.2)] by the sentence that the results of [Ba1, Ba2, Ba3, Ba4] apply to both models (used only for the lower bound at zero wall, item (5b)). All of these steps are proved at block factor 5, in the forms consumed, from Bałaban's printed proofs in [AS], the renormalisation bound for the operator with the scalar wall in the form (P1) of Section 1.4 included, except the six readings (B1)–(B6) at the end of this subsection, proved in [SUP]; the cumulant lemma of Benfatto et al., whose printed form covers the nearest-neighbour Markov field only, is replaced in [SW1] by a cumulant bound with constants free of the shell mass (item (5l)(i)). The repairs at block factor 5 of four printed steps are proved in [CT] (items (5l)(v),

(vi)), and their re-runs for the shell model are (8e). Lemma 2.7 of [Ba2] with (2.113) holds at block factor 5 on the tori of the construction; read on King's cube with the Dirichlet weight, its printed form fails near the Dirichlet face and at the first scale, and the corrected form that King's cube limit uses is proved in [CA].

The step whose proof was completed last is Proposition 2.6 of [Ba2] for the interaction terms at a non-zero background. In the form proved in [AS] it reads, in the notation of [Ba2]: *in the inductive step $k \geq 1$ of the upper bound* [Ba2, §2] *at block factor* 5, *with $t = 5^k\varepsilon$, let $\tilde{B}$ be the background field on $B^k(\Lambda_0^{(k)})$ of* [Ba2, p. 579]*, and consider the interaction terms of that step localised in $B^k(\Lambda_7^{(k)})$, every vertex summation being restricted to $\Lambda_7^{(k)}$; their scalar lines are $G_k(\Omega_1, \tilde{B})$, $\Omega_1 := B^{k-1}(\Lambda_2^{(k-1)})$, the domain of the statement* [Ba2, Prop. 2.6, p. 580]*, their external scalar fields are $a_k G_k(\Omega_1, \tilde{B})f$ with $\|f\|_\infty \leq Cp\,\lambda(t)^{-1/4}$, and their other external fields have norms at most $Cp^a\lambda(t)^{-b}$. Replace $G_k(\Omega_1, \tilde{B})$ by $G_k(\Omega_2, B^{(k+1),\eta})$, $\Omega_2 := B^k(\Lambda_2^{(k)})$, in every scalar line and in every such external field. Then for every $\nu > 0$ there are $C_\nu < \infty$ and $t_0(\nu) > 0$ such that, for $t \leq t_0(\nu)$, the sum over all localisations of the absolute values of all terms containing at least one difference of the two propagators is at most $C_\nu t^\nu |\Lambda_7^{(k)}|$, with constants independent of the initial spacing and of the background $\tilde{B}$, the case $\tilde{B} \neq 0$ included.* In the proof the domain Ω_1 is written $B^k(\Lambda_2^{(k-1)\prime})$, as in [Ba2, (2.56)]; this is the same set, by the definitions of the prime and of B in [Ba1, (1.16)–(1.18)], since $\Lambda_2^{(k-1)}$ is a union of large blocks. The proof regroups the terms with at least one difference into contractions of products of full classes of Bałaban's graphs of the cut components, to which the renormalisation bound [Ba3, (1.33)] at block factor 5 applies at every background; no Ward identity for a fixed ordering is used. The replacement in the remaining part of the step, after [Ba2, (2.107)], is proved without the renormalisation bound. This is the form on the cubic tori and on the domains of the shell estimates; on the skew tori of [CT], which the torus comparison also uses, the decay of the difference lines with their depth is (P7) of Section 1.4, proved there by two independent arguments, and with it the same form holds on the skew tori.

Readings of Bałaban's papers. Six readings of Bałaban's papers that the chain uses, (B1)–(B6), some of them carrying more than one statement, are asserted there without a printed argument, or are read off his construction, and are not covered by [AS]. All are proved in [SUP], in the forms stated:

(B1) the bounds [Ba3, (2.10)–(2.12)] for the propagators of the vector field: "Of course the same inequalities hold for vector field propagators" [Ba3, p. 426]: proved in [SUP], on the periodic torus fitted by blocks, for the sup bound, one first difference per endpoint, the Hölder bound (2.11) with one difference and the contour legs, uniformly in the spacing, the scales and the scale mass; used in (10g)(ii) through the sup and first-difference bounds of the photon fluctuation field, in both regimes, and in (12e) through (KA*), whose norm [Ba3, (1.32)] carries the Hölder quotient;

(B2) the pointwise bounds for the massive nearest-neighbour Green's function $C^\xi = (-\Delta^\xi + 1)^{-1}$ on $\xi\mathbb{Z}^3$ and for the kernels built from it, $|C^\xi(y - y')| \leq O(1)e^{-|y-y'|/2}/|y - y'|$ "and the corresponding inequalities for derivatives" [Ba3, p. 437], with their first two differences and their differences between two scales: proved in [SUP], the kernel bound, the first two differences and the differences between two scales with an unspecified rate $c_0 > 0$ in the exponential (Bałaban's printed rate $\frac{1}{2}$ is not claimed and not used); used in the boundary analysis of King's cube limit (13) and in the cut (P5); the block-spin inequality of the same page, which one boundary bound of (13) uses, is proved as well, at zero shell mass and zero background, through the pointwise line-kernel bounds of (8g) [SUP];

(B3) Bałaban's treatment of the local term of the primitively divergent classes with two scalar legs and one vector leg at zero shell mass [Ba3, (3.33)–(3.38), pp. 443–444]: the coefficient of the local term vanishes for the graphs built from C^ξ, by (3.37), (3.38), the sign pairing

of (2.18) and translation invariance, after a replacement of the propagators that leaves "a convergent expression": proved on the torus in [SUP], for the window coefficient with a bounded potential and for the block term at any window; the coefficient vanishes for the graphs built from C^{ξ} and is bounded, in general non-zero, for the window lines; and the first-moment step of the same kind for the vacuum-polarisation bubble [Ba3, (3.26)–(3.28), p. 441] at zero shell mass with the wall: the first moment of the part built from C^{ξ} vanishes as printed, by the symmetry (3.27)–(3.28), and the remainder carrying the wall is bounded, uniformly in the spacing, the scale, the torus and the wall, with constants depending on the wall height n through $m^2 + n$, proved in [SUP], by two routes. The mixed-class step is used by the bounds for the mixed classes in the shell model, inside (8k)(ii) and in (10g)(iii), (iv), and through them by (8e) = (8i), (P3) and, through the repaired cube step of (1), by (K_0); the vacuum-polarisation step by the κ-Cauchy estimate (10);

(B4) the estimate of the integral over the vector fields in the completion of [Ba2, §3C], which [Ba2, p. 591] obtains "in a way similar to (3.22)", read componentwise: proved in [SUP], from (3.21) as printed and the scalar bound (3.39), on the cubic and the skew tori and for the model with the shell mass inside the covariance, whose vector action is unchanged (the shell mass enters the matter Gaussian only, [SW1]), with the vector part of the constant a number independent of the spacing, the scale, the history and the torus; it gives the vector half of the constants (5l)(vii) and enters (K_0) and (8) through them;

(B5) the regularity, in the sense of [Ba1, Prop. 2.1] and [Ba4, (1.7)], of the gauge background on the blocks of scale k within distance one of each propagator domain Ω of that scale. Two statements: on Ω itself, [Ba2, Lemma 2.3] in the form that holds at block factor 5 (the printed lemma fails there for small couplings), with the regularity of the cut field; and off Ω, on the k-blocks within distance one of it, the regularity of the uncut background $A^{(k)}$ of [Ba2, (2.44), (2.54)], which equals the Ω-background on Ω (p. 567 there); the cut field $\theta_k A^{(k)}$ is not regular on the slice of the cut and is not used off Ω. Both are proved in [SUP], the second from [Ba2, (2.55), (2.7)–(2.8)], the decay of the vector propagator and the rows of (1) on the small-field restrictions, which are imposed on Λ_0 and on its neighbourhood of thickness $r(\varepsilon)$ [Ba2, p. 558, (2.24)–(2.25), (2.95)–(2.98)]; used by (8k)(i) and, through it, by (13), and by (8e) = (8i) and (K_0) through the bound for the mixed classes;

(B6) the way a bound on the scalar block field near the shell, imposed at one step, is carried to the next step: it is inserted before the characteristic functions of that step are bounded by 1, as [Ba2, (2.55)] carries the restrictions given by the characteristic functions to the next step [Ba2, p. 570]: proved in [SUP] for the inherited cap (an insertion lemma and an inherited-cap lemma); the gradient line of [Ba2, (2.55)] is carried at steps $k \geq 2$ on the domain $\Lambda_2^{(k-1)\prime}$ of [Ba2], by a transport theorem whose constant c_1 is split into its supremum, vector and scalar parts, of which the scalar part depends on the couplings as in [Ba2, (2.21)]; the supremum, $|A|$ and $|\partial A|$ lines hold on $\Lambda_0^{(k-1)\prime}$, and the printed domain Λ'_{-1} of the gradient line is not used; used by (8k)(i) and, through it, by (13), and by (8e) = (8i) and (K_0) through the bound for the mixed classes.

The regrouping of the cut families. Bałaban's renormalised classes [Ba3, pp. 417–432] can be read as the classes closed under his completion moves, the moves listed in (P1)(b): the subgraphs are read as sets of lines; the move that forms the classes with two vector legs [Ba3, p. 431] acts at every divergent part with two vector legs, not only at the maximal ones; and in the picture of [Ba3, p. 417], in which the legs of a subgraph are moved to an anchor, the anchor is fixed by a rule intrinsic to the subgraph. No such reading of the printed classes is needed: what the regrouping of the cut families along the orbits of the moves uses is proved in [SUP], namely that the per-label full classes are closed under the completion moves, and every closed family obeys the bound. It is used by the regrouping in (8k)(i) and in (13), through (8k)(i) by (13), and by (8e) = (8i) and

(K_0) through the bound for the mixed classes. Bałaban's sentence that the divergent vacuum graphs are cancelled by the vacuum-energy counterterms [Ba3, p. 445] holds at zero background only; at King's backgrounds the vacuum families with the mismatch vertex are bounded by (P2) of Section 1.4, whose proof does not use it, and those without it by the perturbative bound (8k)(ii), the one-loop determinant of the step excepted, which enters through the determinant clause of (8k)(ii); a bound of the part of these families that depends on the background is not printed in [Ba3], and none is taken from there.

8.3. **King's asserted steps.** *King's cube limit* (13), which defines King's state: [K2, Thm. 2.1] at $h = 0$. Its proof [K2, §3.1] transfers the torus proof of [K1] to Neumann–Dirichlet cubes with written modifications and states that "all the results of [K1] go through" (p. 332). The analysis at the sharp boundary is printed for one counterterm graph ("the other graphs in δm^2 can be treated similarly"), and for the photon it proceeds one ordering of the renormalised graphs at a time (pp. 331–332), which in that reading rests on the per-ordering cancellation discussed below; the replacement (3.54) is omitted (p. 330); the Hölder continuity behind the factor $5^{-\alpha k}$ of (3.13) is not stated; and the image-sum identity (3.9), which the transfer uses, is false as printed together with (3.3) and (3.8) for the weight printed there, and holds with a Dirichlet weight that suffices for the transfer. The bound at a fixed cube used in [K2, (3.41)] is this theorem at real coefficients. These steps are proved, in the forms consumed and on King's cubes C_J^K, in [CA], with the bound at long trees of Proposition 1.2 (item (13)): the reduction of the long trees of the localisations on the cube is the cut (P5), with (P8) of Section 1.4 for the long trees in the difference clause, and the ultraviolet construction on the cube uses the sub-domain form (8k)(i) of the perturbative bound at zero shell mass; there the photon boundary step uses the zeroth moment of the full photon kernel of the cube, with no identity for a fixed ordering.

King's renormalisation estimates (item (5k)). [K1, Thm. 3.1] is proved by a reference to [Ba1, Ba2, Ba3, Ba4] and to the Appendix of [K1]. [K1, Prop. 3.6] is stated on p. 662 and its proof (pp. 663–665) ends in an assertion; its one-flow instance is proved (5g), from the strict degree property of King's renormalised graphs as rewritten by Bałaban, [Ba3, (3.5)], [K1, (3.78)], which King asserts on p. 668 ("(3.38) relies solely on the cancellation of divergences") and which is proved in [SUP], in the componentwise form of [Ba3, (3.5)]; a reading of the print finds [K1, (3.78)] false on certain vacuum graphs, whose contribution is a removed constant and is not used. [K1, Thm. 3.5(i)] is false as printed and is replaced by [RS] (5h); [K1, Thm. 3.4], false as printed, is not used, the form needed at a fixed wall being proved in [H6]. The propagator bounds [K1, Props. 3.7–3.10, Lemma 4.5, (3.72), (3.88)] and the remainder bounds of [K1, Thm. 3.3] enter [CT, H6, RS] through (5k): [K1, Prop. 3.7] is a printed sketch ("follows immediately from Theorem 3.3 and the scaling"); in the proofs of [K1, Props. 3.8, 3.9] only the last line of (3.71) and the first line of (3.73) are printed, the derivative line of (3.73), the contour form (3.74) and the Hölder forms (3.75) being covered by "the other bounds follow in the same way" (p. 675); [K1, Lemma 4.5] rests on [Ba4, §5] and on the step (4.36)–(4.37), which is false as printed (the block average is treated there as a Fourier multiplier), while its conclusion (4.33) holds (Proposition A.4); and the bound [K1, (4.21)] is false as printed for $m > 0$; only its truncation to the alias-free term, in a corrected form, is used, and (4.28) is not used (both belong to the printed proof of the last line of (3.71), which is proved in [AS] by a different route). These bounds are proved in the forms consumed in [AS]; an extension of [K1, Prop. 3.7] beyond its printed scope is false and is used nowhere.

The vacuum polarisation. King states on [K1, p. 666] that the divergences cancel for each ordering of the renormalised graphs, with a reference to [Ba3], whose identities (2.24)–(2.27) are for full propagators and whose organisation sums over the scales first and keeps a convergent remainder. In its per-ordering reading the statement is false, the violation being of order one in exact computations; it is used nowhere on the chain: King's photon boundary step for (13) [K2, pp. 331–332], which works one ordering at a time, is replaced in [CA] by the zeroth moment of

the full photon kernel of the cube. Four places in the main paper state a vacuum-polarisation identity per ordering or per scale: the ordering-restricted corollary for King's estimate in [CT], the vacuum-polarisation step of the strict degree property in [H6], the Ward identities with one insertion in [SW3], and the vanishing zeroth moment of the combined photon two-leg kernel at each scale in [SW1]. In [CT], [H6] and [SW3] the statement consumed is Bałaban's resummed form, and [H6] says so; in [SW1] the form consumed is not stated there, which is the step (F5)(c) of Section 8.5; it is the window identity of (8k)(ii) for full lines.

King's Appendix and sources. The expansion to order 13 with (3.54) "still sufficient" (p. 676) is replaced for the shell model by [SW1, Lemma appclass] (8b) and is proved in [AS] elsewhere (items (5k), (12e)). The photon legs of the per-step constants of the field-independent part of the perturbative bound of the shell model [SW1, Prop. TD(d)] are handled without the moment bounds of the Appendix [K1, (A.2)–(A.6)]: the photon slice lines and King's photon legs are Bałaban's external fields in the norm [Ba3, (1.32)], (KA*), proved in [SUP] (item (12e)). The inclusion of sources ("straightforward", p. 677), the componentwise replacement of the sources (p. 665) and the uniformity of King's constants over the source family (pp. 664–665) are asserted in print; they are used in [H6] (item (4)), in (5f), in Remark 2.9 and in the large-field clause of (1) at the walls (Section 1.5), and they are proved in [SUP] in the forms consumed: the inclusion of sources as the small-field half of [K1, Thm. 3.1] at every fixed source of bounded size, in King's vertex form, for King's flow and for the centred flow, and for the hybrid flow its lower half; the componentwise replacement, p. 665, by a summation by parts for the curvature source and the mean value theorem for the scalar source and the partition of unity, at cost $O(5^{-k})$ for every pair of scales and on the torus as on the cube, the sources entering through their supremum norms only; and the uniformity over the source family through a log-convexity of the walled partition function in a constant scalar source, so that the consumers need King's bound at four fixed sources only. The photon clause of [K1, Thm. 3.3] (p. 656, "of course with $A = 0$") is a specialisation of two lines on the torus that is not written in print; it is used in (5a) and in [K1, (A.5)–(A.6)] (item (12e)) and is proved in [AS] (item (5k)). *King's near-face analysis* [K2, §5.2, (5.41)–(5.49)] is a one-page sketch; its transplant is (8c).

8.4. **Reisz, Brydges, Fröhlich and Seiler.** Reisz's theorems are used in the corrected three-dimensional form (5e): the transfer to other dimensions, which [Re1, p. 86] states can easily be made, is written out in [CT], and the naturalness clause, false as printed, is replaced by a condition proved for the graphs consumed. The forest identity [Re2, Lemma 5.1], whose proof there is a pointer to Zimmermann's paper, is Zimmermann's forest formula, cited by name (Section 8.6). The membership of the subtracted integrands in Reisz's class, asserted at [Re2, (5-1)] and on p. 97 there, is proved in Remark 8.1. The reversed correlation inequalities of [BFS, Thm. 6.2(3)–(4)], sketched in print, and King's [K2, (3.28)], stated without proof, are proved in Lemma 2.12(d).

Remark 8.1 (membership in Reisz's class). In the notation of [Re1, Defs. 2.2–2.5], $\mathcal{C}_m$ is the set of functions $V = a^{-m}F(xa)$ of the momenta x with F smooth, $\mathcal{C}^c_m \subset \mathcal{C}_m$ the subset on which $\lim_{a\to 0} V$ exists for every x, $\mathcal{C}^c$ the finite sums, and Reisz's class consists of the quotients V/C with $V = \sum_i P_i(\mu)V_i$, $V_i \in \mathcal{C}^c_{m_i}$, P_i polynomials in the masses, and $C = \prod_i (e_i(l_i;a) + \mu_i^2)$, $l_i \neq 0$ linear in the loop momenta, $e_i = a^{-2}\eta_i(l_i a) \in \mathcal{C}^c_2$ with η_i periodic, positive off $2\pi\mathbb{Z}^3$, and $\lim e_i = l_i^2$. *Claim:* for the graphs consumed in [CT] and the operators used there (the evaluation of external momenta at 0, the periodic lattice subtraction operator of first order (not a Taylor operator), with $P(q;a) = a^{-1}\sin(qa)$, and derivatives in the masses and in the external momenta), every forest term of the subtracted integrand, and each of its derivatives, lies in Reisz's class and is $2\pi/a$-periodic in every loop momentum, the loop momenta of each subgraph being chosen among those of the graph as in [Re2, (2-24)]. *Proof.* (i) $a^{-m}F(xa)\,a^{-n}G(xa) = a^{-(m+n)}(FG)(xa)$, and limits multiply. $\partial_j[a^{-m}F(xa)] = a^{-(m-1)}(\partial_j F)(xa)$; if $\lim_a a^{-m}F(xa)$ exists for every x and $m \geq 1$, the Taylor terms of F at 0 of order less than m vanish (the least non-vanishing one would make

$a^{-m}F(xa)$ diverge), so $a^{-(m-1)}(\partial_j F)(xa)$ converges to the derivative of the term of order m; for $m \le 0$ it tends to 0. Evaluation at $q = 0$ and integer linear substitutions of the momenta preserve $\mathcal{C}^c_m$; $a^{-1}\sin(qa) \in \mathcal{C}^c_1$. (ii) A derivative of a propagator factor is $\partial(e+\mu^2)^{-1} = -(\partial e)(e+\mu^2)^{-2}$ with $\partial e \in \mathcal{C}^c_1$ by (i), and $\partial_{\mu^2}(e+\mu^2)^{-1} = -(e+\mu^2)^{-2}$; so derivatives of products of propagators have numerators in $\mathcal{C}^c$ over products of the same factors. (iii) On a subgraph γ, the subtraction operators turn a factor $e(k^\gamma+p)+\mu^2$ into $e(k^\gamma)+\mu^2$, which has the required form with $k^\gamma \neq 0$ as a linear form, since γ is one-particle irreducible (a line of γ carrying no loop momentum of γ would be a bridge); the numerators stay in $\mathcal{C}^c$ by (i). (iv) The line momenta have integer coefficients in the loop momenta, and η_i and sin are periodic, so every factor is $2\pi/a$-periodic in each loop momentum; $e+\mu^2 \ge \mu^2 > 0$. (v) $V_1/C_1 + V_2/C_2 = (V_1C_2 + V_2C_1)/(C_1C_2)$, where C_1C_2 is again a product of admissible factors and C_1, C_2 expand into elements of $\mathcal{C}^c$ with coefficients polynomial in the masses, by (i); induction over the finitely many forests gives the claim. □

8.5. **Steps of the main paper completed in other parts of it.** The following steps of [CT, H6, SW1, SW3] are written there without a complete justification; for each, the completion is stated with the place where it is proved, and the items that contain them use the completed forms.

- (F1) In [CT], the decay bound $|\Gamma^F(z,w)| \le C(1+|z-w|)^{-1}e^{-cm_k|z-w|}$ for the coarse covariance $\Gamma^F = H^F B^F H^{F*}$ (the kernel B^F conjugated by the external-line kernels H^F; the bound is checked for the conjugated kernel, the one the chain consumes, not for B^F alone) is derived from the Paley–Wiener theorem, which gives a constant depending on m_k; the product form does not follow, as written, from the minimum of the polynomial and the exponential bound, and the analyticity of the inverse symbol in the strip rests on a lower bound for its real part that is stated without proof. The bound holds in the product form, uniformly in m_k: the kernel is written as the difference of the free propagator and its fine part, and the product bound for the free lattice propagator is obtained by a contour shift and an integration by parts, with no use of the Paley–Wiener theorem or of the strip bound (item (5j)).
- (F2) In the skew-torus and boundary-box re-run of Bałaban's steps behind (1) and (3) in [CT], which [SW1] also uses for its $(\mathrm{BAL})_5$: at the boundary boxes the background is not constant near the Neumann faces, while [Ba4, Lemma 2.2] assumes it constant there, and the open-ended clause of [Ba1, Prop. 2.1] ("for some simple sets") is used for them; the cumulant lemma of Benfatto et al. is applied directly there; and the hybrid kernels need their own domain-difference identity, since [Ba4, (3.9)–(3.10)] cover a change of pinning only. Completions: no step of the re-run needs [Ba4, Lemma 2.2] on a boundary box at a non-constant background, and the boundary clause is re-stated in the restricted form that is proved; the cumulant lemma is replaced by the cumulant bound of [SW1], whose lattice hypotheses hold on the skew torus; the transfer of the rest of its proof to the skew torus is proved, and so is that proof itself (item (5l)(i)); the domain-difference identity for the hybrid kernels is proved. The sign misprint of [Ba2, (2.99)] does not matter, since $W^{(j)}$ is defined there by "=:" and only $C(\tilde B + \tilde A)^{-1} = C(\tilde B)^{-1} - W^{(j)}$ is used, never a term of $W^{(j)}$ (items (1), (3), (8)).
- (F3) In the shell estimates, the Hölder clause of [Ba3, p. 412] is completed by a derivation in which the restriction radius is at least Bałaban's localisation radius (proved, with Bałaban's smooth cutoffs taken in an admissible form whose second differences are bounded uniformly in the scale; items (1), (8), and (5l)(iii), whose proof uses it).
- (F4) In [H6], the Montel–Vitali step from the strip bounds needs their uniformity in the scale. It holds: $|r_K(z)| \le r_K(\mathrm{Re}\, z) \le \exp[C_G(\mathrm{Re}\, z)^2]$ for the walled moment functions r_K, with C_G the supremum over K of half the variance of $F(G)$ under the free photon, finite by the convergence of the lattice variances; this bound is part of the strip-bound lemma of [H6], so the step is complete (item (4)).

(F5) In [SW1]: (a) the clause of the sup bound with block data in L^1 is false uniformly in the spacing; every consumer uses bounded or block-L^2 data, or a point datum at the fixed spacing of the first step, the form that the printed proof proves (proved in [SUP]). (b) The pointwise bound $|\Pi_j^\kappa(x,y)| \le C|x-y|^{-4}$ for the photon two-leg kernel is attributed to the sup and first-difference clauses of the bounded-source estimate, which do not give pointwise kernel bounds; the kernel bounds it needs are of Kato and Combes–Thomas type, and they are (8g), proved. (c) The vanishing zeroth moment of the combined photon two-leg kernel is stated at each scale, and the text does not say that the resummed form of Section 8.3 is the one consumed. Completion: the vanishing is used only for the window class of a divergent subgraph with two photon legs, all its internal lines summed over the scales below the window, through an exact window identity whose cut members are bounded in (8k)(ii); no identity for a fixed ordering or scale assignment is used. (d) The sup form of the light-side bounds at distance $\ge M$ from the faces of a sub-domain rests on a localisation statement that claims King's [K1, (3.7)] at all points of a union of large blocks, which fails near re-entrant edges; its restricted form, at distance at least Bałaban's radius from the faces, holds; whether the sub-domains used satisfy the corresponding face condition need not be decided, since where it fails, near the edges and corners of H', the localisation lemma of (8j) supplies the forms that the consumers use, the payment of the far blocks (8k)(vi) included. (e) The use of the external field of the Appendix class in the proof of the finite-domain perturbative bound is asserted to match (item (8b)). (f) Citations: the uniform exponent $\frac{1}{4}-\beta$ is attributed to [Ba1, (3.58)], which prints $\frac{1}{2}-\beta$, while Bałaban prints $\frac{1}{4}$ for one vertex [Ba2, (2.21)–(2.22)] and asserts the uniform exponent at [Ba3, p. 421] with the count written for a few vertices only; its uniform form over all vertex families is proved, its input from [SW1] being the norm form of the perturbative bound ((8k)(ii)). The diamagnetic inequality is used with a site-dependent mass, an extension of three lines of [BFS, §3.3] that is not printed there (proved in [SUP]: the random-walk representation of the printed proof holds for every non-negative site-diagonal mass that is not identically zero, so that the massive operator is invertible (in the singular case the inequality is trivial); [SW1] writes it out at the place of use). [K1, (A.3)–(A.6)] are photon identities, cited there for the matter background; they carry no load, the matter content at those places being [Ba2, Lemma 2.4] (item (8k)). The constants of [Ba4, Lemma 2.2, Prop. 2.3] depend on the block factor, which is harmless at the fixed block factor 5. The image sums on the torus do not improve the kernel bounds for the massless kernel in three dimensions, where the bound is obtained by lifting to the infinite lattice (item (8)).

(F6) In [SW3], in the face term of the tail theorem: the ballot bound for the outer pieces of a walk is used without its uniformity in the starting distance, which fails in general and holds for a start next to the half-line, the case used; the edges of a lifted coordinate walk above the axis are identified with its visits to U, which holds as two inequalities; and the moment bound $(Cq)^{2q}$ for the martingale part is attributed to Burkholder's inequality with constant $O(q)$, which gives only $(Cq)^{3q}$, while the stated bound holds by an exponential-supermartingale estimate for martingales with predictable increment bounds. Each completion is proved, and the exponent c in (10a), which (10) uses only through $\tau(\kappa)\to 0$, is unchanged (item (10a)).

The configuration produced by Theorem 7.8 must satisfy at once the class (6), the block alignment of [SW1, SW3] and the provisos of [H6] on the torus size and the wall. This is an argument of this supplement, proved in Remark 5.5: the provisos of [H6] are the clauses of the class (6), the wall condition is linear in the height at a fixed torus, and the torus, the block alignment and the grid offset of the corners of H' are choices made in Theorem 7.10.

8.6. **Textbook results.** Cited by name, with their exact hypotheses checked against the use: the local limit law and the Green's function asymptotics of the simple random walk [LL], the theorems of Montel and Vitali, the two-constants theorem, Kato's inequality, the Combes–Thomas estimate, the Paley–Wiener theorem (whose use in [CT] is the step (F1) above), the Chung–Feller theorem, the discrete Nash inequality and the heat-kernel bounds of Carlen, Kusuoka and Stroock and of Davies, the Neumann eigenvalues of the discrete cube, and Zimmermann's forest formula. Burkholder's inequality is cited in [SW3] for a bound that it does not give in the form stated; this is the step (F6) above. The multiple-reflection bound of Glimm and Jaffe behind [K2, (3.43)] is not consumed: at $h = 0$ that bound is trivial.

9. The inputs and where they are proved

The table lists every input of the theorem, where and how it is proved, its source, and the inputs on which it rests; the item numbers are those of Section 1.3.

Input	Proof	Source	Rests on
Proved inputs and readings (Sections 1.4, 8.2, 8.3)			
(DOM) the first-order term of the step determinants, composed over the earlier steps, on the propagator domains of Bałaban's upper bound and on the torus, uniformly in the shell mass and with no bound on $\|B\|$: (a) the per-site form at every block; (b) the current in L^1 at the blocks of the vertices; (c) the torus	proved, by two independent arguments; used by the determinant clause of (8k)(ii)	[SUP]; [Ba2, (2.86), (2.98), (2.104)–(2.107)]; [Ba3, p. 412]; [Ba4, (1.8)]	(1); for (b) also (8g), (8k)(iii), (F3), (8d), (P3)
(P1) the renormalisation bound for the operator with the scalar wall, (a) at every regular background for Bałaban's families, (b) at zero background for the node-closed classes; (P2) the vacuum families with the mismatch vertex at King's backgrounds, on the torus and on the propagator domains; (P3) Bałaban's enumeration block by block at higher orders	proved; (P2), (P3) through the cube step of (8e), carrying the readings (B3), (B5), (B6) on which (8e) rests, proved; the difference form of (P2) on the histories with proper propagator domains through (P9)	[AS] for (P1); [SUP] for (P2), (P3); [Ba3, Prop. 1, Prop. 2.2, §§2–3]	(1), (8d), (8e), (B3); (P2) also (P9)
(P5) King's cut of the long lines on King's cubes	proved; on the torus in [AS]; its use at the long trees in the difference clause of (13) is (P8)	[CA]; [K1, (3.38), p. 666]; [RS]	(B2)

Input	Proof	Source	Rests on
(P7) the change of domain of Bałaban's propagators on the skew tori: decay of the difference with the depth of its two points; its inputs, Bałaban's bounds for the Green's functions and the block operators and the scope statements of his construction, on the skew tori	proved on the skew tori by two independent arguments: the transfer of the proof on the cubic tori, which uses no symmetry of the cubic lattice, and a reduction to the aligned cubic torus that covers the skew torus; proved on the cubic tori and on the domains of the shell estimates; enters (1) on the skew tori, hence (K_0), only	[AS]; [CT]; [Ba2, Prop. 2.6]; [Ba4, §2]	(5l)(iv)
(P8) the long trees in the difference clause of the split degree count on King's cubes, at the faces and away from them, at each of two cutoffs	proved, the long trees by three proofs; enters (13), hence the object and ($TIGHT_0$)	[CA]; [K1, (3.41)]	(P5), (5l)(x)
(P9) the sub-domain form (8k)(i) of the perturbative bound at order 14, for the families of the removal of the tail of the background, with and without the mismatch vertex	proved by two arguments; enters the upper bound of (8) only, hence (RES_0) and ($TIGHT_0$)	[SUP]; [SW1]; [Ba2, (2.89)–(2.90)]	(8k)(i), (8e), (8d)

Input	Proof	Source	Rests on
(B1)–(B6) readings of Bałaban's papers: the vector-field propagators on the torus; the massive lattice Green's function and its differences (unspecified rate $c_0 > 0$); the local coefficient of the classes with two scalar legs and one vector leg (torus, window coefficient with a bounded potential, block term at any window) and the first moment of the vacuum-polarisation bubble at zero shell mass with the wall (constants depending on the wall height); the vector part of the completion of [Ba2, §3C]; the regularity of the gauge background on the propagator domain ([Ba2, Lemma 2.3] at block factor 5) and of the uncut background within distance one of it; the carrying of the scalar block-field bound to the next step (inherited cap; gradient line at $k \geq 2$ on $\Lambda_2^{(k-1)\prime}$); and the closure of his per-label classes under the completion moves behind the regrouping of the cut families	proved	[SUP]; [Ba3, pp. 426, 417–432, 437, 441, 443–444]; [Ba2, pp. 558, 562, 567, 570, 591, (2.21), (2.55)]	(1), (8), (10)
(KA*) the photon slice lines and King's photon legs as Bałaban's external fields in the norm [Ba3, (1.32)], for the per-step constants of the field-independent part of the perturbative bound of the shell model	proved from the bounds for the vector-field propagators on the periodic torus, the reading (B1), itself proved, componentwise and uniformly in the spacing, the scale and the scale mass, in that norm	[SUP]; [SW1, Prop. TD(d)]; [Ba3, (1.32), p. 426]	(B1)
The four further statements, and the odd-block clause of the degree census (Section 1.5)			
Block vertices at long trees, on King's cubes, in the families with one inserted external field (Proposition 1.2)	proved: (P8)'s argument with the three rows the inserted field requires; enters (13), hence the object and (TIGHT$_0$)	[SUP]; [K1, (3.41)]	(P5), (P8)

Input	Proof	Source	Rests on
The fluctuation estimates of [Ba1, Ba2, Ba3] and [K1, Thm. 3.1] for the in-block contours rooted at the block centres and for the composite contours of the hybrid flow (Proposition 1.3)	proved, by a census of the occurrences of the contours and of the operators built from them in the printed proofs and, for the hybrid family, a direct proof of [Ba2, (2.65)–(2.66)]; enters (K_0) only	[SUP]; [Ba1, (2.1)–(2.3)]; [K1, (2.12)]; [Ba4, p. 573]	(1)
The positivity condition on the constant Δ_T of the convention bridge, on the three pairs of a lattice and a torus, with M_1 in King's range (Proposition 1.4)	proved by three independent arguments at every point of King's range outside the set E_W; at E_W Theorem A by right-continuity of King's state in the scalar mass (item (15)), proved; enters (1) at a non-zero wall, hence (H3), (4), (K_0) and (TIGHT$_0$)	[SUP]; [K1, (3.32)–(3.33)]	(1), (5f)
The final-scale cap $\varepsilon_0 \le \varepsilon_{\mathrm{sys}}(M, R, \rho, \delta_0)$ on the coarsest scale, J-free and closed downward, under which the large-field clause of (1) at the wall holds on the skew torus for every J of the class (6)	proved; a proviso of the torus theorem on its parameters, met by the class (6); not a hypothesis of Theorem A; enters (H3) on D_J and (SK), hence (K_0)	[SUP]; [Ba1, (1.14)]; [Ba2, (1.3)]; Remark 5.5	–
The odd-block clause of the degree census of the shell model, inside the κ-Cauchy estimate (item (10))	proved; enters (10) only, hence (RES$_0$) and (TIGHT$_0$)	[SUP]; [SW3]; Section 1.5	–
The object and the closure			
Geometry: $5R$ integral, descent to D_J, $\lvert Rx\rvert_\infty \le \frac{7}{5}\lvert x\rvert_\infty$, density of G_0 (Lemma 2.2)	proved	here; [K2, pp. 337–338]	–
Correlation inequalities (Lemma 2.12)	(a)–(c) printed with proof; (d) proved here (sketched in [BFS]; [K2, (3.28)] stated without proof); the killing clauses proved here, with no limit	[BFS, Thms. 6.1, 6.2, pp. 275–281]; [K2, Prop. 3.1, (3.24)–(3.29)]; here	–
Current representation, Gaussian domination (Lemma 2.13)	proved	here	–

Input	Proof	Source	Rests on
Killing as a limit of masses (Lemma 2.7); cutting lemma (Lemma 2.14); passive covariance (Lemma 2.10)	proved	here; [K2, Thm. 3.2]	–
King's cubes and the boxes: interlacing, monotonicity in J, iterated limits (Lemma 2.16, Corollary 2.17)	proved	here; [G]; [K2, (3.3)–(3.9), pp. 328–334]	–
King's cube limit with curvature sources on King's cubes C_J^K, own-frame 2-form reading (item (13))	proved, at zero shell mass, the long trees by (P5) and (P8), with Proposition 1.2; (B2), (B5), (B6) and the regrouping of the cut families proved; asserted in print without a complete argument (proof transferred from [K1] by assertion; the per-ordering photon boundary step and (3.9) replaced)	[CA]; [K2, Thm. 2.1, §3.1, pp. 331–332, (3.9)]; [K1, Thm. 3.1, Prop. 3.6, p. 665]	(1), (5k), (10f), (8k), (P5), (P8), Proposition 1.2
King's state on curvature sources, [K2, Thms. 2.2, 2.3(iii)] and the parts of [K2, (3.40)–(3.44)] used, at $h = 0$, and the continuity lemma (Theorem 3.3(i)–(iv), Lemma 3.2), on King's cubes	proved given (13)	here; [K2, Thm. 3.2]	(13), Lemma 2.16
Closure from the generators to $O(3)$, reflection (Lemma 6.2, Theorem 6.3)	proved (at $g = 0$ on King's cubes, given (13))	here; [G]	(13), Lemma 2.16
Translations (Theorem 3.3(v) at $h = 0$; item (14))	proved given (13)	here	(13)
Right-continuity of King's state in the scalar mass at the points of E_W (item (15)): monotonicity in the bare mass coefficient ([BFS, Thm. 6.1], [K2, (3.25)]) and in the cube ([K2, Thm. 3.2]), real-analyticity of the limiting difference of the bare mass coefficients, the cube against the dead-shell torus at a fixed stopping scale with the endpoint step of [SW1]	proved; used for Theorem A at E_W only, for the $E(3)$ clause as for the rotations	[SUP]; [BFS, Thm. 6.1]; [K2, (3.25), Thm. 3.2]; [SW1]	(13), (8b), (12a)
The torus comparison (K_0)			

Input	Proof	Source	Rests on
Involution S (Lemma 4.3); closing at one spacing (Theorem 4.5); Corollary 4.6	proved from (F_0)	here	–
(F_0) (the one-sided forward bound in total form on the mass-increasing family, with its source-linear constant on D_J of upper limit at most zero and the constant on the rotated torus $R^{-1}D_J$ equal to it up to $o(1)$) and the existence on the axis lattice for walled curvature ratios	supplied by Theorem 4.8	[CT]	(1)–(5)
(1) $(BAL)_5$ with the large-field clause of [K1, (3.4)], the walled form (P1) included	proved (asserted in print without a complete argument), the readings (B1)–(B6) proved; on the skew tori the replacement of the propagators of [Ba2, Prop. 2.6] through (P7); the repaired cube step through the cube step of (8e) at zero shell mass, with (B3), (B5), (B6) and the regrouping of the cut families, proved; its block-factor-5 repairs proved, (5l)(v), (vi) included; for the contour families of the flows of [CT] with Proposition 1.3; the large-field clause at a non-zero wall proved with $c_* < 1$, with Proposition 1.4 outside E_W, for the rotated lattice also Proposition 1.3, on the skew torus under the final-scale cap, with King's p. 677 sentence in its proved form (5f)	[AS]; [SUP]; [Ba1, Ba2, Ba3, Ba4]; [CT]	(8e); (P7) on the skew tori; Propositions 1.3, 1.4
(2) (H1*): the two-flow comparison bound in per-flow normal form, with a positive power of the scale ratio	proved	[CT]; [Ba3, Prop. 1]; [K1, p. 665]	(2a), (2b), (3), (5)

Input	Proof	Source	Rests on
(2a) first-layer slice-kernel bound (cubic-Hermite extension)	proved	[CT]	–
(2b) restricted contraction rule; local-term chain under it	proved	[CT]; [SUP]	–
(3) (B_Δ): hybrid block-spin kernels	proved, with (F2) in its completed form, (5l)(viii), (ix) included	[CT]; [K1, Thm. 3.3]	–
(H3), (SK): King's upper bound, skew instances	proved from (1), (5k) (RD_J) and relative to (1) (D_J), with (5l)(i); [SKEW-U] with (P7); with Propositions 1.3, 1.4 (outside E_W), on D_J under the final-scale cap of the class (6); posed in the Cameron–Martin form of the effective action, reached from King's vertex form by a proved bridge (5i)	[CT]; [K1, Thm. 3.1]	(1), (5f), (5k), (P7), Propositions 1.3, 1.4
(H4): hybrid lower bound (item (5i))	proved from (1), (3), (CT), [SKEW-L], with its own steps (5l)(i)–(iii), the lower half of (5f) for the hybrid flow and the bridge of (5i), with Proposition 1.3	[CT]; compare [K2, Prop. 4.1]	(1), (3), (5f), Proposition 1.3
(H5): polynomial lower bound, at $h = 0$ and at the wall	proved ($h = 0$, RD_J, item (5b), under (1); D_J; wall); (5b) asserted in print without a complete argument	[Ba1, (1.14)]; [AS]; [CT]	(1); at the wall also (4c) and the skew instances (SK) under the final-scale cap
(CT): convention bridge	(a) proved; (b) proved in the consumed form; (c) not an input	[CT]; [K1, (3.33), (2.8), (3.21), (3.35)]; [K2, p. 330]	(5f); its rate (4c) through (4) only
(4) (H6W): existence of the walled value on the axis lattice (the limit input of Theorem 7.8)	proved, (4a)–(4d) included, with Proposition 1.4 outside E_W through the large-field clause of (1) at the wall (not used at E_W); King's strict degree property and his treatment of the sources, which it uses, proved	[H6]; [SUP]; [RS]; [K1, Thm. 2.1, Prop. 3.6, pp. 664–665, 668, 677]; Remark 7.9	(1), (5), (5k), Proposition 1.4

Input	Proof	Source	Rests on
(5a), (5c), (5e) block-spin operators, counterterm conditions and Ward identities, Reisz in corrected form	(5a), (5c) printed with proof, the three asserted steps of (5a) proved under (1), (5k); for the transfer of [Ba4, §§2–4] see (5l)(iv); (5e) proved from the printed proofs (forest identity = Zimmermann's formula, cited; membership proved, Remark 8.1)	[K1, Thm. 3.3, (3.32)–(3.33)]; [Ba4, Lemmas 2.1, 2.2, 2.4]; [Ba3, (2.18)–(2.28), Prop. 2.1]; [Re1, Re2]; [CT]	(1), (5k)
(5b) [Ba1, (1.14)]; (5f) source-shift uniformity of King's constant	(5b) proved under (1); (5f) proved under (1), (5k), King's p. 677 sentence included, in the form of the small-field half of [K1, Thm. 3.1] at fixed sources (King's and the centred flow; lower half for the hybrid flow); asserted in print without a complete argument	[Ba1, (1.14)]; [K1, Thm. 3.1]; [AS]; [SUP]	(1); (5f) also (5k)
(5d) local limit law	textbook, cited	[LL, Chap. 2]	–
(5g) [K1, Prop. 3.6], one-flow instance	proved (its printed proof ends in an assertion); its degree count proved: the strict degree property in the componentwise form of [Ba3, (3.5)] for King's flow, the unweakened form ($D \geq 0$, bounded zero-degree count) for the hybrid flow; the strict form $D' > 0$ not consumed	[CT, H6]; [SUP]; [Ba3, (3.5)]; [K1, pp. 662–665]	(4), (5k)
(5h) [K1, Thm. 3.5(i)]: false as printed; restricted form	proved	[RS]	(4b)
(5i), (5j) hybrid lower bound; free-propagator bound, Bałaban-side completions, coarse-covariance decay	proved (the last with the completion (F1))	[CT]; [SUP]	(1)

Input	Proof	Source	Rests on
(5k) King's asserted steps: [K1, Thm. 3.1]; [K1, Props. 3.7–3.10, Lemma 4.5, (3.72), (3.88)] and the remainder bounds of [K1, Thm. 3.3]; its photon clause; the source readings	proved in the forms consumed, the source readings included; asserted in print without a complete argument; (4.33) proved, Proposition A.4	[AS]; [K1, pp. 656, 664–677]; here	(1)
(5l) the own steps of [CT]: the cumulant bound on cubic and skew tori; the absorption in (H4); [Ba3, Prop. 1] for the hybrid flow; the transfer of [Ba4, §§2–4], read at the printed page; two repair steps, the determinants included; the constants of [Ba2, §3C] at block factor 5; two statements of (3); four steps of lower weight; the degree counts for block-vertex classes	proved; the vector half of (vii) with the reading (B4), proved	[CT]; [SUP]; [Ba2, Ba3, Ba4]	(1), (3)
(K_0): the torus comparison, curvature sector, on the class (6)	proved (closing implication and own steps (5l)), with (P7) through (1) on the skew tori, with Propositions 1.3, 1.4, outside E_W, under the final-scale cap of the class (6); its literature inputs that are asserted in print without a complete argument are all proved: (B4), King's two statements and, through the end-functional bound of (8e) in the cube step of (1), (B3), (B5), (B6)	Theorem 4.8	(1)–(5), (P7), Propositions 1.3, 1.4
Walled values, free direction, orientation independence at $g = 0$ (Definition 5.1, Lemmas 5.2, 5.3)	proved given (K_0)	here	(K_0), (13)
Theorems 6.1, 6.4, Corollary 6.5	proved, on King's cubes	here	(K_0), (TIGHT_0), (13), Lemma 2.16
The closing hypothesis (TIGHT_0)			
Photon boundary switch (Lemma 7.1)	proved	here; [LL, Chap. 4]	–

Input	Proof	Source	Rests on
The dead-shell bound (7.1) (Theorem 7.5)	proved	here	(13)
Two-level sandwich (Lemma 7.6)	proved	here; [K2, (3.25)]	–
$(\mathrm{RES}_0) \Rightarrow (\mathrm{TIGHT}_0)$ (Theorem 7.8)	proved	here	(K_0), (10c)
(7) the heavy-shell theorem: (8)–(11) $\Rightarrow (\mathrm{RES}_0)$ at one configuration (Theorem 7.10)	proved	here (Section 7.4)	(8)–(11)
(8) $(\mathrm{SW1}^\circ)_{\mathrm{int}}$: two-sided bounds on the capped stopping-scale integral, uniformly in $\kappa \geq \kappa_0$, for the free-boundary model	proved, with (F2), (F3), (F5) in their completed forms, the determinant clause of (8k)(ii) with (DOM), its upper bound on the histories with proper propagator domains with (P9); the photon legs of its per-step constants by (KA*), proved	[SW1]; [SUP]	(8a)–(8k), (1), (12), (P1)–(P3), (P9), (DOM), (KA*)
(8a) scale-free sub-domain bound; the earlier form false at small c_0	proved	[SW1]	–
(8f) counterterm offset $(\mathrm{CT})_{\mathrm{sh}}$	proved	[SW1, Lemma M(d)]	Δ_*
(8b) the short extensions written in [SW1]; (8c) the near-face sketch transplanted; (8e) the shell-model re-runs of the repaired steps	proved, (8e) with the mixed-class bound and (8k)(i), and with (B3), (B5), (B6) and the regrouping of the cut families, proved; the input [Ba2, Lemma 2.7] of (8c) proved under (1)	[SW1]; [SUP]; [K2, §5.2]	(1); (8e) also (P1), (8k)(i), (B3)
(8d) Bałaban's enumeration as a frame	proved (asserted in print without a complete argument)	[AS]; [Ba3, §§2–3]; [SW1]	(1)
(8g) (K-pt): pointwise first and mixed second difference bounds of the step-j line kernels	proved (two arguments on the torus; on the sub-domains in the restricted form used there)	[SW1, Lemma ward(b), Prop. TD(b)]	(8k)(iii)
(8h) regularity of the reflected gauge field in the sense of [K1, (3.5)]: not an input	false as printed; the form used (regular after a gauge; coercivity of the doubled operator for every field) proved	[SW1, Lemma refl(a),(b)]	–

Input	Proof	Source	Rests on
(8i) $(\mathrm{BAL})_5^{\kappa}$: the clause of (1) for the shell model (= (8e))	proved, all clauses, as (8e); its steps of the lower and the upper bound use (DOM) through (8k)(ii); on the histories of the upper bound with proper propagator domains its removal of the tail of the background with (P9)	[SW1]; [SUP]	(1), (P1), (8k)(i), (P9)
(8j) (LOC-R): light-side estimates on sub-domains at deep points where (L_x) holds; near the edges and corners of H' the localisation lemma, (8k)(vi) included	proved	[LOC, Prop. global]; [LT]; [SW1]	(8k)(iv)
(8k) the further steps of [SW1]: the perturbative bound on sub-domains; the cut members and the mismatch clause; the random-walk gradient bound; the wall reading in [LOC]; the far-block payment	proved; the determinant clause of (ii) proved on the torus and on the propagator domains, with (DOM); (i) for the M of Definition 2.1 and the corners of H' in (6), with (B5), (B6) and the regrouping of the cut families, proved, at order at most 13 (at order 14 it is (P9)); (ii) with (P1)–(P3), and (B3), proved	[SW1]; [LOC]; [SUP]	(5l)(i), (vii), (F3), (P1)–(P3), (DOM)
(9) $(\mathrm{SW2}^{\min})$: the disc bound, at $g = 0$	proved (matter factor 1)	here (Lemma 2.13; Theorem 7.10, Step 0)	–
(10) $(\mathrm{SW3}^{\circ})_{\mathrm{int}}$: κ-Cauchy property of the reduced small-field action	proved, with (F6) in its completed form	[SW3]	(10a), (10b), (10d)–(10g), (12); (8g), (8i)–(8k) through [SW1]
(10a) tail of the block-free determinant factor, torus form with the winding bracket	proved; its winding bookkeeping proved; the form without the winding bracket false on the torus	[SW3]	–
(10b) heavy-kernel bounds: kernel, mass, far pairs, heavy far field, inserted line, differentiated light estimate, near pairs in the dist^{-2} form	proved; the Yukawa heavy-pair clause and the full-covariance sup bound next to ∂U false, not used	[SW3]	(8)

Input	Proof	Source	Rests on
(10c) dead-shell chain for $U = T \setminus H'$	proved	here (Theorem 7.5)	–
(10d) the κ-derivative bound for deep clusters closed in matter	proved	[SW3, Lemma Σ(iv′)(d)]	(8g), (8i), (10g)(i), (vi), (5l)(x), (10f)
(10e) the further inputs of [SW3]; (10f) Bałaban's enumeration with one inserted vertex	(10e) proved; (10f) proved (asserted in print without a complete argument)	[SW3]; [AS]; [Ba3, §§2–3]	(1)
(10g) the further steps of [SW3]: the restricted [K1, (3.38)] for the shell model; the one-loop z^2-term with the block fluctuation line; the Hölder remainder with an open matter component; the photon-leg classes with an insertion; the lattice regime of the Hölder-leg scale sum; the first-moment bound in place of the regrouped Ward remainder; the source-linear members at the light local vertex; field-dependent members there	proved; (ii) uses [Ba3, p. 426], the reading (B1), proved; (iii), (iv) use (B3), proved	[SW3]; [SUP]; [Ba3, pp. 426, 441]	(1), (8)
(11) (SWT): threshold clause	proved (empty on the capped object)	[SW1, SW3]	–
(12) literature inputs of the shell estimates	(12a) printed with proof; (12b) not an input (Theorem 3.3); (12c), (12d), (12e) asserted in print without a complete argument, proved through (1), (5k), (8b), (8c), and (13) for (12d) on King's cubes; King's p. 677 sentence, (KA*) in (12e) and (B1)–(B6) in (12c) proved	[K1, K2, BFS, Ba1, Ba2, Ba3, Ba4]; [AS]	(1), (5k); (13) only for (12b) and for (12d) on King's cubes, not used by (8)–(11)
(RES_0) at one configuration	proved, through (8)–(11), the upper bound of (8) with (P9)	Theorem 7.10	(8)–(11)
(TIGHT_0)	proved, through (K_0), (13) and (8), with the statements of the literature printed with their proofs, outside E_W	Theorems 7.10, 7.8	(K_0), (8)–(12), (13)
Appendix			

Input	Proof	Source	Rests on
Large-field term in [K1, Thm. 3.1] (Proposition A.1)	proved	here	–
King's torus conjecture, scalar reading (Proposition A.2)	proved at $e = 0$: false as printed	here; [K2, (2.12), Thm. 2.4]	–
The Wick logarithm (Proposition A.3)	proved in [CT]; computation sketched here	here; [CT]	–
King's bound (4.33) (Proposition A.4)	proved	here; [K1, (4.5), p. 674]	–
Membership in Reisz's class (Remark 8.1)	proved	here; [Re1, Defs. 2.2–2.5]; [Re2, (2-24), (5-1)]	–

In summary. Every implication in the chain

$$(1)\text{–}(5) \;\Rightarrow\; (\mathrm{K}_0), \qquad (8)\text{–}(11) \;\Rightarrow\; (\mathrm{RES}_0) \xrightarrow{(\mathrm{K}_0),\ (10c)} (\mathrm{TIGHT}_0),$$

$$(\mathrm{K}_0) + (\mathrm{TIGHT}_0) \;\Rightarrow\; \text{invariance under } \mathcal{R}_0 \;\Rightarrow\; O(3) \xrightarrow{(14)} E(3)$$

is proved: Theorem A follows from the inputs of the list of Section 1.3, at every point of King's range, at the points of E_W by right-continuity in the scalar mass. These inputs are proved here or in the main paper, the bound (DOM) of Section 1.4 for the step determinants on the propagator domains and the statements (P7)–(P9) behind item (1) on the skew tori, King's cube limit (13) and the upper bound of (8) included, or they are statements of the literature printed with their proofs; the steps (F1)–(F6) of the main paper are used in their completed forms (Section 8.5). The four statements of Section 1.5, behind King's cube limit (13), the contour families of the flows of [CT] and the large-field clause of (1) at the walls, and the odd-block clause of the degree census of the shell model, inside the κ-Cauchy estimate (10), are proved there. Section 8 states every input in plain words.

10. Open problems

Theorem A concerns curvature sources. The following problem is not solved in this supplement or in the main paper; it is not a hypothesis of Theorem A.

Scalar sources. Extend Theorem A to King's state with scalar sources: the generating functional of the Wick squares $:\phi^2:(h)$, $h \in C_c^\infty$ real, together with the curvature sources, on the cone $\mathcal{K}$ of Definition 2.8 and beyond it. Three statements are needed, none of which enters the curvature sector. (a) *The renormalisation of the Wick square.* With the free Wick constant alone the one-point function of $:\phi^2:$ diverges at order e^2 in perturbation theory (Proposition A.3), so a further vacuum counterterm, linear in h and depending on the lattice spacing only, has to be added to King's action, and the existence of King's cube limit and of the infinite-volume limit has to be proved for the functionals so renormalised, with a bound on the limit uniform in the cube for non-negative scalar sources (the analogues of [K2, Thm. 2.1, (3.43)]); King's renormalisation group, with the counterterm carried as an additional h-linear vacuum vertex, is the natural route. (b) *The torus comparison for scalar sources*: the equality of the two lattice limits of Hypothesis 4.1 for walled tori carrying a scalar source as well; by [CT] it follows from the inputs (H1)–(H5) of Section 4.3 and one further input of that part of the main paper, the convergence on the torus of the Wick-source functionals so renormalised. (c) *The closing statement for scalar sources*: that King's state on the pairs $(f, g) \in \mathcal{K}$ is the lower envelope of the walled tori, as (TIGHT_0) says for curvature sources. The reduction of (c) to a heavy-shell decoupling estimate at one configuration,

as in Section 7, is expected not to depend on $g = 0$; the heavy-shell estimate of Theorem 7.10 is proved at $g = 0$ only.

Appendix A. Corrections to the literature

We list the points of the literature that affect its use in this supplement, with the corrected or restricted forms used; the use sites say so in one sentence each. None of them changes the statement of King's cube limit at $h = 0$ that we use, item (13), proved in [CA] (Section 8).

A.1. **The large-field term in Theorem 3.1 of** [K1]**.** Theorem 3.1 of [K1] compares the lattice generating functional on a torus T with the small-field integral $I^{(k)} := \int (dA_k)(d\phi_k)\chi_k \exp[-S^{(k),1}]$ of the effective action after k block-spin steps, $\eta := 5^k \varepsilon_K$. As printed on p. 655 the upper bound reads

$$Z^{\varepsilon_K}(T_{\varepsilon_K}, g, h) \le \int (dA_k)(d\phi_k)\chi_k \exp\big[- S^{(k),1} + C\eta^\sigma |T|\big] + \exp\big[- p(\eta)^2 + C|T|\big], \qquad (3.4)$$

with $p(\eta) = b_0(1 + \log \eta^{-1})^p$, followed by "Once again, C depends on g, h and $|T|$". *Convention P.* The exponent p of the small-field function is a fixed integer, $p = 6$ throughout; any fixed integer $p \ge 4$ would do. [K1, (3.41)] prints "for any σ, since $p > 1$", King's standing inequality on p; $p \ge 2$ is forced by [Ba1, (3.54)] (kept by [K1, p. 676]) and $p \ge 4$ by [Ba2, pp. 557, 594] ("$p > 2$"; "$2p \ge (d+1)r$", $r \ge 2$, $d = 3$). The consumers of the large-field clause use p only through $p(\delta_{r(k)})^2$ growing at least quadratically in k; [K1, (3.41)] uses $p > 1$. The lower bound (3.3) has the small-field integral with the exponent $\mp C\eta^\sigma|T|$ (in a lower bound the sign must be $-$). The second term of (3.4) is an additive large-field remainder; it is used in [K1] only at fixed volume ((3.10)–(3.13) there), and in the same way in King's torus comparison [K2, (4.7)], where it is harmless.

Proposition A.1. *Suppose (3.3)–(3.4) hold with C independent of $|T|$, and $I^{(k)} \ge e^{-C''|T|}$. Then*

$$-C\eta^\sigma \le |T|^{-1} \log\big(Z/I^{(k)}\big) \le \max\{C\eta^\sigma,\ C + C'' - p(\eta)^2/|T|\} + |T|^{-1}\log 2.$$

At fixed η the right side tends to $C + C''$ as $|T| \to \infty$, not to $O(\eta^\sigma)$.

Proof. $Z/I \le e^{C\eta^\sigma|T|} + e^{-p^2 + (C+C'')|T|}$ and $\log(x+y) \le \max(\log x, \log y) + \log 2$. □

Hence the printed bounds do not give a two-sided representation of the free energy per unit volume by a small-field effective integral at a fixed coarse scale, uniformly in the volume, and arguments that need such a representation cannot be based on them. In this supplement the volume is fixed wherever these bounds are used (in (F)), and the large-field term is absorbed on the left by a polynomial lower bound, as in [CT, Prop. 7.1]. The constant of [K1, Thm. 2.1(ii)] likewise depends on $|T|$, so King's theorems give no bound on torus functionals uniform in the torus, and none is used here.

A.2. **King's torus conjecture: Theorem 2.4 of** [K2] **as printed.** In [K1, K2] the curvature source is $F(g) = \sum_p \varepsilon^3 g(p) F(p)$ with one function g on all plaquettes [K2, (2.12)], and in Theorem 2.4 of [K2] the source is rotated as a scalar.

Proposition A.2 (the scalar reading)**.** *Let $e = 0$ and $F^{sc}_\Lambda(g) := \sum_p \varepsilon^3 g(c(p))F(p)$ with a single real function g on all plaquettes. Let $R = R_{12}(\theta_0)$, $\ell = 50MJ$ and $g(x) = \cos(2\pi x_1/\ell)$. Then* $\lim_k \langle e^{F^{sc}(g)}\rangle_{\Lambda_k, D_J} \ne \lim_k \langle e^{F^{sc}(g)}\rangle_{R\Lambda_k, D_J}$.

Proof. At $e = 0$ the photon is the centred Gaussian with covariance $\delta_{\alpha\beta}(-\Delta_\varepsilon + \mu_0^2)^{-1}$ on the torus, so $\langle e^{F^{sc}(g)}\rangle = e^{\frac12 \operatorname{Var} F^{sc}(g)}$. On a lattice with frame (f_α), $\sum_{\alpha<\beta} F(f_\alpha, f_\beta) = \langle u, B\rangle$ for a continuum field strength $F = dA$, $B = \star F$, where $u := \sum_{\alpha<\beta} f_\alpha \times f_\beta$. For the plane wave g only the momenta $\pm k_0$, $k_0 = (2\pi/\ell)e_1$, contribute, and the lattice symbols converge, so $\lim_k \operatorname{Var} F^{sc}(g) =$

$\frac{\ell^3}{2}\,|k_0|^2(|u|^2-u_1^2)/(|k_0|^2+\mu_0^2)$. For Λ_k, $u=(1,-1,1)$ and $|u|^2-u_1^2=2$; for $R\Lambda_k$, $u=R(1,-1,1)=(\frac{7}{5},-\frac{1}{5},1)$ and $|u|^2-u_1^2=\frac{26}{25}$. The limiting variances differ by the factor $\frac{25}{13}$. □

With 2-form sources sampled in each lattice's own frame (Definition 2.8) the limit on either lattice is the variance of $\int\langle G,F\rangle$, which does not depend on the frame, and the obstruction disappears. This reading is consistent with King's proof, which treats the source components separately (Remark 2.9); it is the form of Hypothesis 4.1 and of Theorem 4.8. The second error in the printed statement concerns the Wick constant and is the subject of the next subsection. Since the printed statement is false on both counts, we refer to it as King's torus conjecture; the statement that is true is Theorem 4.8, proved in [CT] from the inputs of Section 4.3. We note also that Theorem 2.4 is not, by itself, a statement about King's infinite-volume state: the passage from tori to the state [K2, (2.24)–(2.25)] is explicitly conjectural there ("we can expect that a cluster expansion would give the following result"), and neither of the two routes of this supplement uses it (Remark 4.10).

A.3. **The Wick square and the constant** $\ln 5$**.**

Proposition A.3 (the logarithm)**.** *In formal perturbation theory, King's model on $\varepsilon\mathbb{Z}^3$ with the free Wick constant C_ε alone satisfies, at order $e^2\lambda^0$,*

$$\langle{:}\phi^2{:}_{\varepsilon\mathbb{Z}^3}(h)\rangle=\frac{e^2}{4\pi^2}\,\ln(1/\varepsilon)\int h+O(1).$$

The coefficient does not depend on m or μ_0. (A complete proof, with the lattice regularisation carried through, is [CT, Thm. 9.2, App. B]*; we indicate the computation.)*

Computation. In the normalisation of Definition 2.4 each real component of ϕ has propagator $(p^2+m^2)^{-1}$ at $e=0$. Write the one-loop correction to the inverse propagator as $\Sigma(p)$; the mass counterterm removes $\Sigma(0)$. The photon-exchange part is $-e^2\int\frac{d^3k}{(2\pi)^3}\frac{(2p-k)^2}{((p-k)^2+m^2)(k^2+\mu_0^2)}$; writing $(2p-k)^2=2p^2+2(p-k)^2-k^2$ and using $\int\frac{d^3k}{(2\pi)^3}\frac{1}{k^2(p-k)^2}=\frac{1}{8|p|}$, one finds $\Sigma(p)-\Sigma(0)=-\frac{e^2}{4}|p|+O(m+\mu_0)$ for $1\ll|p|\ll\varepsilon^{-1}$, the lattice regularisation changing only momentum-independent terms and terms of order εp^2. Inserting, $\langle|\phi(0)|^2\rangle-2\mathsf{C}_\varepsilon=-2\int\frac{d^3p}{(2\pi)^3}\frac{\Sigma(p)-\Sigma(0)}{(p^2+m^2)^2}+O(e^4)=\frac{e^2}{2}\cdot\frac{1}{2\pi^2}\ln(1/\varepsilon)+O(1)$ at order e^2. □

Consequences.

(a) For h with $\int h\neq0$ the printed generating functionals with scalar sources have no finite continuum limit beyond the curvature sector, at least in perturbation theory: for $h\geq0$, $h\not\equiv0$, Jensen's inequality gives $\langle e^{{:}\phi^2{:}(h)}\rangle\geq e^{\langle{:}\phi^2{:}(h)\rangle}\to\infty$. This affects [K1, Thm. 2.1] and [K2, Thms. 2.1–2.3, (3.43), (4.2)] for scalar sources with $\int h\neq0$. In the cube, King subtracts the Dirichlet constant [K2, (2.13)] and, at distance >1 from the boundary, the free one, which differ by a convergent amount only. The statement of [K1, p. 677] that the sources are proportional to $(5^k\varepsilon)^\alpha$, $\alpha>0$, in the small-field region does not hold for the marginal graph that produces the logarithm. Walled ratios and the curvature sector are unaffected, since the h-linear c-numbers cancel in them; for scalar sources a further vacuum counterterm, linear in h, is needed (Section 10).

(b) King's proof of [K2, Thm. 2.4] compares the effective action generated from spacing ε by a first block-spin step with rotated blocks with the effective action generated from the rotated lattice of spacing 5ε [K2, (4.3)–(4.9)]. If the logarithm is removed by a vacuum counterterm $a_\varepsilon\mathsf{S}(h)$, linear in h, the two generating functionals carry the multipliers $e^{-a_\varepsilon\mathsf{S}(h)}$ and $e^{-a_{5\varepsilon}\mathsf{S}(h)}$ with $a_\varepsilon-a_{5\varepsilon}\to\frac{e^2}{4\pi^2}\ln5$, so the comparison [K2, (4.6)] must carry the c-number $\frac{e^2}{4\pi^2}\ln5\int h$, and as printed it cannot hold for $\int h\neq0$. Since a walled ratio contains the

wall source $-w$ with $\int w > 0$, this enters the proof of walled comparisons even though their statement is free of the counterterm. In the equal-spacing formulation of Section 4 and of [CT] the two lattices carry the same counterterm, and the issue does not arise.

A.4. **The bound (4.33) of** [K1]**.** In the proof of [K1, Lemma 4.5] (p. 674) the positivity (4.33) is derived through (4.36)–(4.37), which treat the block average as a Fourier multiplier; that step is false as printed. The conclusion holds. We use King's setting at the unit scale of step k: Ω is a periodic box of the unit lattice $\mathbb{Z}^3$ that is a union of blocks of side 5, the background vanishes, $\Delta^{(k)}$ and $\Delta^{(k+n)}$ are the translation-invariant operators of the flow on Ω, Q is the block average, so that $Q^*Q = P$ is the orthogonal projection onto block-constant functions, and $-\Delta$ is the lattice Laplacian of Ω. King's coefficients satisfy $a_j \geq \frac{24}{25}a$, and $\Delta^{(j)}(p)^{-1} = a_j^{-1} + \sum_l |u(p+l)|^2/\Delta^\eta(p+l)$ [K1, (4.5)], where u is the symbol of the average over a unit block of the fine lattice of spacing η, $\sum_l |u(p+l)|^2 = 1$, the sum runs over $l \in 2\pi\mathbb{Z}^3$ with $p+l$ in the Brillouin zone of the fine lattice, and $\Delta^\eta(q) = \sum_\mu 4\eta^{-2}\sin^2(\eta q_\mu/2) + m_k^2$ with $m_k^2 \geq 0$.

Proposition A.4 (the bound (4.33))**.** *For $s \in [0,1]$, all k, n and $m_k^2 \geq 0$, the operator $X(s) := s\Delta^{(k)} + (1-s)\Delta^{(k+n)} + \frac{a}{25}Q^*Q$ on $\ell^2(\Omega)$ satisfies $X(s) \geq \gamma_0$, where $\gamma_0 := \min\left(4c_\Delta \sin^2\frac{\pi}{10}, \frac{a}{25}\right)$ and $c_\Delta := \left(\frac{25\cdot 3\pi^2}{24a} + \frac{\pi^2}{4}\right)^{-1}$.*

Proof. (i) For $|t| \leq \pi$, $4\sin^2(t/2) \geq 4t^2/\pi^2$, so $\Delta^\eta(q) \geq 4|q|^2/\pi^2$ on the fine Brillouin zone; and $|p_\mu + l_\mu| \geq |p_\mu|$ for $|p_\mu| \leq \pi$, $l_\mu \in 2\pi\mathbb{Z}$. With $\sum_l |u(p+l)|^2 = 1$ this gives $\Delta^{(j)}(p)^{-1} \leq a_j^{-1} + \frac{\pi^2}{4}|p|^{-2}$, and since $|p|^2 \leq 3\pi^2$ and $a_j \geq \frac{24}{25}a$, $\Delta^{(j)}(p) \geq c_\Delta|p|^2 \geq c_\Delta \sum_\mu 4\sin^2(p_\mu/2)$, the symbol of $-\Delta$. So $s\Delta^{(k)} + (1-s)\Delta^{(k+n)} \geq c_\Delta(-\Delta)$. (ii) On one block $B = \{0,\dots,4\}^3$ the Neumann Laplacian is a sum of three path-graph Laplacians, whose smallest non-zero eigenvalue is $\lambda_1 = 4\sin^2\frac{\pi}{10}$; hence $\sum_{x\in B}|\phi(x) - \phi_B|^2 \leq \lambda_1^{-1}\sum_{\{x,x'\}\subset B}|\phi(x) - \phi(x')|^2$, the sum over nearest-neighbour pairs, ϕ_B the block mean. Summing over the blocks, whose inner bonds are bonds of Ω, gives $\|(1-P)\phi\|^2 \leq \lambda_1^{-1}\langle\phi, -\Delta\phi\rangle$. (iii) Therefore $\langle\phi, X(s)\phi\rangle \geq c_\Delta\lambda_1\|(1-P)\phi\|^2 + \frac{a}{25}\|P\phi\|^2 \geq \gamma_0\|\phi\|^2$. For the photon the same holds componentwise. □

A.5. **Further points.** Each of the following is false as printed, or is used in print beyond what is proved there; the corrected or restricted form used is named, with the item of Section 1.3 at which it enters.

(i) [K1, Thm. 3.5(i), (3.38)] fails on the h-linear vacuum family of order $(2,0)$; the restricted form, Theorems R-i–R-iii of [RS], is used (item (5h)). [K1, Thm. 3.4] fails as printed and is not used; the form needed at a fixed wall is proved in [H6] (item (4)).

(ii) The even-n line of [K1, (3.96)] and the middle line of [K1, (3.98)] are false as printed; King's transfer of the quadratic form and of the determinants to the cube is used in the form of the difference at two cutoffs, without these lines (items (4d), (13)).

(iii) [K1, (3.78)] fails on certain vacuum graphs, whose contribution is a removed constant; the strict degree property of King's renormalised graphs is used in the componentwise form of [Ba3, (3.5)] (items (4), (5g)).

(iv) King's credit of one degree for each photon field contracted outside a subgraph [K1, p. 664, (3.68)–(3.70)] is false for the counts of subgraphs; a count that splits the degree replaces it (item (5l)(x)).

(v) The statement [K1, p. 666] that the divergences cancel for each ordering of the renormalised graphs is false in its per-ordering reading, the violation being of order one in exact computations; the identities that hold are those for full propagators [Ba3, (2.24)–(2.27)] and Bałaban's resummed organisation, and no identity for a fixed ordering or scale assignment of King's graphs is used anywhere on the chain (Section 8.3).

(vi) The bound [K1, (4.21)] is false as printed for $m > 0$; only its truncation to the alias-free term, in a corrected form, is used, and (4.28) is not used (item (5k)). The step (4.36)–(4.37) in the proof of [K1, Lemma 4.5] is false as printed; its conclusion (4.33) is Proposition A.4.

(vii) The statement [K1, p. 677] that the sources are proportional to $(5^k\varepsilon)^\alpha$ in the small-field region does not hold for the marginal graph that produces the logarithm of Proposition A.3; the inclusion of the sources is used as the small-field half of [K1, Thm. 3.1] at fixed sources, proved in [SUP] (items (4), (5f), (5k)).

(viii) The image-sum identity [K2, (3.9)] is false as printed, together with (3.3) and (3.8) there, for the weight printed there; it holds with the Dirichlet weight f_D, with which King's flow on C_J^K is run (Definition 2.15, item (13)). The bound [K2, (3.43)] fails in King's Wick ordering at $\int h_+ > 0$ at order e^2 (Section A.3); the printed [K2, (4.5)] has a sign slip and is replaced in [CT] (item (5i)).

(ix) In [Ba2], the count (3.45) and the first inequality of (3.46) are false as printed at every block factor, and a repaired count of the cubes replaces them (item (5l)(vii)); Lemma 2.3 fails at block factor 5 for small couplings and is used in the form that holds there ((B5) of Section 8.2); (2.99) has a sign misprint that does not matter, since only $C(\tilde{B}+\tilde{A})^{-1} = C(\tilde{B})^{-1} - W^{(j)}$ is used ((F2) of Section 8.5); Lemma 2.7 with (2.113), read on King's cube with the Dirichlet weight, fails near the Dirichlet face and at the first scale, and the corrected form is proved in [CA] (Section 8.2).

(x) In [Ba3], the sentence on p. 445 that the divergent vacuum graphs are cancelled by the vacuum-energy counterterms holds at zero background only; at King's backgrounds the vacuum families with the mismatch vertex are bounded by (P2) of Section 1.4. A bound on the remainder of the regrouped Ward identity of p. 441 with a gain in degree is false as so posed; the first-moment bound (10g)(vi) is used instead.

(xi) In [Ba4, (2.27)] the lower bound π^2 for the gap is false as printed; the correct value $4N^2\sin^2(\pi/2N)$ suffices (item (12c)).

(xii) In Reisz's theorems [Re1, Re2] the naturalness clause is false as printed and is replaced by a condition proved for the graphs consumed, and three printed statements need repair in the three-dimensional form (item (5e), Section 8.4).

(xiii) The following statements of the main paper are false as written there and are not used: the regularity of the reflected gauge field in the sense of [K1, (3.5)], [SW1, Lemma refl(a)] (item (8h)); the sub-domain clause with the constant 16 and a rate in the penetration depth, which fails at small c_0 (item (8a)); the tail bound without the bracket for the loops winding through U and with a right side proportional to C_A^2, which fails on the torus (item (10a)); the Yukawa form of the heavy-pair clause, which fails for first-layer pairs, and a sup bound for the full covariance next to ∂U in the lattice regime (item (10b)); the clause of the sup bound with block data in L^1 ((F5)(a)); a localisation statement of the first written proof of (8k)(i), replaced by a weaker form; the lift-distance clause of (5l)(ix); the decaying form of the sharp box-zone extension of the slice kernels (item (2a)); the hybrid-flow lemma as printed in [CT], which has a gap (item (5l)(iii)); and one display of [SW1] for a non-constant wall, which has an exact replacement (item (8k)).

A.6. **Minor points.** (i) The inequalities [K2, (3.24)–(3.29)] are stated for ratios of expectations of $e^{r^2(h)+\mathrm{i}A(g)}$; we use them in the form of [BFS, Thms. 6.1, 6.2], which also covers killed sites and the two-level walls. (ii) King's vertex list [K1, (3.23)] is printed "for $2 < m$"; the action [K1, (2.1)–(2.4)], from which [K1, (3.14)–(3.17)] build the list, gives it for every even $m \geq 2$, so the seagull vertex of order two belongs to the model, and with the list as printed one renormalisation statement for the photon two-point graphs fails. (iii) In the proof of [K2, Thm. 3.2] (p. 334) the sentence that the remaining expression "is exactly $\langle \mathcal{O}(ig,h)\rangle_{N,D}^{C_J}$" names the state of the box $C_J \cap \Lambda_k$ of Definition 2.5(a), in which every site of the smaller cube is alive; between two of King's

renormalisation-group cubes it would be one layer off, and the inequalities (3.32)–(3.33) between them follow instead from Lemma 2.16 (Corollary 2.17).

References

[AS] Bałaban's and King's asserted steps at block factor 5: (I) Bałaban's same-scale estimates, with Proposition 2.6 of [Ba2] at a non-zero background and the change of domain of his propagators behind it, on the cubic and on the skew tori, and the renormalisation bound for the operator with a scalar wall; (II) Bałaban's enumeration of renormalised graphs as a frame, with one inserted vertex, and King's cut of long lines on the torus; (III) King's propagator and remainder bounds in the forms consumed. Proofs from the printed proofs, for King's in-block contours; for the contour families of the flows of [CT] the estimates are Proposition 1.3 of Section 1.5, proved in [SUP]. Parts of the main paper, in their versions of 19 October 2026.

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[CA] King's cube limit at zero scalar source on the renormalisation-group cube: the Cauchy argument of [K2, §3.1] in ratio form, with the ultraviolet construction on the cube, the scalar boundary remainders, the line bounds of the flow with the Dirichlet weight, the transfer [K1, (3.90)–(3.98)] to the cube, the classes with curvature vertices, the cut of the long lines on the cube, and the long trees in the difference of two cutoffs, at the faces of the cube and away from them, with the bound at long trees for the families with one inserted field whose graphs contain block vertices (Proposition 1.2, proved in [SUP]). Parts of the main paper, in their versions of 19 October 2026.

[CT] The corrected torus theorem for the three-dimensional $U(1)$ Higgs model: the curvature sector, as one chain. Part of the main paper, version 4.8, as of 19 October 2026.

[G] The two lattice conventions for King's Neumann–Dirichlet cube: an interlacing lemma, with its corollaries. Manuscript, as of 17 October 2026.

[H6] Existence of the walled torus limit on the axis lattice for curvature sources. Part of the main paper, version 3.12; its main statement is unchanged since version 3.8.

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[LOC] Bałaban's local inverses with the mass profile $m^2+n\psi+\kappa\mathbf{1}_U$ at a flat face, light-side half. Manuscript, version 3.

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[RS] The restricted form of King's renormalisation theorem: Theorems R-i–R-iii with the degree count of the merged counterterm vertices. Part of the main paper, version 2.9.

[SUP] Completions of [CT, H6, SW1, SW3]: the residues named in Section 1.3, among them the sub-domain form of the perturbative bound, at order fourteen included, its cut members and mismatch clause at King's backgrounds, the vacuum families with the mismatch vertex, Bałaban's enumeration block by block at higher orders and the steps of the κ-Cauchy estimate; the large-field clause of King's upper bound at the walls; and the non-polynomial terms of King's effective action at the frozen wall. Proofs of the large-field clause at the walls with Propositions 1.4 and 1.3 of Section 1.5, on the skew torus under the final-scale cap; the proofs of Propositions 1.2, 1.3, 1.4, of the final-scale cap, of the right-continuity of King's state in the scalar mass, of

the readings (B1)–(B6), of the regrouping of the cut families, of (KA*) and of King's two statements in their consumed forms. Parts of the main paper, in their versions of 19 October 2026.

[SW1] King I Theorem 3.1 with the shell mass inside the covariance: a direct proof of $(\mathrm{SW1}^\circ)_{\mathrm{int}}$, both lines. Part of the main paper, version 2.8.20. It cites the two proofs of (8g), states the reflection lemma in the form used (8h), and states (LOC-R) with its flat-face condition (8j); it names the estimates near the edges and corners of H' as a further input, whose consumed forms (8j) supplies through [LT], (8k)(vi) included.

[SW3] The κ-Cauchy property of King's reduced small-field action on the capped stopping-scale set: a proof of $(\mathrm{SW3}^\circ)_{\mathrm{int}}$. Part of the main paper, version 2.5x; it contains the proof of (10d). It uses the conventions and numbered results of [SW1] in version 2.8.7, which names the inputs of its main theorem explicitly and states its first-difference clauses on sub-domains at points away from the complement of the sub-domain; the statements of [SW1] cited in [SW3] are read with the inputs (8g), (8i)–(8k) (Section 8.1).

[TE] One configuration of the heavy-shell estimate implies the closing hypothesis. Manuscript, version 3.